



Fermi Liquids and Luttinger Liquids

From quasiparticles to collective modes

School of Physics, Nanjing University

Quantum Transport Theory 2

1D and mesoscopic quantum physics

Lecture notes for the course taught by

Associate Professor Gu Zhang

Adiabatic continuity and spectral weight; density response and nesting; bosonization and finite-size algebra; interacting ground states; correlation functions, boundary tunnelling, duality, and renormalization.

September 2026

Graduate lecture notes

Contents

How to read these notes **iii**
Reading orientation . . . iii

1 Fermi liquids and quasiparticles **1**
1.1 Adiabatic motivation and the question of interactions . . . 1
1.2 What a quasiparticle hypothesis requires . . . 1
1.3 Energy shift, residue, and the pole expansion . . . 2
1.4 The interacting Fermi surface and effective mass . . . 3
1.5 Spectral weight and the meaning of a lifetime . . . 4
1.6 Why the phase space for decay shrinks near the Fermi surface . . . 5

2 Density response and the one-dimensional problem **7**
2.1 Causality, the drive, and the density operator . . . 7
2.2 Fourier normalization and the bilinear commutator . . . 8
2.3 Momentum selection in the equilibrium response . . . 9
2.4 The Lindhard function and its particle–hole interpretation . . . 9
2.5 Static response and the geometry of nesting . . . 10
2.6 The two- and three-dimensional static functions . . . 11
2.7 Why one dimension is different . . . 12
2.8 Self-consistent spin-resolved response . . . 13
2.9 Pairing response, fluctuations, and the route to a Luttinger liquid . . . 14

3 Density modes and bosonization **16**
3.1 Why particle–hole density modes? . . . 16
3.2 Linearizing the two Fermi points . . . 17
3.3 Outline of the bosonization construction . . . 18
3.4 Normal ordering and the density central term . . . 19
3.5 Canonical oscillators and their vacuum . . . 20
3.6 The energy carried by a density oscillator . . . 21
3.7 From a density commutator to a fermion vertex . . . 22
3.8 The local density and the Klein factors . . . 24
3.9 Canonical fields and finite-size zero modes . . . 25
3.10 The canonical commutator from the mode sum . . . 26
3.11 The physical meaning of the canonical fields . . . 27
3.12 Checking the density–vertex commutator . . . 28
3.13 Vertex reordering and equal-branch statistics . . . 29
3.14 Splitting the phase and summing the finite-size contraction . . . 30
3.15 The mixed anticommutator and the ultraviolet normalization . . . 31

4 The interacting Luttinger Hamiltonian **33**
4.1 The free Hamiltonian in canonical fields . . . 33
4.2 Forward interactions and the Luttinger parameters . . . 37
4.3 From the interaction Hamiltonian to the ground state . . . 40

5 Bogoliubov diagonalization and the ground state 42
5.1 Explicit Bogoliubov diagonalization . . . 42
5.2 The vacuum conditions in the original oscillator basis . . . 44
5.3 Solving and normalizing the paired state . . . 46
5.4 Physical content of the interacting ground state . . . 47

6 Correlations, temperature, and propagating modes 49
6.1 Why transport requires electron correlation functions . . . 49
6.2 Gaussian reduction of a fermionic vertex correlator . . . 51
6.3 Canonical-field correlation functions at zero temperature . . . 53
6.4 The bulk electron has two light-cone factors . . . 56
6.5 Finite temperature: thermal contractions before analytic continuation . . . 57
6.6 Checking propagation directly in the oscillator expansion . . . 59

7 Reflecting boundaries and weak-link tunnelling 62
7.1 A reflecting endpoint and its unfolded field . . . 62
7.2 Moving the bias into a tunnelling phase . . . 63
7.3 Weak-link current: evaluating the Fourier integral . . . 65

8 Weak backscattering, duality, and the domain of expansion 66
8.1 The opposite limit: weak backscattering in a connected wire . . . 66
8.2 The actual small parameter of a tunnelling expansion . . . 67
8.3 Temperature regularization and the linear conductance . . . 68
8.4 Boundary duality and the domain of the two expansions . . . 69

9 Boundary renormalization: from fluctuations to transport 71
9.1 A local scatterer still feels the entire wire . . . 71
9.2 Two boundary operators and two scaling dimensions . . . 72
9.3 One RG step, including its normalization . . . 74
9.4 What is generated beyond the linear flow? . . . 75
9.5 Crossover scales and when to stop the flow . . . 76
9.6 Boundary stability and the meaning of duality . . . 78
9.7 From the running coupling back to conductance . . . 78

A Notation and useful consistency checks 80

References 81

How to read these notes

The central question is how an interacting Fermi system should be described at low energies. A Fermi liquid retains long-lived excitations with a nonzero overlap with a single fermion. A one-dimensional Luttinger liquid instead organizes its low-energy Hilbert space into collective density modes. The second description follows from explicit operator algebra, not from a change of statistics of microscopic electrons.

The first two sections develop the quasiparticle and response tests. The bosonization sections construct the density modes, fermion vertices, and their zero modes before adding forward scattering. Diagonalization then determines the interacting vacuum and correlation exponents. The final sections distinguish a bulk fermion, a reflecting endpoint, a weak link, and weak backscattering. Figure P.1 gives the dependencies.

Basic second quantization, equilibrium statistical mechanics, and Fourier analysis are assumed. Units are $\hbar = k_B = 1$, unless restored for interpretation; $e > 0$ is the magnitude of the electronic charge, so electrons carry $-e$. An energy has units of inverse time, a velocity has units of energy times length, and the branch fermion has units of length$^{-1/2}$.

The early spin-resolved response example has two spin species. From the bosonization construction onward the homogeneous LL has one spinless channel, with right and left branches $r = +1$ and $r = -1$. The normal-ordering Fermi-sea reference $|0_{\mathrm{FS}}\rangle$ is distinct from the interacting vacuum $|\Omega_{\mathrm{LL}}\rangle$. Local density formulas use excess particle numbers relative to the declared reference. Finite-size calculations retain their separate number sectors and boundary prescription.

The symbol $a > 0$ is a short-distance cutoff, L is system size, and $\tau_a = a/u$ is a cutoff time. The canonical fields (ϕ, θ) obey $[\phi(x), \partial_{x'}\theta(x')/\pi] = i\delta(x-x')$ on the infinite line. Their physical branch phases $\varphi_r = r\phi - \theta$ are different from the propagating eigenfields $\chi_r = \phi/\sqrt{K} - r\sqrt{K}\theta$.

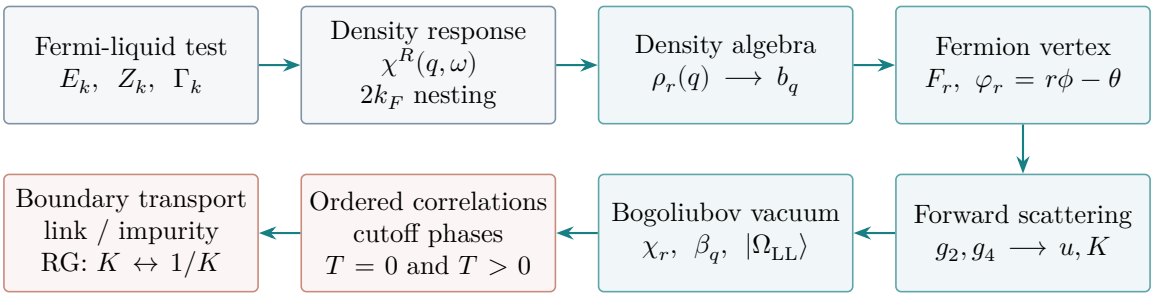


Figure P.1: Concepts and dependencies. The failure of a regular quasiparticle expansion motivates collective modes; their algebra fixes the vertices, the Hamiltonian fixes propagation and the vacuum, and the geometry fixes which correlator enters transport.

Reading orientation

The Fermi-liquid discussion can be read alongside Henrik Bruus and Karsten Flensberg, *Many-Body Quantum Theory in Condensed Matter Physics: An Introduction* [1]. For Luttinger liquids, the first reading is Thierry Giamarchi, *Quantum Physics in One Dimension* [2]. The detailed finite-size construction is developed by Jan von Delft and Herbert Schoeller in *Bosonization for Beginners—Refermionization for Experts* [3]. The reading list also names C. L. Kane and M. P. A. Fisher in a cancelled, unfinished entry beginning "Trans…"; it does not specify an identifiable complete title. The boundary-transport discussion below stands on its displayed derivations.

1 Fermi liquids and quasiparticles

The problem is whether an interacting metal retains resolvable particle-like excitations. We use adiabatic continuity as motivation, then test it through the retarded self-energy, spectral weight, and available decay phase space.

1.1 Adiabatic motivation and the question of interactions

Begin with noninteracting fermions in two dimensions. A translationally invariant one-particle state can be labelled by (k_x, k_y), and independent occupations follow the Fermi–Dirac distribution. Individual particles do not scatter from one another. Figure 1.1 shows the geometric starting point and the proposed continuation of levels. Once particles interact, close encounters can scatter them between states. The question is whether a description in terms of particle-like excitations survives this mixing. A one-dimensional well gives a preliminary example of adiabatic continuity.

$$H(t) = H_{\text{kin}} + V(x,t). \tag{1.1}$$

$$V(x,t) = -V_0(t)\exp[-x^2/(2x_0^2)], \qquad x_0 > 0, \tag{1.2}$$

where $V_0(t) > 0$ is an energy and x_0 sets the well's width.

$$i\partial_t \left|\psi(t)\right\rangle = H(t)\left|\psi(t)\right\rangle, \qquad i\partial_t \psi(x,t) = \left[-\frac{\partial_x^2}{2m} + V(x,t)\right]\psi(x,t). \tag{1.3}$$

For an instantaneous bound level n, in a gauge with vanishing Berry connection along this one-parameter path, the adiabatic approximation is

$$\left|\psi_{\text{ad},n}(t)\right\rangle \approx \exp\left[-i\int_{t_0}^{t} E_n(t')\,dt'\right]\left|n;t\right\rangle. \tag{1.4}$$

$$H(t)\left|n;t\right\rangle = E_n(t)\left|n;t\right\rangle. \tag{1.5}$$

Here $\psi(x,t) = \langle x|\psi(t)\rangle$, m is the particle mass, and $\hbar = 1$. For a normalized nondegenerate isolated level, a sufficient local slowness condition is

$$\frac{\left|\left\langle m;t\right|\dot{H}(t)\left|n;t\right\rangle\right|}{|E_m(t) - E_n(t)|^2} \ll 1 \quad (m \neq n). \tag{1.6}$$

In another gauge the solution also contains $\exp[i\gamma_n(t)]$, with $\gamma_n = i\int_{t_0}^{t}\langle n;t'|\partial_{t'}n;t'\rangle\,dt'$. The absence of this written phase therefore requires a gauge choice, not the absence of geometric phases in quantum mechanics. The approximation suppresses transitions between the instantaneous levels. An isolated level must remain separated from the levels with which the drive can mix it; the crossing sketch illustrates a failure of this simple argument. For the many-body motivation, continuation also presumes that no phase transition intervenes. A macroscopic metal is gapless, so the isolated-level theorem is a motivation for, rather than a proof of, the Fermi-liquid hypothesis. The operative test will be a narrow low-energy fermion resonance with nonzero spectral weight.

1.2 What a quasiparticle hypothesis requires

At fixed k_x, imagine turning on interactions while following the dispersion in k_y, as in Fig. 1.2. The proposed correspondence labels each low-energy interacting excitation by a state of the free system and leaves a well-defined Fermi surface. Corresponding states should be connectable under a continuous change of the interaction. Their lifetime must be long enough to resolve their

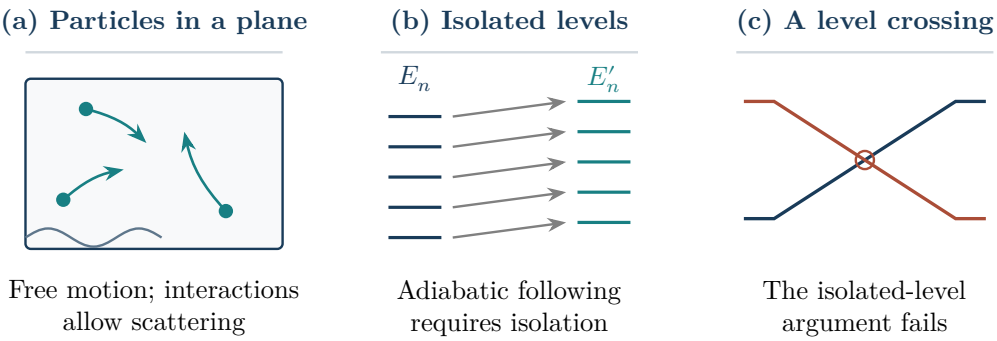


Figure 1.1: Schematic motivation for adiabatic continuity. Particle trajectories in two dimensions motivate the interaction question; the level diagrams distinguish smooth following from a closing separation. The arrows map instantaneous levels and do not represent an energy measurement.

energy. Weak interaction is a useful sufficient starting regime for perturbation theory, though small bare coupling is not a definition of every Fermi liquid. For one specified fermion spin component, evolve with $\mathcal{H} = H - \mu N$, define $\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - \mu$, and set

$$G^R(\mathbf{k}, t) = -i\Theta(t)\langle\{c_{\mathbf{k}}(t), c^\dagger_{\mathbf{k}}(0)\}\rangle, \tag{1.7}$$

$$G^R(\mathbf{k}, \omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} G^R(\mathbf{k}, t). \tag{1.8}$$

The frequency ω is an energy in these units. Θ denotes the step function, and the equilibrium expectation includes the interacting ground state or thermal state as specified.

$$G^R_0(\mathbf{k}, \omega) = \frac{1}{\omega - \xi_{\mathbf{k}} + i0^+}. \tag{1.9}$$

$$G^R(\mathbf{k}, \omega) = \frac{1}{\omega - \xi_{\mathbf{k}} + i0^+ - \Sigma^R(\mathbf{k}, \omega)}. \tag{1.10}$$

$$\omega = \xi_{\mathbf{k}} \qquad \text{(free resonance energy)}. \tag{1.11}$$

$$\omega = \xi_{\mathbf{k}} + \operatorname{Re}\Sigma^R(\mathbf{k}, \omega) \qquad \text{(real resonance condition)}. \tag{1.12}$$

For a free eigenstate the amplitude decay time is $1/\eta \to \infty$. An interacting resonance has a finite decay time scaling inversely with the magnitude of its imaginary self-energy; the retarded sign and residue will be included explicitly below. The self-energy must be regular enough near the Fermi surface to permit the expansion used next. The assumed continuation should not encounter a ground-state phase transition or a crossing that destroys the chosen low-energy correspondence. Finally, the response to an external perturbation must remain controlled. This motivates the independent density-response test after the lifetime calculation.

1.3 Energy shift, residue, and the pole expansion

$$\Sigma^R(\mathbf{k}, \omega) = \Sigma_1(\mathbf{k}, \omega) + i\Sigma_2(\mathbf{k}, \omega). \tag{1.13}$$

$$G^R(\mathbf{k}, \omega) = \frac{1}{\omega - \xi_{\mathbf{k}} - \Sigma_1(\mathbf{k}, \omega) - i\Sigma_2(\mathbf{k}, \omega)}. \tag{1.14}$$

The real self-energy shifts the resonance energy, whereas a negative imaginary part broadens it and gives a finite lifetime. Define the shifted energy directly by

$$E_{\mathbf{k}} - \xi_{\mathbf{k}} - \Sigma_1(\mathbf{k}, E_{\mathbf{k}}) = 0. \tag{1.15}$$

$$\omega - \xi_{\mathbf{k}} - \Sigma_1(\mathbf{k}, \omega) = 0. \tag{1.16}$$

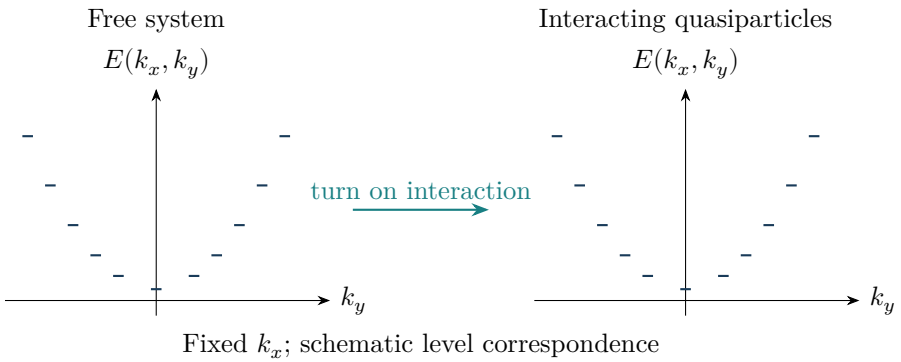


Figure 1.2: The source's fixed-k_x level correspondence, shown schematically. The points label states in the k_y direction. In the interacting system, their energy, spectral weight, and width must be obtained from the self-energy; the drawing does not assume that these quantities stay unchanged.

If the real solution is $E_{\mathbf{k}}$, then

$$E_{\mathbf{k}} - \xi_{\mathbf{k}} - \Sigma_1(\mathbf{k}, E_{\mathbf{k}}) = 0. \tag{1.17}$$

$$\omega - \xi_{\mathbf{k}} - \Sigma_1(\mathbf{k}, \omega) \approx \omega - \xi_{\mathbf{k}} - \Sigma_1(\mathbf{k}, E_{\mathbf{k}}) - \partial_\omega \Sigma_1|_{E_{\mathbf{k}}} (\omega - E_{\mathbf{k}}) \tag{1.18}$$

$$= (\omega - E_{\mathbf{k}}) \left[1 - \partial_\omega \Sigma_1|_{E_{\mathbf{k}}}\right]. \tag{1.19}$$

$$\omega - \xi_{\mathbf{k}} - \Sigma^R(\mathbf{k}, \omega) \approx (\omega - E_{\mathbf{k}}) \left[1 - \partial_\omega \Sigma_1|_{E_{\mathbf{k}}}\right] - i\Sigma_2(\mathbf{k}, E_{\mathbf{k}}). \tag{1.20}$$

The imaginary term is retained at its on-shell value: at a finite-energy resonance it need not vanish when the real denominator does. Its variation across a narrow peak is neglected in this approximation.

$$Z_{\mathbf{k}} = \left[1 - \partial_\omega \Sigma_1(\mathbf{k}, \omega)|_{\omega = E_{\mathbf{k}}}\right]^{-1}. \tag{1.21}$$

$$G^R_{\mathrm{qp}}(\mathbf{k}, \omega) \approx \frac{Z_{\mathbf{k}}}{\omega - E_{\mathbf{k}} + i\Gamma_{\mathbf{k}}}. \tag{1.22}$$

$$\Gamma_{\mathbf{k}} = -Z_{\mathbf{k}} \Sigma_2(\mathbf{k}, E_{\mathbf{k}}) \geq 0. \tag{1.23}$$

Thus the linewidth depends on both the imaginary self-energy and the real self-energy derivative through $Z_{\mathbf{k}}$.

$$\frac{Z_{\mathbf{k}}}{\omega - E_{\mathbf{k}} + i\Gamma_{\mathbf{k}}} = \frac{1}{(\omega - E_{\mathbf{k}})/Z_{\mathbf{k}} - i\Sigma_2(\mathbf{k}, E_{\mathbf{k}})}. \tag{1.24}$$

The expansion presumes a smooth real part, $Z_{\mathbf{k}} > 0$, and sufficiently weak frequency variation of the imaginary part over the peak. A well-resolved low-energy quasiparticle additionally requires $\Gamma_{\mathbf{k}}/|E_{\mathbf{k}}| \to 0$ as $E_{\mathbf{k}} \to 0$. The full Green function also has an incoherent contribution; Eq. (1.22) does not describe its entire frequency range.

1.4 The interacting Fermi surface and effective mass

The shifted energy still obeys

$$E_{\mathbf{k}} - \xi_{\mathbf{k}} - \Sigma_1(\mathbf{k}, E_{\mathbf{k}}) = 0. \tag{1.25}$$

The lifetime scales as $\tau_{\mathbf{k}} \sim \Gamma_{\mathbf{k}}^{-1}$; a precise decay convention is specified in the spectral discussion. Two questions remain: does turning on the interaction preserve the local ordering of levels, and are the resulting excitations long lived? For the first question, focus near a Fermi surface away from the band's top and bottom. These low energies are especially relevant to transport.

$$E_{\mathbf{k}_F} = 0. \tag{1.26}$$

$$\xi_{\mathbf{k}_F} + \Sigma_1(\mathbf{k}_F, 0) = 0. \tag{1.27}$$

Zero energy is measured from the chemical potential; it is not a second subtraction of μ. For the following scalar mass formula assume an isotropic dispersion and write $k = |\mathbf{k}|$ and $k_F = |\mathbf{k}_F|$. Implicit differentiation of Eq. (1.17) gives

$$\frac{dE_k}{dk} = \frac{d\xi_k}{dk} + \partial_k \Sigma_1(k, E_k) + \partial_\omega \Sigma_1(k, E_k) \frac{dE_k}{dk}. \tag{1.28}$$

$$[1 - \partial_\omega \Sigma_1(k, E_k)] \frac{dE_k}{dk} = \frac{d\xi_k}{dk} + \partial_k \Sigma_1(k, E_k). \tag{1.29}$$

$$E_k \approx (k - k_F) Z_{k_F} \frac{\partial}{\partial k} [\xi_k + \Sigma_1(k, 0)]\Big|_{k_F} \tag{1.30}$$

$$= \frac{k_F}{m^*}(k - k_F). \tag{1.31}$$

For a bare parabolic band $\epsilon_k = k^2/(2m)$,

$$\frac{m}{m^*} = Z_{k_F} \left[1 + \frac{m}{k_F} \, \partial_k \Sigma_1(k, 0)|_{k_F} \right]. \tag{1.32}$$

Locally, if $v_F^* = k_F/m^* > 0$, increasing k still increases the excitation energy, while interactions renormalize its slope or effective mass. This local positive slope does not prove a global no-crossing theorem. In an anisotropic band, the useful statement is instead $\mathbf{v}_F^* = Z_{\mathbf{k}_F}[\nabla_{\mathbf{k}} \xi_{\mathbf{k}} + \nabla_{\mathbf{k}} \Sigma_1(\mathbf{k}, 0)]_{\mathbf{k}_F}$. The residue alone determines the mass only when the relevant momentum derivative vanishes.

1.5 Spectral weight and the meaning of a lifetime

$$A(\mathbf{k}, \omega) \equiv -2 \operatorname{Im} G^R(\mathbf{k}, \omega) = \frac{-2\Sigma_2(\mathbf{k}, \omega)}{[\omega - \xi_{\mathbf{k}} - \Sigma_1(\mathbf{k}, \omega)]^2 + \Sigma_2(\mathbf{k}, \omega)^2}. \tag{1.33}$$

Keeping the derivative factor in the real denominator gives

$$A_{\mathrm{qp}} \approx \frac{-2\Sigma_2(\mathbf{k}, E_{\mathbf{k}})}{(\omega - E_{\mathbf{k}})^2 / Z_{\mathbf{k}}^2 + \Sigma_2(\mathbf{k}, E_{\mathbf{k}})^2}. \tag{1.34}$$

$$A_{\mathrm{qp}}(\mathbf{k}, \omega) = -2 \operatorname{Im} G_{\mathrm{qp}}^R(\mathbf{k}, \omega) \tag{1.35}$$

$$= \frac{2 Z_{\mathbf{k}} \Gamma_{\mathbf{k}}}{(\omega - E_{\mathbf{k}})^2 + \Gamma_{\mathbf{k}}^2}. \tag{1.36}$$

The half-width at half maximum is $\Gamma_{\mathbf{k}} = -Z_{\mathbf{k}} \Sigma_2$, as shown in Fig. 1.3. The full width is $2\Gamma_{\mathbf{k}}$. The unrenormalized estimate $|\Sigma_2|$ applies only if $Z_{\mathbf{k}}$ is replaced by one. Closing the inverse-frequency contour below the real axis for $t > 0$ gives

$$G_{\mathrm{qp}}^R(\mathbf{k}, t) = \int \frac{d\omega}{2\pi} \frac{Z_{\mathbf{k}} e^{-i\omega t}}{\omega - E_{\mathbf{k}} + i\Gamma_{\mathbf{k}}} = -i Z_{\mathbf{k}} \Theta(t) e^{-iE_{\mathbf{k}} t - \Gamma_{\mathbf{k}} t}, \tag{1.37}$$

$$\tau_{\mathrm{amp}} = \Gamma_{\mathbf{k}}^{-1}, \qquad \tau_{\mathrm{prob}} = (2\Gamma_{\mathbf{k}})^{-1}. \tag{1.38}$$

The latter describes the decay of the squared pole amplitude; neither is automatically a transport relaxation time. Restoring units puts $\hbar$ in both lifetime numerators.

$$A(\mathbf{k}, \omega) = A_{\mathrm{qp}}(\mathbf{k}, \omega) + A_{\mathrm{inc}}(\mathbf{k}, \omega). \tag{1.39}$$

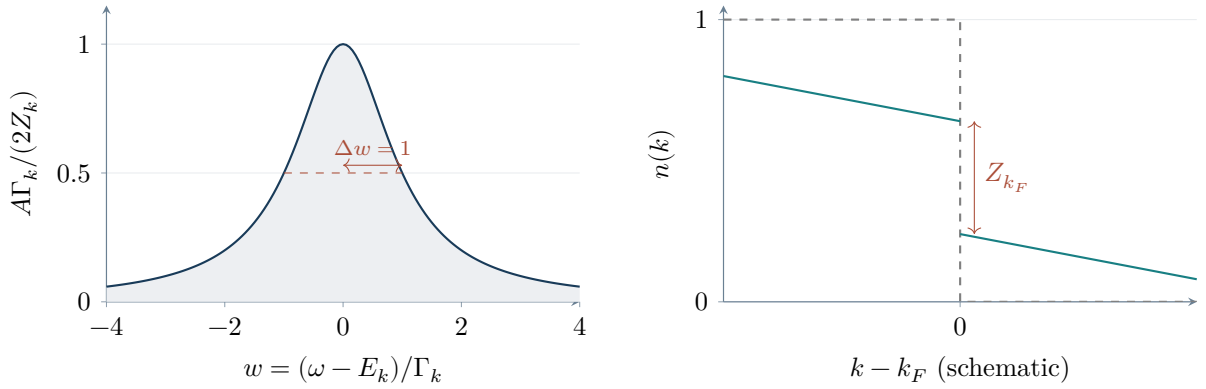


Figure 1.3: Left: the normalized coherent Lorentzian from Eq. (1.36); the shaded profile visualizes the coherent peak, and its half-maximum is reached at $|\omega - E_k| = \Gamma_k$. Right: a schematic momentum distribution, showing a free unit step (dashed) and the reduced interacting jump Z_{k_F} (solid). The smooth background is illustrative and carries the incoherent contribution.

$$\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} A_{\mathrm{qp}}(\mathbf{k}, \omega) = Z_{\mathbf{k}}. \tag{1.40}$$

If the residue varies weakly over the momentum window of interest, replace it by $Z_{\mathbf{k}_F}$. For a narrow peak, its energy half-width must also be small compared with the energy scale over which the pole parameters vary. If that scale is specified as a momentum width k_0, the dimensionally consistent condition is $\Gamma_{\mathbf{k}} \ll v_F^* k_0$.

$$\int \frac{d\omega}{2\pi} A_{\mathrm{qp}}(\mathbf{k}, \omega) \approx Z_{\mathbf{k}_F}. \tag{1.41}$$

$$\int \frac{d\omega}{2\pi} [A_{\mathrm{qp}}(\mathbf{k}, \omega) + A_{\mathrm{inc}}(\mathbf{k}, \omega)] = 1. \tag{1.42}$$

$$\int \frac{d\omega}{2\pi} A_{\mathrm{inc}}(\mathbf{k}, \omega) = 1 - Z_{\mathbf{k}}. \tag{1.43}$$

The unit sum follows from the equal-time canonical anticommutator, equivalently from $G^R(\omega) \sim 1/\omega$ at large frequency. A low-energy projection does not itself obey the full sum rule. The coherent/incoherent split is useful near a well-resolved pole; extrapolating a fitted Lorentzian far beyond its regime need not leave an everywhere-positive residual spectrum. At zero temperature,

$$n(\mathbf{k}) = \int_{-\infty}^{0} \frac{d\omega}{2\pi} A(\mathbf{k}, \omega), \qquad n(k_F^-) - n(k_F^+) = Z_{k_F}, \tag{1.44}$$

provided the incoherent part is continuous across the Fermi surface. The free unit step becomes a reduced jump, also illustrated in Fig. 1.3.

1.6 Why the phase space for decay shrinks near the Fermi surface

At $T = 0$, states below the Fermi surface are occupied and those above it are empty. Add a particle at energy $\varepsilon = \xi_{\mathbf{k}} > 0$ and ask which scattering processes can destroy it. Figure 1.4 shows why Pauli blocking becomes restrictive near the surface.

$$\mathcal{H} = \sum_{\mathbf{k}} \xi_{\mathbf{k}} c_{\mathbf{k}}^\dagger c_{\mathbf{k}} + \frac{1}{2\mathcal{V}} \sum_{\mathbf{k},\mathbf{p},\mathbf{q}} U(\mathbf{q}) c_{\mathbf{k}+\mathbf{q}}^\dagger c_{\mathbf{p}-\mathbf{q}}^\dagger c_{\mathbf{p}} c_{\mathbf{k}}. \tag{1.45}$$

Here $\mathcal{V} = L^d$ is the volume, and $\mathbf{q}$ is the momentum transferred between the particles. For one spin component the matrix element must be antisymmetrized; additional spin channels must be

summed explicitly when present. With $\mathcal{M}_{\mathbf{kp}\to\mathbf{k}'\mathbf{p}'}$ the antisymmetrized matrix element and each final state counted once, the probability decay rate is

$$\frac{1}{\tau_{\mathrm{prob},\mathbf{k}}} = 2\Gamma_{\mathbf{k}} = 2\pi \sum_{\mathbf{p};\{\mathbf{k}',\mathbf{p}'\}} |\mathcal{M}_{\mathbf{kp}\to\mathbf{k}'\mathbf{p}'}|^2 n_{\mathbf{p}}(1-n_{\mathbf{k}'})(1-n_{\mathbf{p}'})\delta_{\mathbf{k}+\mathbf{p},\mathbf{k}'+\mathbf{p}'} \times \delta(\xi_{\mathbf{k}}+\xi_{\mathbf{p}}-\xi_{\mathbf{k}'}-\xi_{\mathbf{p}'}). \tag{1.46}$$

For the interaction above, $\mathcal{M}$ includes $\mathcal{V}^{-1}$ and direct minus exchange amplitudes. The Kronecker delta imposes momentum conservation; the Dirac delta imposes energy conservation. The occupation factors are $n_{\mathbf{p}} = \Theta(-\xi_{\mathbf{p}})$ at $T = 0$.

$$\varepsilon + \xi_{\mathbf{p}} = \xi_{\mathbf{k}'} + \xi_{\mathbf{p}'} > 0. \tag{1.47}$$

$$\xi_{\mathbf{p}} < 0. \tag{1.48}$$

$$\xi_{\mathbf{k}'} + \xi_{\mathbf{p}'} < \varepsilon. \tag{1.49}$$

$$0 < \xi_{\mathbf{k}'} + \xi_{\mathbf{p}'} < \varepsilon \longrightarrow 0. \tag{1.50}$$

Thus all available final energies lie in a shrinking region. If the angular/matrix-element factor can be approximated by a finite constant C, the energy phase space becomes

$$\mathcal{J}(\varepsilon) = C\int_{\mathbb{R}^3} d\varepsilon_1 d\varepsilon_2 d\varepsilon_3\, \Theta(-\varepsilon_1)\Theta(\varepsilon_2)\Theta(\varepsilon_3)\delta(\varepsilon+\varepsilon_1-\varepsilon_2-\varepsilon_3) \tag{1.51}$$

$$= C\int_{\mathbb{R}^2} d\varepsilon_1 d\varepsilon_2\, \Theta(-\varepsilon_1)\Theta(\varepsilon_2)\Theta(\varepsilon+\varepsilon_1-\varepsilon_2) \tag{1.52}$$

$$= C\int_{-\varepsilon}^{0} d\varepsilon_1 \int_0^{\varepsilon+\varepsilon_1} d\varepsilon_2 \tag{1.53}$$

$$= \frac{C}{2}\varepsilon^2. \tag{1.54}$$

The geometric factor requires its own check. A representative local energy-shell integral written in the notes is

$$C_d(\varepsilon) \;\propto\; \int_{|\mathbf{q}_\perp|<\Lambda_\perp} \frac{d^{d-1}q_\perp}{\sqrt{q_\perp^2 + (\varepsilon/v_F)^2}}. \tag{1.55}$$

Here the parallel momentum shift is of order ε/v_F, the separation of nearby energy shells; $\mathbf{q}_\perp$ spans transverse directions. This is a geometric estimate, not a complete scattering rate. When C tends to a finite nonzero constant,

$$\tau_{\mathrm{prob},\mathbf{k}} \propto \varepsilon^{-2}, \tag{1.56}$$

so the lifetime diverges at the Fermi surface. This constant-factor assumption is dimension dependent. Setting $b = |\varepsilon|/v_F$ gives, for the displayed estimate,

$$C_3 \propto 2\pi \int_0^{\Lambda_\perp} \frac{q\,dq}{\sqrt{q^2+b^2}} = 2\pi(\sqrt{\Lambda_\perp^2+b^2} - b), \tag{1.57}$$

$$C_2 \propto 2\int_0^{\Lambda_\perp} \frac{dq}{\sqrt{q^2+b^2}} = 2\,\mathrm{arsinh}(\Lambda_\perp/b). \tag{1.58}$$

The second expression has a logarithm as $b \to 0$. Consequently a generic two-dimensional Fermi liquid can have logarithmic corrections to the ε^2 rate. In one dimension there is no transverse integration to justify replacing this factor by a constant. The elementary phase-space argument must not be used to prove Fermi-liquid behavior there. These statements concern a regular low-energy dispersion and weak, nonsingular effective scattering; different interactions or critical fluctuations require a separate analysis.

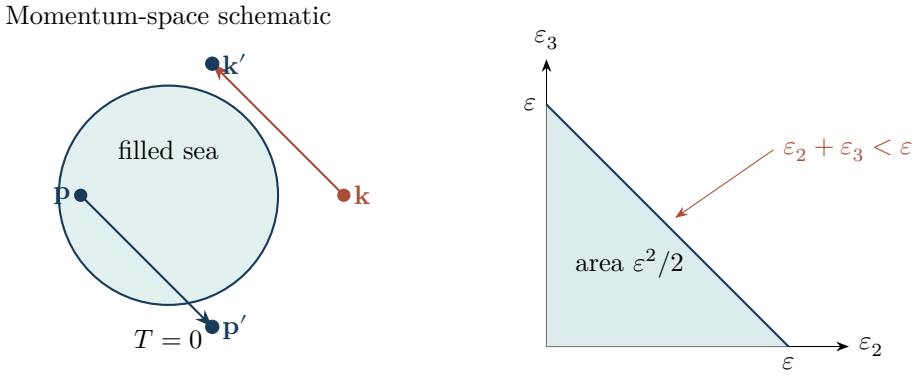


Figure 1.4: A particle above the sea scatters with an occupied state into two unoccupied states (left, schematic). Energy conservation and Pauli blocking restrict the positive final energies to a triangle (right). The triangle's area is $\varepsilon^2/2$; angular phase space and the interaction matrix element supply the factor C, which need not be constant in every dimension.

2 Density response and the one-dimensional problem

A long-lived fermion is only one part of the low-energy description. We now derive the density response of the free reference, examine how Fermi-surface geometry enters, and explain why the one-dimensional limit calls for a different set of elementary modes.

2.1 Causality, the drive, and the density operator

The density response asks how easily an external potential polarizes the Fermi sea. It will expose a limitation of perturbation theory that a single-particle pole calculation alone does not settle.

$$H_{\mathrm{p}}(t) = H_0 + \int d^d x\, h(\mathbf{x},t) B(\mathbf{x}). \tag{2.1}$$

Here h is an external source and B is the operator to which it couples.

$$\langle A(\mathbf{x},t)\rangle_{H_{\mathrm{p}}} = \langle A(\mathbf{x},t)\rangle_0 + \int d^d x'\, dt'\, \chi^R_{AB}(\mathbf{x}-\mathbf{x}', t-t') h(\mathbf{x}',t') + O(h^2). \tag{2.2}$$

$$\chi^R_{AB}(\mathbf{x}-\mathbf{x}', t-t') = -i\Theta(t-t')\langle [A(\mathbf{x},t), B(\mathbf{x}',t')]\rangle_0. \tag{2.3}$$

The step function enforces causality. Examples include current or noise as the quantity of interest, and charge density or magnetic moment as the coupled operator. Appropriate choices produce conductance and thermoelectric response coefficients. The noncommutativity measures how the source can change the observable, as in the elementary relation between position and a momentum translation operator. Indeed, the interaction-picture expansion $U_I(t) \approx 1 - i\int^t H_{\mathrm{drive},I}(t')dt'$ gives $\delta\langle A(t)\rangle = -i\int^t dt' \langle [A(t), H_{\mathrm{drive}}(t')]\rangle_0$. Difference coordinates in Eq. (2.3) require a homogeneous stationary reference; otherwise retain both positions and times. For a spatially nonuniform potential energy per particle,

$$H_{\mathrm{drive}}(t) = + \int d^d x\, U_{\mathrm{ext}}(\mathbf{x},t)\rho(\mathbf{x}) = \frac{1}{\mathcal{V}} \sum_{\mathbf{q}} U_{\mathrm{ext}}(\mathbf{q},t)\rho(-\mathbf{q}). \tag{2.4}$$

$$\rho(\mathbf{q}) = \int_{\mathcal{V}} d^d x\, e^{i\mathbf{q}\cdot\mathbf{x}} \rho(\mathbf{x}) = \sum_{\mathbf{k}} c^\dagger_{\mathbf{k}+\mathbf{q}} c_{\mathbf{k}}, \qquad \rho(\mathbf{q})^\dagger = \rho(-\mathbf{q}). \tag{2.5}$$

The bilinear moves a particle from $\mathbf{k}$ to $\mathbf{k}+\mathbf{q}$ and leaves a hole at $\mathbf{k}$. The Fourier mode is not itself an elementary boson oscillator. ρ here is particle-number density; for electron charge

density $\rho_Q = -e\rho$, an electrostatic voltage couples through $U_{\mathrm{ext}} = -eV_{\mathrm{el}}$. The intensive response is

$$\chi^R_{\rho\rho}(\mathbf{q},t) = -\frac{i\Theta(t)}{\mathcal{V}}\langle[\rho(\mathbf{q},t),\rho(-\mathbf{q},0)]\rangle_0. \tag{2.6}$$

In a noninteracting equilibrium reference,

$$c_{\mathbf{k}}(t) = e^{-i\xi_{\mathbf{k}}t}c_{\mathbf{k}}. \tag{2.7}$$

$$c^\dagger_{\mathbf{k}}(t) = e^{i\xi_{\mathbf{k}}t}c^\dagger_{\mathbf{k}}. \tag{2.8}$$

$$\rho(\mathbf{q},t) = \sum_{\mathbf{k}} c^\dagger_{\mathbf{k}+\mathbf{q}}(t)c_{\mathbf{k}}(t) \tag{2.9}$$

$$= \sum_{\mathbf{k}} e^{i(\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t}c^\dagger_{\mathbf{k}+\mathbf{q}}c_{\mathbf{k}}. \tag{2.10}$$

$$\chi^R_{\rho\rho}(\mathbf{q},t) = -\frac{i\Theta(t)}{\mathcal{V}}\sum_{\mathbf{k},\mathbf{p}} e^{i(\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t}\langle[c^\dagger_{\mathbf{k}+\mathbf{q}}c_{\mathbf{k}}, c^\dagger_{\mathbf{p}-\mathbf{q}}c_{\mathbf{p}}]\rangle_0. \tag{2.11}$$

The bracket is a commutator of even fermion bilinears, although each elementary c obeys anti-commutation relations.

2.2 Fourier normalization and the bilinear commutator

Start before imposing translation invariance:

$$\chi^R(\mathbf{x},\mathbf{x}';t,t') = -i\Theta(t-t')\langle[\rho(\mathbf{x},t),\rho(\mathbf{x}',t')]\rangle_0. \tag{2.12}$$

$$\chi^R(\mathbf{x},\mathbf{x}';t,t') = -i\Theta(t-t')\langle[\rho(\mathbf{x}-\mathbf{x}',t-t'),\rho(0,0)]\rangle_0. \tag{2.13}$$

Stationarity and translation invariance justify the second form. They do not by themselves imply that the response is even in space or time. In particular, retarded time ordering is asymmetric. With the project spatial Fourier sign,

$$\chi^R(\mathbf{q},\omega) = \int_{\mathcal{V}} d^dx \int_{-\infty}^{\infty} dt\, e^{i\mathbf{q}\cdot\mathbf{x}+i\omega t}\chi^R(\mathbf{x},t) \tag{2.14}$$

$$= -i\int_{\mathcal{V}} d^dx \int_0^{\infty} dt\, e^{i\mathbf{q}\cdot\mathbf{x}+i\omega t}\langle[\rho(\mathbf{x},t),\rho(0,0)]\rangle_0. \tag{2.15}$$

$$\rho(\mathbf{x},t) = \frac{1}{\mathcal{V}}\sum_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{x}}\rho(\mathbf{p},t). \tag{2.16}$$

In the continuum this is

$$\rho(\mathbf{x},t) = \int \frac{d^dp}{(2\pi)^d} e^{-i\mathbf{p}\cdot\mathbf{x}}\rho(\mathbf{p},t). \tag{2.17}$$

$$\chi^R(\mathbf{q},\omega) = -\frac{i}{\mathcal{V}^2}\int d^dx \int_0^{\infty} dt \sum_{\mathbf{p},\mathbf{p}'} e^{i(\mathbf{q}-\mathbf{p})\cdot\mathbf{x}+i\omega t}\langle[\rho(\mathbf{p},t),\rho(\mathbf{p}',0)]\rangle_0 \tag{2.18}$$

$$= -\frac{i}{\mathcal{V}}\int_0^{\infty} dt \sum_{\mathbf{p}'} e^{i\omega t}\langle[\rho(\mathbf{q},t),\rho(\mathbf{p}',0)]\rangle_0. \tag{2.19}$$

The spatial integral gave $\mathcal{V}\delta_{\mathbf{p},\mathbf{q}}$. For orthonormal single-particle labels a,b,c,d,

$$[c^\dagger_a c_b, c^\dagger_c c_d] = [c^\dagger_a c_b, c^\dagger_c]c_d + c^\dagger_c[c^\dagger_a c_b, c_d] \tag{2.20}$$

$$= c^\dagger_a\{c_b, c^\dagger_c\}c_d - c^\dagger_c\{c^\dagger_a, c_d\}c_b \tag{2.21}$$

$$= \delta_{bc}c^\dagger_a c_d - \delta_{ad}c^\dagger_c c_b. \tag{2.22}$$

$$\rho(\mathbf{q},t)=\sum_{\mathbf{k}} c^{\dagger}_{\mathbf{k}+\mathbf{q}}(t)c_{\mathbf{k}}(t)=\sum_{\mathbf{k}} e^{i(\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t}c^{\dagger}_{\mathbf{k}+\mathbf{q}}c_{\mathbf{k}}. \tag{2.23}$$

$$\rho(\mathbf{p}',0)=\sum_{\mathbf{k}'} c^{\dagger}_{\mathbf{k}'+\mathbf{p}'}c_{\mathbf{k}'}. \tag{2.24}$$

$$[\rho(\mathbf{q},t),\rho(\mathbf{p}',0)]=\sum_{\mathbf{k},\mathbf{k}'} e^{i(\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t}[c^{\dagger}_{\mathbf{k}+\mathbf{q}}c_{\mathbf{k}},c^{\dagger}_{\mathbf{k}'+\mathbf{p}'}c_{\mathbf{k}'}] \tag{2.25}$$

$$=\sum_{\mathbf{k},\mathbf{k}'} e^{i(\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t}\left(\delta_{\mathbf{k},\mathbf{k}'+\mathbf{p}'}c^{\dagger}_{\mathbf{k}+\mathbf{q}}c_{\mathbf{k}'}-\delta_{\mathbf{k}+\mathbf{q},\mathbf{k}'}c^{\dagger}_{\mathbf{k}'+\mathbf{p}'}c_{\mathbf{k}}\right) \tag{2.26}$$

$$=\sum_{\mathbf{k}} e^{i(\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t}\left(c^{\dagger}_{\mathbf{k}+\mathbf{q}}c_{\mathbf{k}-\mathbf{p}'}-c^{\dagger}_{\mathbf{k}+\mathbf{q}+\mathbf{p}'}c_{\mathbf{k}}\right). \tag{2.27}$$

2.3 Momentum selection in the equilibrium response

For the free homogeneous reference, $\langle c^{\dagger}_{\mathbf{k}}c_{\mathbf{p}}\rangle_0=\delta_{\mathbf{k},\mathbf{p}}n_F(\xi_{\mathbf{k}})$, where $n_F(\xi)=(e^{\beta\xi}+1)^{-1}$ and $\beta=1/T$.

$$\langle[\rho(\mathbf{q},t),\rho(\mathbf{p}',0)]\rangle_0=\delta_{\mathbf{p}',-\mathbf{q}}\sum_{\mathbf{k}} e^{i(\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t}[n_F(\xi_{\mathbf{k}+\mathbf{q}})-n_F(\xi_{\mathbf{k}})]. \tag{2.28}$$

$$\chi^R(\mathbf{q},\omega)=-\frac{i}{\mathcal{V}}\int_0^{\infty} dt\, e^{i\omega t}\sum_{\mathbf{p}'}\langle[\rho(\mathbf{q},t),\rho(\mathbf{p}',0)]\rangle_0 \tag{2.29}$$

$$=-\frac{i}{\mathcal{V}}\int_0^{\infty} dt\sum_{\mathbf{k}} e^{i(\omega+\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t}[n_F(\xi_{\mathbf{k}+\mathbf{q}})-n_F(\xi_{\mathbf{k}})]. \tag{2.30}$$

A drive changes the density distribution by creating particle–hole excitations. The equilibrium two-point function pairs $\mathbf{q}$ with $-\mathbf{q}$, because its total inserted momentum must vanish. This selection rule follows from translation invariance of the reference. It says that the two inserted momenta sum to zero; an equilibrium state need not have zero absolute momentum in every possible geometry.

2.4 The Lindhard function and its particle–hole interpretation

To retain the direct calculation, apply the same algebra once more:

$$[c^{\dagger}_a c_b,c^{\dagger}_c c_d]=[c^{\dagger}_a c_b,c^{\dagger}_c]c_d+c^{\dagger}_c[c^{\dagger}_a c_b,c_d] \tag{2.31}$$

$$=c^{\dagger}_a\{c_b,c^{\dagger}_c\}c_d-c^{\dagger}_c\{c^{\dagger}_a,c_d\}c_b \tag{2.32}$$

$$=\delta_{bc}c^{\dagger}_a c_d-\delta_{ad}c^{\dagger}_c c_b. \tag{2.33}$$

$$\sum_{\mathbf{p}}[c^{\dagger}_{\mathbf{k}+\mathbf{q}}c_{\mathbf{k}},c^{\dagger}_{\mathbf{p}-\mathbf{q}}c_{\mathbf{p}}]=\sum_{\mathbf{p}}\left(\delta_{\mathbf{k},\mathbf{p}-\mathbf{q}}c^{\dagger}_{\mathbf{k}+\mathbf{q}}c_{\mathbf{p}}-\delta_{\mathbf{k}+\mathbf{q},\mathbf{p}}c^{\dagger}_{\mathbf{p}-\mathbf{q}}c_{\mathbf{k}}\right) \tag{2.34}$$

$$=c^{\dagger}_{\mathbf{k}+\mathbf{q}}c_{\mathbf{k}+\mathbf{q}}-c^{\dagger}_{\mathbf{k}}c_{\mathbf{k}}. \tag{2.35}$$

This is the difference of occupations of modes separated by $\mathbf{q}$.

$$\chi^R(\mathbf{q},t)=-\frac{i\Theta(t)}{\mathcal{V}}\sum_{\mathbf{k}} e^{i(\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t}[n_F(\xi_{\mathbf{k}+\mathbf{q}})-n_F(\xi_{\mathbf{k}})]. \tag{2.36}$$

$$\chi^R(\mathbf{q},\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t}\chi^R(\mathbf{q},t) \tag{2.37}$$

$$= -\frac{i}{\mathcal{V}}\sum_{\mathbf{k}}[n_F(\xi_{\mathbf{k}+\mathbf{q}}) - n_F(\xi_{\mathbf{k}})]\int_0^{\infty} dt\, e^{i(\omega+\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}})t-0^+t} \tag{2.38}$$

$$= \frac{1}{\mathcal{V}}\sum_{\mathbf{k}}\frac{n_F(\xi_{\mathbf{k}+\mathbf{q}}) - n_F(\xi_{\mathbf{k}})}{\omega+\xi_{\mathbf{k}+\mathbf{q}}-\xi_{\mathbf{k}}+i0^+}. \tag{2.39}$$

The last integral uses $-i\int_0^{\infty} e^{iXt-\eta t}dt = 1/(X+i\eta)$. For an inversion-symmetric band, changing the dummy variable to $-\mathbf{k}-\mathbf{q}$ gives the equivalent form

$$\chi^R(\mathbf{q},\omega) = \frac{1}{\mathcal{V}}\sum_{\mathbf{k}}\frac{n_F(\xi_{\mathbf{k}}) - n_F(\xi_{\mathbf{k}+\mathbf{q}})}{\omega+\xi_{\mathbf{k}}-\xi_{\mathbf{k}+\mathbf{q}}+i0^+}. \tag{2.40}$$

No inversion assumption was needed for Eq. (2.39). Below the bands are parabolic and inversion symmetric. Each displayed sum so far is for one spin component, and $\mathcal{V}^{-1}\sum_{\mathbf{k}} \to \int d^dk/(2\pi)^d$. At zero temperature a positive-frequency absorption process removes a particle from an occupied state below the Fermi surface and places it in an empty state above it. The denominator resolves the energy cost of that particle–hole pair. The opposite ordered process supplies the other part of the full response. The response measures the ease of polarizing the reference sea. A finite regular response permits a perturbative starting point for a sufficiently weak interaction in the channel being examined. A divergent response warns that a naive expansion about the free sea is not uniformly controlled. An instability can require a different starting point, but a singular bare response alone does not prove an ordered phase. The notes illustrate this distinction with a one-channel barrier parameterized by a nonnegative dimensionless strength γ_B:

$$I_{N,e} = \frac{e}{h}\int d\omega\, \frac{4\gamma_B}{(1+\gamma_B)^2}[f_1(\omega) - f_2(\omega)]. \tag{2.41}$$

$$I_{N,e} \approx \frac{e}{h}\int d\omega\, 4\gamma_B[f_1(\omega) - f_2(\omega)] \qquad (\gamma_B \ll 1). \tag{2.42}$$

Here $I_{N,e}$ means e times particle transfer from reservoir 1 to 2, and $h = 2\pi$ in the working units. The electric current in that direction is $-I_{N,e}$. The transmission $4\gamma_B/(1+\gamma_B)^2$ defines this particular barrier parametrization; it is distinct from the later velocity-valued contact amplitude γ. γ_B is not the spectral linewidth $\Gamma_{\mathbf{k}}$. A leading term is reliable only in its stated small parameter regime. The same caution will matter for boundary tunnelling.

2.5 Static response and the geometry of nesting

For the inversion-symmetric band under consideration,

$$\chi^R(\mathbf{q},\omega) = \frac{1}{\mathcal{V}}\sum_{\mathbf{k}}\frac{n_F(\xi_{\mathbf{k}}) - n_F(\xi_{\mathbf{k}+\mathbf{q}})}{\omega+\xi_{\mathbf{k}}-\xi_{\mathbf{k}+\mathbf{q}}+i0^+}. \tag{2.43}$$

A static external potential probes $\omega = 0$.

$$H_{\text{drive}}(t) = \frac{1}{\mathcal{V}}\sum_{\mathbf{q}} U_{\text{ext}}(\mathbf{q},t)\rho(-\mathbf{q},t). \tag{2.44}$$

$$\delta\langle\rho(\mathbf{q},\omega)\rangle = \chi^R(\mathbf{q},\omega)U_{\text{ext}}(\mathbf{q},\omega). \tag{2.45}$$

$$U_{\text{ext}}(\mathbf{q},\omega) = 2\pi U_{\text{ext}}(\mathbf{q})\delta(\omega). \tag{2.46}$$

$$\chi^R(\mathbf{q},0) = \frac{1}{\mathcal{V}}\sum_{\mathbf{k}}\frac{n_F(\xi_{\mathbf{k}}) - n_F(\xi_{\mathbf{k}+\mathbf{q}})}{\xi_{\mathbf{k}}-\xi_{\mathbf{k}+\mathbf{q}}+i0^+}. \tag{2.47}$$

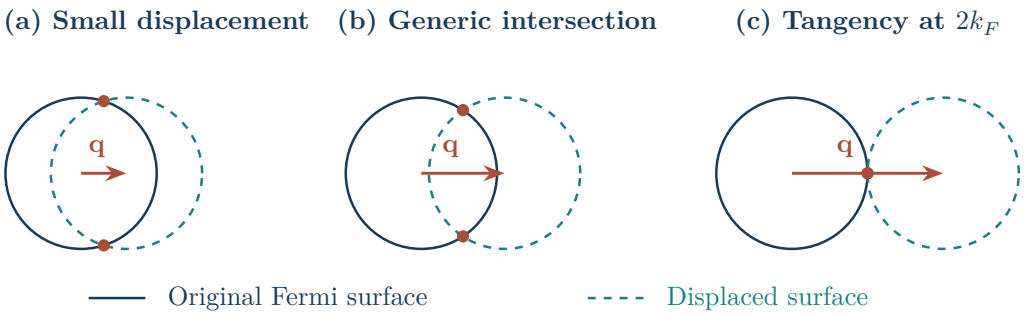


Figure 2.1: Schematic displaced circular Fermi surfaces. For generic displacements below $2k_F$, only two points coincide. At $2k_F$, the circles touch; moving away from the tangency direction destroys perfect matching. In three dimensions the corresponding generic intersection is a circle. The geometry restricts the measure of nearly degenerate particle–hole pairs.

Potential enhancement comes from many nearly degenerate occupied–empty pairs. At exactly equal energies the finite-temperature numerator vanishes too. A singularity must therefore be determined from the integral and the limiting occupation discontinuity, not from the denominator in isolation. At small q, only a thin shell can be promoted across a smooth Fermi surface. The shrinking numerator phase space compensates the small energy denominator, giving a finite static compressibility in the regular free gas. A displacement near $2k_F$ connects antipodal points. On a curved surface, changing the initial direction generally spoils this matching: nesting is restricted to a small subset, as in Fig. 2.1. For generic $0 < q < 2k_F$, two displaced circular Fermi surfaces intersect at two points in two dimensions; two displaced spherical surfaces intersect along a one-dimensional circle in three dimensions. At $q = 2k_F$ the intersection degenerates to tangency, and for $q > 2k_F$ it disappears. This limited matching is compatible with a controlled weak-interaction description away from an actual instability. The response can nevertheless have nonanalytic derivatives. The order of limits is essential: for conserved total particle number, $\chi^R(\mathbf{q} = 0, \omega \neq 0) = 0$, whereas taking the static limit first and then $q \to 0$ gives $-\partial n/\partial\mu$. The static formulas below take $\omega \to 0$ at fixed q, then examine q; all are at $T = 0$ and per spin.

2.6 The two- and three-dimensional static functions

Use $\epsilon_k = k^2/(2m)$ and $\xi_k = \epsilon_k - \mu$, with $\hbar = 1$. Reindexing the occupied and empty terms gives an occupied-ball expression,

$$\chi_d(q, 0) = -2\,\mathrm{PV} \int_{|\mathbf{k}|<k_F} \frac{d^d k}{(2\pi)^d} \frac{1}{q^2/(2m) + \mathbf{k}\cdot\mathbf{q}/m}. \tag{2.48}$$

The factor two here combines the two terms in the numerator; it is not a spin degeneracy. In two dimensions, the angular principal value vanishes for $2k > q$ and gives $2\pi/[q\sqrt{q^2 - 4k^2}]$ for the denominator $q^2 + 2kq\cos\alpha$ when $2k < q$. Consequently,

$$\chi_2(q, 0) = -\frac{2m}{\pi q} \int_0^{\min(k_F, q/2)} \frac{k\,dk}{\sqrt{q^2 - 4k^2}} \tag{2.49}$$

$$= -\frac{m}{2\pi} \left[1 - \sqrt{1 - \frac{4\min(k_F, q/2)^2}{q^2}} \right]. \tag{2.50}$$

$$\chi_2(q, 0) = -\frac{m}{2\pi} \begin{cases} 1, & 0 \le q \le 2k_F, \\ 1 - \sqrt{1 - (2k_F/q)^2}, & q > 2k_F. \end{cases} \tag{2.51}$$

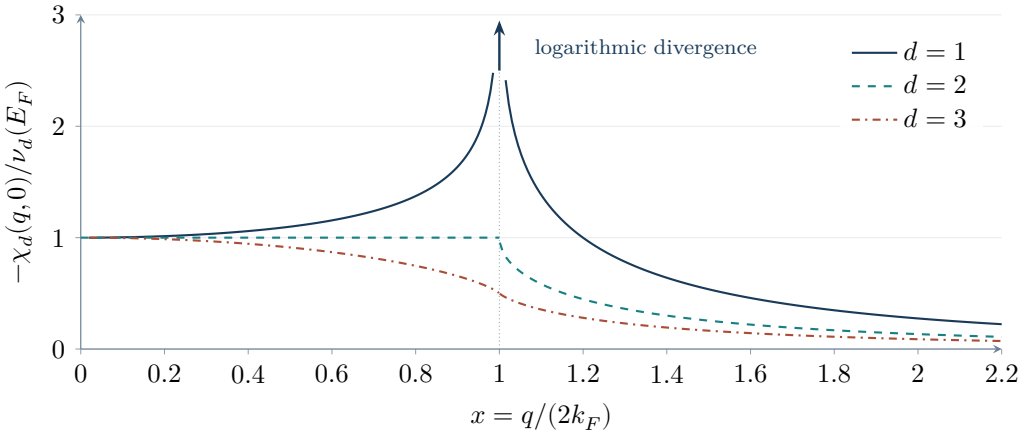


Figure 2.2: Static free-fermion density response at $T = 0$, per spin, for parabolic bands. Curves are computed from Eqs. (2.58), (2.51), and (2.54), normalized by each dimension's density of states $\nu_d(E_F)$. The one-dimensional curve diverges at $x = 1$ (a narrow interval is omitted); the two- and three-dimensional values remain finite. Axes are dimensionless; these curves contain no experimental data.

Both one-sided limits equal $-m/(2\pi)$. The response is continuous at $2k_F$, but its derivative from above diverges as $(q - 2k_F)^{-1/2}$. In three dimensions, put $z = \cos\alpha$. The angular integration yields

$$\chi_3(q,0) = -\frac{m}{2\pi^2 q}\int_0^{k_F} k\,dk \ln\left|\frac{q+2k}{q-2k}\right|, \tag{2.52}$$

$$\int_0^{k_F} k\,dk \ln\left|\frac{q+2k}{q-2k}\right| = \frac{qk_F}{2} + \frac{4k_F^2 - q^2}{8}\ln\left|\frac{q+2k_F}{q-2k_F}\right|. \tag{2.53}$$

The second line follows by integration by parts, with $d\ln|(q+2k)/(q-2k)|/dk = 4q/(q^2-4k^2)$. Writing $x = q/(2k_F)$,

$$\chi_3(q,0) = -\frac{mk_F}{4\pi^2}\left[1 + \frac{1-x^2}{2x}\ln\left|\frac{1+x}{1-x}\right|\right]. \tag{2.54}$$

The value is continuous, including at $x = 1$ where $(1-x^2)\ln|(1+x)/(1-x)| \to 0$; its slope has a logarithmic singularity. The limits $\chi_2(0,0) = -m/(2\pi)$ and $\chi_3(0,0) = -mk_F/(2\pi^2)$ are minus the density of states per spin at the Fermi level. The units are number per volume per energy. Multiplying by a spin degeneracy g_s is appropriate only when all g_s components respond identically. Figure 2.2 compares these functions with the one-dimensional result derived next.

2.7 Why one dimension is different

In one dimension particles cannot bypass one another as they can in a plane. Scattering between the two Fermi points can therefore reorganize the low-energy response. This geometric distinction is shown in Fig. 2.3. The relevant response remains

$$\chi_1(q,\omega) = \int \frac{dk}{2\pi}\frac{n_F(\xi_k) - n_F(\xi_{k+q})}{\omega + \xi_k - \xi_{k+q} + i0^+}. \tag{2.55}$$

Near the two Fermi points, a momentum transfer $2k_F$ links particle states on one branch to hole states on the other, and vice versa, within the linearized low-energy window. The sketch can be viewed as retaining the longitudinal matching present at antipodal points of a higher-dimensional Fermi surface. In higher dimensions, transverse directions spoil that matching for

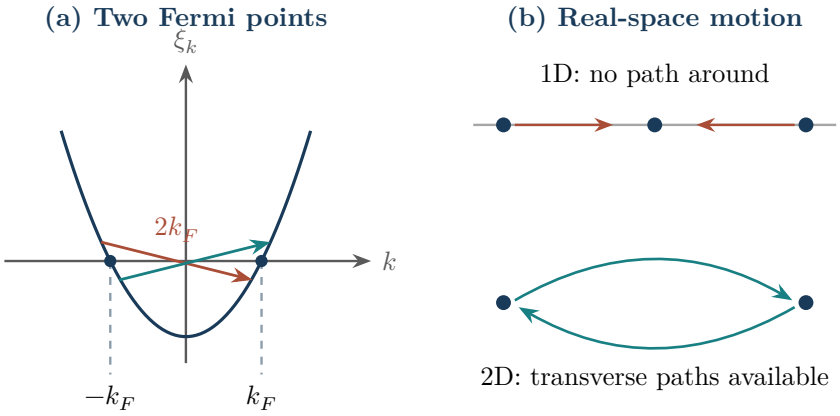


Figure 2.3: Near $\pm k_F$, a displacement by $2k_F$ links the low-energy particle and hole neighborhoods. Transverse directions that weaken this matching in higher dimensions are absent. The real-space trajectory analogy illustrates constrained motion; it does not by itself derive bosonization.

most states. In one dimension there are no such directions. This is a geometric analogy; the actual dimensional integration measures are different. At $T = 0$, for one spin species,

$$\chi_1(q,0) = -2\,\mathrm{PV}\int_{-k_F}^{k_F} \frac{dk}{2\pi}\frac{2m}{q^2+2kq} \tag{2.56}$$

$$= -\frac{m}{\pi q}\left[\ln|q+2k|\right]_{-k_F}^{k_F}. \tag{2.57}$$

$$\chi_1(q,0) = -\frac{m}{\pi q}\ln\left|\frac{2k_F+q}{2k_F-q}\right|. \tag{2.58}$$

It diverges logarithmically as $q \to 2k_F$. An arbitrarily weak perturbation in this channel can receive large low-energy corrections, so a naive regular Fermi-liquid expansion is not a reliable starting point. At $q \to 0$, in contrast, $\chi_1 \to -1/(\pi v_F)$. For the response example that follows, explicitly retain two spin species and a short-range interaction between opposite spins:

$$H_{\mathrm{int}} = U\int dx\,\rho_\uparrow(x)\rho_\downarrow(x). \tag{2.59}$$

$$\rho_\sigma(x) = \bar\rho_\sigma(x) + \delta\rho_\sigma(x), \qquad \bar\rho_\sigma = \langle\rho_\sigma\rangle, \quad \sigma = \uparrow,\downarrow. \tag{2.60}$$

For weak density fluctuations, neglect their product:

$$H_{\mathrm{int}}^{\mathrm{MF}} = U\int dx\,[\bar\rho_\uparrow\bar\rho_\downarrow + \bar\rho_\uparrow\delta\rho_\downarrow + \bar\rho_\downarrow\delta\rho_\uparrow]. \tag{2.61}$$

This is a spin-resolved mean-field/RPA illustration. It is distinct from the spinless Luttinger model introduced in the next main section.

2.8 Self-consistent spin-resolved response

$$H_{\mathrm{int}}^{\mathrm{MF}} = U\int dx\,[\bar\rho_\uparrow\bar\rho_\downarrow + \bar\rho_\uparrow(\rho_\downarrow - \bar\rho_\downarrow) + \bar\rho_\downarrow(\rho_\uparrow - \bar\rho_\uparrow)] \tag{2.62}$$

$$= U\int dx\,[-\bar\rho_\uparrow\bar\rho_\downarrow + \bar\rho_\uparrow\rho_\downarrow + \bar\rho_\downarrow\rho_\uparrow]. \tag{2.63}$$

Dropping the state-independent energy term for the linear-response equations,

$$H_{\mathrm{ind}} = U\int dx\,[\bar\rho_\uparrow\rho_\downarrow + \bar\rho_\downarrow\rho_\uparrow]. \tag{2.64}$$

$$U_{\uparrow}^{\mathrm{eff}}(q,\omega) = U_{\uparrow}^{\mathrm{ext}}(q,\omega) + U\delta\bar{\rho}_{\downarrow}(q,\omega), \tag{2.65}$$
$$U_{\downarrow}^{\mathrm{eff}}(q,\omega) = U_{\downarrow}^{\mathrm{ext}}(q,\omega) + U\delta\bar{\rho}_{\uparrow}(q,\omega). \tag{2.66}$$

The uniform reference Hartree potential is absorbed in the equilibrium band; the $\delta\bar{\rho}_\sigma$ here are the induced deviations. U is the contact interaction coefficient, while $U_\sigma^{\mathrm{ext/eff}}$ are potential energies; in one dimension their units are respectively velocity and energy.

$$\delta\bar{\rho}_{\uparrow} = \chi_0^R U_{\uparrow}^{\mathrm{eff}}, \tag{2.67}$$
$$\delta\bar{\rho}_{\downarrow} = \chi_0^R U_{\downarrow}^{\mathrm{eff}}. \tag{2.68}$$

$$\begin{pmatrix}\delta\bar{\rho}_{\uparrow}\\ \delta\bar{\rho}_{\downarrow}\end{pmatrix} = \begin{pmatrix}0 & U\chi_0^R\\ U\chi_0^R & 0\end{pmatrix}\begin{pmatrix}\delta\bar{\rho}_{\uparrow}\\ \delta\bar{\rho}_{\downarrow}\end{pmatrix} + \chi_0^R\begin{pmatrix}U_{\uparrow}^{\mathrm{ext}}\\ U_{\downarrow}^{\mathrm{ext}}\end{pmatrix}. \tag{2.69}$$

The inverse of the matrix on the left after moving the induced term is

$$\begin{pmatrix}1 & -U\chi_0^R\\ -U\chi_0^R & 1\end{pmatrix}^{-1} = \frac{1}{1-U^2(\chi_0^R)^2}\begin{pmatrix}1 & U\chi_0^R\\ U\chi_0^R & 1\end{pmatrix}. \tag{2.70}$$

$$\delta\bar{\rho}_{\uparrow} = \frac{\chi_0^R[U\chi_0^R U_{\downarrow}^{\mathrm{ext}} + U_{\uparrow}^{\mathrm{ext}}]}{1-U^2(\chi_0^R)^2}, \tag{2.71}$$
$$\delta\bar{\rho}_{\downarrow} = \frac{\chi_0^R[U\chi_0^R U_{\uparrow}^{\mathrm{ext}} + U_{\downarrow}^{\mathrm{ext}}]}{1-U^2(\chi_0^R)^2}. \tag{2.72}$$

The response is strongly modified by feedback. Its singularities must be read from the complete fractions: divergence of χ_0^R alone does not imply divergence of every interacting channel. For a spin-antisymmetric (magnetic) drive,

$$U_{\uparrow}^{\mathrm{ext}} = -U_{\downarrow}^{\mathrm{ext}}. \tag{2.73}$$

$$\delta\bar{\rho}_{\uparrow} = \chi_s^R U_{\uparrow}^{\mathrm{ext}} = \frac{\chi_0^R}{1+U\chi_0^R}U_{\uparrow}^{\mathrm{ext}}. \tag{2.74}$$

$$\chi_s^R = \frac{\chi_0^R}{1+U\chi_0^R}. \tag{2.75}$$

For a common potential, each component instead responds with $\chi_c^R = \chi_0^R/(1-U\chi_0^R)$. The total number-density response to that common potential is $2\chi_c^R$. These results are the random phase approximation (RPA), built from the spin-resolved free susceptibility. A zero of an RPA denominator diagnoses failure or instability of that approximation; establishing the actual ordered state requires fluctuations and its dimensional constraints.

2.9 Pairing response, fluctuations, and the route to a Luttinger liquid

An analogous resummation in an attractive pairing channel has the form

$$\chi_{\mathrm{pair}}(q,\omega) = \frac{\chi_{\mathrm{pair}}^0(q,\omega)}{1+U\chi_{\mathrm{pair}}^0(q,\omega)}. \tag{2.76}$$

The opposite-spin pair creation operator is

$$\Delta_q^\dagger = \sum_k c_{k+q/2,\uparrow}^\dagger c_{-k+q/2,\downarrow}^\dagger. \tag{2.77}$$

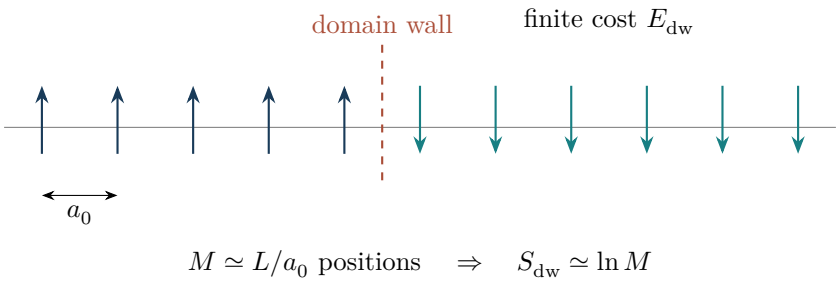


Figure 2.4: Schematic domain wall in a short-range Ising-like chain. Its energy cost stays finite while the number of possible positions grows with L. The entropy term therefore destabilizes uniform order for $T > 0$ in the thermodynamic limit. This discrete-symmetry example has different defects from a continuous superconducting phase.

$$U < 0 \qquad \text{(attractive contact interaction)}. \tag{2.78}$$

This pairing susceptibility has its own source convention: add $H_{\mathrm{source}} = -\eta\Delta^\dagger - \eta^*\Delta$, and differentiate the pair amplitude with respect to η. Its static normal-state bubble is positive:

$$\chi^0_{\mathrm{pair}}(0,0;T) = \int_{-E_c}^{E_c} d\xi\, \nu(\xi) \frac{\tanh(\xi/2T)}{2\xi}. \tag{2.79}$$

Here E_c is an energy window for the attractive model and ν is the single-spin density of states per length. This response must not be identified with the negative density response to a positive potential energy. Within this resummation a candidate pairing scale satisfies

$$1 + U\chi^0_{\mathrm{pair}}(0,0;T_{\mathrm{pair}}) = 0. \tag{2.80}$$

This is a mean-field instability temperature. Whether it is a thermodynamic transition requires a separate fluctuation analysis. For the two-point one-dimensional Fermi surface, the $2k_F$ matching persists throughout the linearized low-energy neighborhoods of the two branches. For the short-range one-dimensional examples considered here, thermal fluctuations invalidate the naive finite-temperature ordering inference from RPA. Perturbation theory around an ordinary Fermi-liquid starting point is therefore insufficient to capture the one-dimensional low-energy physics. To see one fluctuation mechanism explicitly, consider a short-range Ising-like chain with open ends. A domain wall between two oppositely aligned domains has a finite energy E_{dw} independent of chain length, as in Fig. 2.4. If there are $M \simeq L/a_0$ possible wall positions, with lattice spacing a_0, its positional entropy is

$$S_{\mathrm{dw}} \simeq \ln(L/a_0). \tag{2.81}$$

$$F_{\mathrm{dw}} = E_{\mathrm{dw}} - TS_{\mathrm{dw}} \simeq E_{\mathrm{dw}} - T\ln(L/a_0) \longrightarrow -\infty \quad (L \to \infty, T > 0). \tag{2.82}$$

Entropy favors introducing walls in an arbitrarily long chain at any fixed positive temperature. The finite-temperature state therefore does not sustain the corresponding uniform Ising order. A ring requires walls in pairs, but their finite creation cost and extensive positional opportunities lead to the same conclusion. This argument uses a discrete order parameter and a finite wall cost from short-range interactions. Pairing has a continuous phase and is disrupted by phase fluctuations; the domain-wall argument is an instructive example, not a proof for every symmetry or every interaction range. Long-range couplings, finite-size crossovers, and zero-temperature ordering require separate analysis. We now turn to a spinless low-energy model in which density waves, rather than long-lived individual fermions, form the natural elementary modes.

3 Density modes and bosonization

The next model is spinless. Linearization isolates the two Fermi branches; normal ordering about a specified sea then produces the density central term. We construct bosonic oscillators and reconstruct the physical fermion, including its number-changing sector.

3.1 Why particle–hole density modes?

The natural variables for a one-dimensional interacting system are collective particle–hole excitations. To see the motivation, first compare freely moving particles with particles whose motion is coupled by scattering. An isolated particle has an individual trajectory. In a coupled chain, a displacement disturbs its neighbours: a region with fewer particles and a region with more particles form a density disturbance. Figure 3.1 redraws this schematic comparison.

$$H_{\rm cl} = \sum_i \frac{P_i^2}{2m} + \frac{\kappa}{2}\sum_i (X_{i+1} - X_i)^2, \qquad P_i = m\dot{X}_i. \tag{3.1}$$

Here X_i is a displacement from equilibrium, P_i its conjugate momentum, and κ a spring constant. The chain is only a mechanical analogy: its normal modes suggest collective coordinates, but fermionic statistics and the density algebra below must still be derived. The anticipation is therefore to describe eigenstates in terms of collective excitations.

$$H_{\rm int} \propto \int_0^L dx\, {:}\rho(x)^2{:} = \frac{1}{L}\sum_q {:}\rho(q)\rho(-q){:}. \tag{3.2}$$

$$\rho_r(q) = \sum_k {:}c^\dagger_{r,k+q}c_{r,k}{:}, \qquad r = +1\ (R),\ -1\ (L). \tag{3.3}$$

Each bilinear transfers momentum q, while preserving particle number. Thus a density–density interaction is naturally expressed in particle–hole variables. The inserted outline below summarizes the construction and the subsequent route to the interacting ground state. A term $c^\dagger_{r,k+q}c_{r,k}$ creates a particle–hole pair from the reference Fermi sea only when k is occupied and $k+q$ is empty. The branch label r specifies the left or right Fermi point, as illustrated in Fig. 3.2. To make the proposed bosonic description precise, we must check its commutator, its action on the reference state, and whether it carries a well-defined excitation energy. From here onward the model is spinless. Put $\psi_r(x) = L^{-1/2}\sum_k e^{ikx}c_{r,k}$, where k measures momentum relative to rk_F, and use $\{c_{r,k}, c^\dagger_{r',k'}\} = \delta_{rr'}\delta_{kk'}$. The slow fields obey $\Psi(x) \simeq \sum_r e^{irk_Fx}\psi_r(x)$; the rapid Fermi phases are not part of ψ_r. Density momenta on the ring are $q = 2\pi n/L$, $n \in \mathbb{Z}$.

$$[\rho_r(p), \rho_r(-p')] = \sum_{k_1,k_2} [c^\dagger_{r,k_1+p}c_{r,k_1}, c^\dagger_{r,k_2-p'}c_{r,k_2}]. \tag{3.4}$$

$$\begin{aligned}[\rho_r(p), \rho_r(-p')] = \sum_{k_1,k_2} (&c^\dagger_{r,k_1+p}c_{r,k_2}\delta_{k_1,k_2-p'} \\ &- c^\dagger_{r,k_2-p'}c_{r,k_1}\delta_{k_1+p,k_2}).\end{aligned} \tag{3.5}$$

$$= \sum_k \left(c^\dagger_{r,k+p-p'}c_{r,k} - c^\dagger_{r,k-p'}c_{r,k-p}\right). \tag{3.6}$$

The two delta functions impose $k_2 = k_1 + p'$ and $k_2 = k_1 + p$, respectively. A formal shift of the momentum sum would seem to cancel the two terms. This is unsafe for an infinitely filled sea: subtract the reference expectation before shifting a convergent normal-ordered sum.

$${:}AB{:} = AB - \langle 0_{\rm FS}|AB|0_{\rm FS}\rangle. \tag{3.7}$$

For the bilinears used here, normal ordering subtracts the filled Fermi-sea reference, rather than the empty Fock vacuum.

$$\rho_r(x) = :\psi_r^\dagger(x)\psi_r(x): = \frac{1}{L}\sum_p e^{-ipx}\rho_r(p). \tag{3.8}$$

$$\rho_r(p) = \begin{cases} \sum_k c_{r,k+p}^\dagger c_{r,k}, & p \neq 0, \\ \sum_k (c_{r,k}^\dagger c_{r,k} - n_r^0(k)) = \hat{N}_r, & p = 0. \end{cases} \tag{3.9}$$

$$c_{r,k}^\dagger c_{r,k}|0_{\mathrm{FS}}\rangle = n_r^0(k)|0_{\mathrm{FS}}\rangle, \qquad n_r^0(k) = \begin{cases} 1, & rk < 0, \\ 0, & rk > 0. \end{cases} \tag{3.10}$$

The reference is the noninteracting Fermi sea. The interacting bosonic ground state has not yet been found and must not be identified with this normal-ordering reference. An occupation convention for a possible exactly zero-energy mode must be fixed separately. Changing it shifts the definition of $\hat{N}_r$, but does not change the nonzero-mode central term derived below. A finite ring also requires a fermionic boundary phase; we keep this separate from the periodic density modes.

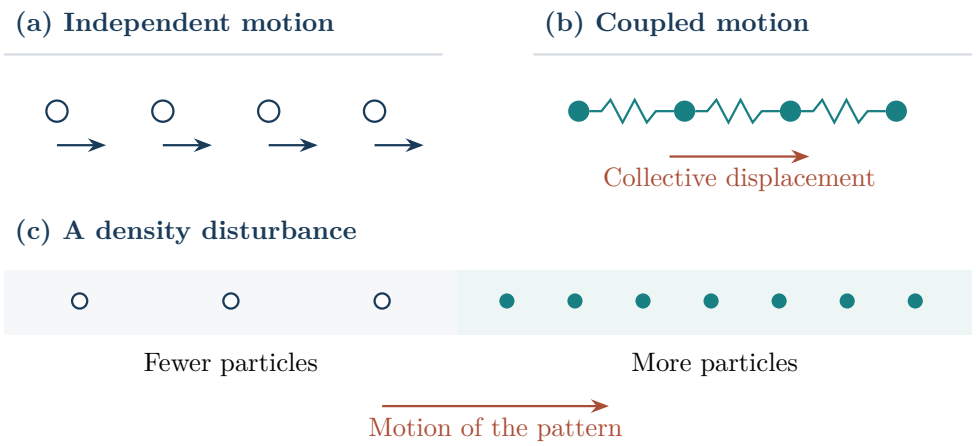


Figure 3.1: Schematic motivation for collective coordinates. Independent trajectories are replaced by coupled displacements in a spring analogy. A density disturbance pairs a depleted region with an excess region; it preserves the total particle number. The arrows in the last panel indicate propagation of the pattern, not a rigid translation of every particle.

3.2 Linearizing the two Fermi points

The low-energy construction starts from a band with quadratic dispersion.

$$\epsilon(k) = \frac{k^2}{2m}, \qquad \mu = \frac{k_F^2}{2m}, \qquad v_F = \frac{k_F}{m}. \tag{3.11}$$

The chemical potential lies well above the band bottom, by an energy $\Delta E = \mu - \epsilon_{\min}$. Linearization is useful when $\Delta E \gg T, |eV|, \Gamma$, where Γ denotes the energy broadening relevant to the experiment. Figure 3.2 retains the band, Fermi level, and the two linear branches.

$$\xi(k) = \epsilon(k) - \mu \simeq \begin{cases} v_F(k - k_F), & k \text{ near } +k_F, \\ -v_F(k + k_F), & k \text{ near } -k_F. \end{cases} \tag{3.12}$$

$$\xi(rk_F + \varkappa) = rv_F\varkappa + \frac{\varkappa^2}{2m} \simeq rv_F\varkappa, \qquad |\varkappa| \ll k_F. \tag{3.13}$$

The discarded curvature is smaller than the linear term by $|\varkappa|/(2k_F)$. In the remainder of the bosonization calculation the relative momentum $\varkappa$ is denoted by k. There are two branches, L and R, with opposite slopes. The labels refer to group velocity, not to the sign of every momentum within an extended branch. The mathematical linear branches extend to arbitrarily negative and positive energies. This extension is an effective low-energy construction, justified only for observables insensitive to the remote band bottom and regulated at short distances.

$$:\hat{O}: = \hat{O} - \langle 0_{\mathrm{FS}}|\hat{O}|0_{\mathrm{FS}}\rangle \qquad \text{for the bilinear operators used here.} \tag{3.14}$$

Normal ordering subtracts the reference contribution of deeply occupied states; it does not literally remove those states from the Hilbert space. Its role becomes explicit in the central-term calculation below.

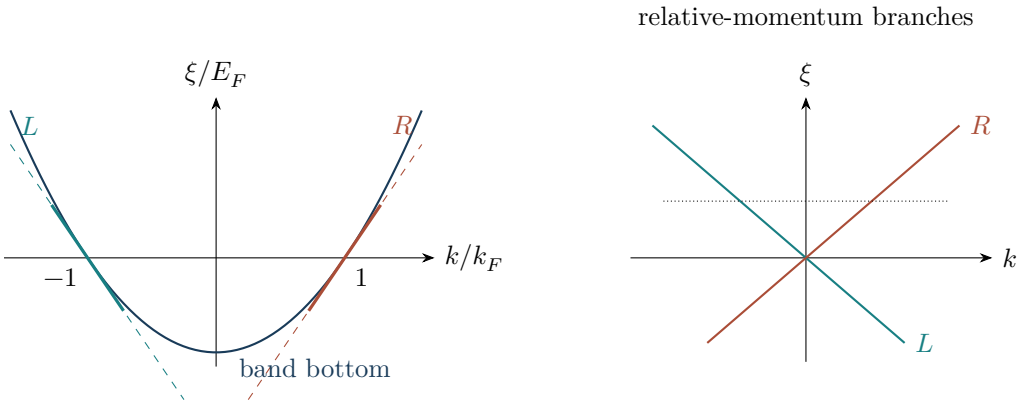


Figure 3.2: A parabolic band and its low-energy linearization. Left: the plotted formulas are $\xi/E_F = (k/k_F)^2 - 1$ and the tangents $2(k/k_F - 1)$, $-2(k/k_F + 1)$, with $E_F = k_F^2/(2m)$. Thick tangent segments indicate the low-energy windows; dashed extensions are an effective-model construction. Right: after shifting each Fermi point to its own origin, the two branches have opposite slopes $\pm v_F$ (schematic).

3.3 Outline of the bosonization construction

The construction has four stages. First introduce particle–hole excitation operators and express density fluctuations in terms of them.

$$\rho_r(q) = \sum_k :c^\dagger_{r,k+q}c_{r,k}:. \tag{3.15}$$

Linearization and the filled-sea reference make it possible to establish a bosonic density commutator.

$$b_p^\dagger = \sqrt{\frac{2\pi}{L|p|}}\left[\Theta(p)\rho_R(p) + \Theta(-p)\rho_L(p)\right], \qquad p \neq 0. \tag{3.16}$$

Second, select the branch appropriate to the sign of p, verify canonical commutation relations, and diagonalize H_0. The interaction can subsequently be diagonalized in a new oscillator basis. Third, express the physical fermion operators through dynamical bosonic fields ϕ, θ. This allows electron transmission and tunnelling operators to be treated within the bosonic description.

$$\rho_R = -\frac{1}{2\pi}\partial_x(\phi - \theta), \qquad \rho_L = -\frac{1}{2\pi}\partial_x(\phi + \theta). \tag{3.17}$$

The canonical algebra and the relation to particle-number density must be checked. Electric charge density is $-e(\rho_R + \rho_L)$. The fermionic anticommutation relations are a necessary check

on the bosonic representation, including its Klein factors. Fourth, use the bosonized Hamiltonian to determine the Luttinger-liquid ground state and its low-energy excitations. This is a dynamical step beyond establishing the operator dictionary. The logical dependency is algebra → representation → Hamiltonian → state and observables. The density algebra alone does not determine either the interacting vacuum or its correlation functions.

3.4 Normal ordering and the density central term

Keep normal-ordering symbols explicit until the infinite-sea cancellation has been justified.

$$\begin{aligned}[\rho_r(p),\rho_r(-p')] = \sum_k \big[&:c^\dagger_{r,k+p-p'}c_{r,k}: + \langle c^\dagger_{r,k+p-p'}c_{r,k}\rangle_0 \\ &- :c^\dagger_{r,k-p'}c_{r,k-p}: - \langle c^\dagger_{r,k-p'}c_{r,k-p}\rangle_0\big].\end{aligned} \tag{3.18}$$

$$\begin{aligned}= \sum_k \big[&(:c^\dagger_{r,k+p-p'}c_{r,k}: - :c^\dagger_{r,k-p'}c_{r,k-p}:) \\ &+ (\langle c^\dagger_{r,k+p-p'}c_{r,k}\rangle_0 - \langle c^\dagger_{r,k-p'}c_{r,k-p}\rangle_0)\big].\end{aligned} \tag{3.19}$$

In the normal-ordered part, shifting the second sum by $k \mapsto k+p$ cancels the first sum on finite-energy excitations of the sea. The separate unsubtracted sums are infinite; that cancellation cannot be used to discard the reference contribution.

$$[\rho_r(p),\rho_r(-p')] = \delta_{p,p'}\sum_k [n^0_r(k) - n^0_r(k-p)]. \tag{3.20}$$

$$k = \frac{2\pi n_k}{L}, \qquad n_k \in \mathbb{Z}, \qquad \langle c^\dagger_{r,k}c_{r,k}\rangle_0 = n^0_r(k) \in \{0,1\}. \tag{3.21}$$

The subscript 0 means the filled reference sea, not the interacting LL ground state.

$$\sum_k [n^0_R(k) - n^0_R(k-p)] = -\frac{pL}{2\pi}, \qquad \sum_k [n^0_L(k) - n^0_L(k-p)] = +\frac{pL}{2\pi}. \tag{3.22}$$

For $p > 0$, precisely $pL/(2\pi)$ momentum levels cross each Fermi point; the occupation difference is negative on the right branch and positive on the left. The result for $p < 0$ follows by reversing the shift.

$$[\rho_r(p),\rho_r(-p')] = -\delta_{p,p'}\frac{rpL}{2\pi}. \tag{3.23}$$

$$[\rho_r(p),\rho_{r'}(-p')] = -\delta_{rr'}\delta_{p,p'}\frac{rpL}{2\pi}. \tag{3.24}$$

The second density carries momentum $-p'$. Equivalently, the unaltered-momentum form contains $\delta_{p+p',0}$.

$$\rho_r(q)^\dagger = \sum_k :c^\dagger_{r,k}c_{r,k+q}: = \rho_r(-q). \tag{3.25}$$

The density modes have the required bosonic commutator after a momentum-dependent normalization. They are collective particle–hole modes, rather than individual fermionic quasiparticles.

$$\rho_L(p)|0_{\mathrm{FS}}\rangle = 0 \qquad (p > 0). \tag{3.26}$$

$$\rho_R(p)|0_{\mathrm{FS}}\rangle = 0 \qquad (p < 0). \tag{3.27}$$

These signs follow from which side of each linear branch is filled; see Fig. 3.3.

$$b^\dagger_p = \sqrt{\frac{2\pi}{L|p|}}\sum_{r=\pm1}\Theta(rp)\rho_r(p), \qquad p \neq 0. \tag{3.28}$$

It adds a particle–hole excitation on the right branch for $p > 0$ and on the left branch for $p < 0$.

$$b_p = \sqrt{\frac{2\pi}{L|p|}} \sum_{r=\pm 1} \Theta(rp)\rho_r(-p). \tag{3.29}$$

It removes the corresponding particle–hole excitation.

$$p > 0: \qquad b_p^\dagger = \sqrt{\frac{2\pi}{Lp}}\rho_R(p), \qquad b_p = \sqrt{\frac{2\pi}{Lp}}\rho_R(-p). \tag{3.30}$$

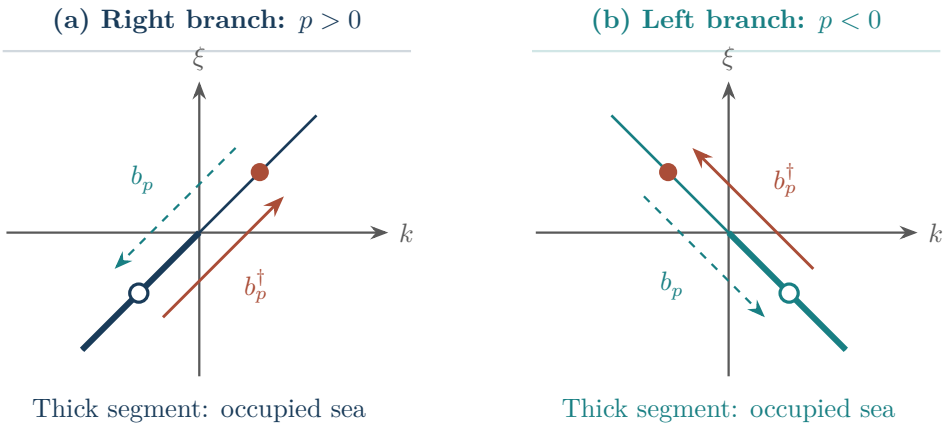


Figure 3.3: Particle–hole excitations of the linearized reference sea (schematic). Thick lower-energy segments are occupied. A creation operator $b_p^\dagger$ removes a fermion at the lower circle and places it at the filled upper circle; red arrows raise energy on either branch. Dashed arrows represent the reverse process. The sign of the momentum transfer is positive on R and negative on L, whereas the excitation energy is always $v_F|p|$.

3.5 Canonical oscillators and their vacuum

We now verify both oscillator commutators rather than inferring them from the name "bosonic".

$$[b_p^\dagger, b_q] = \frac{2\pi}{L\sqrt{|pq|}} \sum_{r,r'} \Theta(rp)\Theta(r'q)[\rho_r(p), \rho_{r'}(-q)]. \tag{3.31}$$

$$= -\frac{2\pi}{L|p|} \sum_{r} \Theta(rp)\frac{rpL}{2\pi}\delta_{p,q}. \tag{3.32}$$

$$\sum_r rp\Theta(rp) = \begin{cases} p = |p|, & p > 0 \ (r = R), \\ -p = |p|, & p < 0 \ (r = L). \end{cases} \tag{3.33}$$

$$[b_p^\dagger, b_q] = -\delta_{p,q}, \qquad [b_p, b_q^\dagger] = \delta_{p,q}. \tag{3.34}$$

$$[b_p^\dagger, b_q^\dagger] = \frac{2\pi}{L\sqrt{|pq|}} \sum_{r,r'} \Theta(rp)\Theta(r'q)[\rho_r(p), \rho_{r'}(q)]. \tag{3.35}$$

$$= -\frac{2\pi}{L|p|} \sum_{r} \Theta(rp)\Theta(rq)\frac{rpL}{2\pi}\delta_{p,-q} = 0. \tag{3.36}$$

$$\Theta(rp)\Theta(-rp) = 0 \quad (p \neq 0), \qquad [b_p^\dagger, b_q^\dagger] = [b_p, b_q] = 0. \tag{3.37}$$

The annihilation conditions are $\rho_L(p>0)|0_{\rm FS}\rangle = 0$ and $\rho_R(p<0)|0_{\rm FS}\rangle = 0$. They apply to the opposite momentum signs from those selected by $b_p^\dagger$. Consequently the creation operator does not annihilate the sea.

$$b_p^\dagger|\mathcal{N}\rangle = \sqrt{\frac{2\pi}{L|p|}}[\Theta(p)\rho_R(p) + \Theta(-p)\rho_L(p)]|\mathcal{N}\rangle. \tag{3.38}$$

$$= \sqrt{\frac{2\pi}{L|p|}} \begin{cases} \rho_R(p)|\mathcal{N}\rangle, & p>0, \\ \rho_L(p)|\mathcal{N}\rangle, & p<0. \end{cases} \tag{3.39}$$

Each nonvanishing bilinear raises a fermion from an occupied level to an empty level in the reference picture.

$$b_p|\mathcal{N}\rangle = \sqrt{\frac{2\pi}{L|p|}}[\Theta(p)\rho_R(-p) + \Theta(-p)\rho_L(-p)]|\mathcal{N}\rangle. \tag{3.40}$$

$$= \sqrt{\frac{2\pi}{L|p|}} \begin{cases} \rho_R(-p)|\mathcal{N}\rangle, & p>0, \\ \rho_L(-p)|\mathcal{N}\rangle, & p<0. \end{cases} \tag{3.41}$$

It reverses the particle–hole excitation, as indicated by the reverse arrows in Fig. 3.3.

$$b_p|0_{\rm FS}\rangle = 0 \qquad \text{for every } p \neq 0. \tag{3.42}$$

The reference sea is the oscillator vacuum: it contains no particle–hole excitations.

3.6 The energy carried by a density oscillator

A density oscillator also has a definite energy in the linearized free model. Take $p_0 > 0$, so that only the right branch contributes.

$$H_0 = \sum_{r,k} r v_F k :c_{r,k}^\dagger c_{r,k}:, \qquad C_{p_0} \equiv \sqrt{\frac{2\pi}{Lp_0}}. \tag{3.43}$$

All momenta k below are relative to the corresponding Fermi point. The normal-ordering reference remains $|0_{\rm FS}\rangle$.

$$[b_{p_0}, H_0] = C_{p_0} \sum_k v_F k[\rho_R(-p_0), c_{R,k}^\dagger c_{R,k}]. \tag{3.44}$$

$$= C_{p_0} \sum_{k,k_1} v_F k[c_{R,k_1-p_0}^\dagger c_{R,k_1}, c_{R,k}^\dagger c_{R,k}]. \tag{3.45}$$

$$= C_{p_0} \sum_{k,k_1} v_F k(c_{R,k_1-p_0}^\dagger c_{R,k}\delta_{k_1,k} - c_{R,k}^\dagger c_{R,k_1}\delta_{k_1-p_0,k}). \tag{3.46}$$

$$= C_{p_0} \sum_k v_F k(c_{R,k-p_0}^\dagger c_{R,k} - c_{R,k}^\dagger c_{R,k+p_0}). \tag{3.47}$$

As before, normal ordering makes the shift of an infinitely extended sea well defined.

$$\begin{aligned} = C_{p_0} \sum_k v_F k[&:c_{R,k-p_0}^\dagger c_{R,k}: - :c_{R,k}^\dagger c_{R,k+p_0}: \\ &+ \langle c_{R,k-p_0}^\dagger c_{R,k}\rangle_0 - \langle c_{R,k}^\dagger c_{R,k+p_0}\rangle_0]. \end{aligned} \tag{3.48}$$

For $p_0 \neq 0$ both reference contractions vanish. At $p_0 = 0$, the two contractions would cancel each other, but that mode is a number zero mode and was excluded from b_p. Shift the second normal-ordered sum to the same bilinear.

$$= C_{p_0} \sum_k v_F[k:c_{R,k-p_0}^\dagger c_{R,k}: - (k-p_0):c_{R,k-p_0}^\dagger c_{R,k}:]. \tag{3.49}$$

$$= C_{p_0} v_F p_0 \sum_k c^\dagger_{R,k-p_0} c_{R,k}. \tag{3.50}$$

The bilinear is off-diagonal in momentum, so its reference contraction is zero and its normal-ordering symbols may now be suppressed.

$$[b_{p_0}, H_0] = v_F p_0 b_{p_0}, \qquad [b_p, H_0] = v_F |p| b_p \quad (p \neq 0). \tag{3.51}$$

$$H_0 = \sum_{p\neq 0} v_F |p| b^\dagger_p b_p + H_{0,\mathrm{zm}} + E_0. \tag{3.52}$$

Here $H_{0,\mathrm{zm}}$ fixes the branch-number sectors and E_0 is the reference energy. Within a fixed number sector the last two terms are constant. The commutator establishes the oscillator energies; completeness of the oscillator states is a separate requirement. A forward density interaction is also quadratic in density modes, $H_{\mathrm{int}} \propto \sum_q \rho(q)\rho(-q)$. Therefore the resulting LL Hamiltonian is quadratic in bosonic oscillators even though its fermionic interaction is quartic.

$$|\mathcal{N}\rangle = f\big(\{c^\dagger_{r,k} c_{r,k'}\}\big)\, |N_R, N_L; 0_{\mathrm{osc}}\rangle. \tag{3.53}$$

The polynomial f may contain products of bilinears, hence higher particle–hole excitations. This statement concerns a fixed pair of branch numbers.

$$c^\dagger_{r,k} c_{r,k'} = \frac{1}{L} \int_{-L/2}^{L/2} dx \int_{-L/2}^{L/2} dx'\, e^{ikx - ik'x'} \psi^\dagger_r(x) \psi_r(x'). \tag{3.54}$$

To establish completeness, one needs to express these bilinears in the oscillator representation. In general the expression contains products of oscillators, not merely a linear combination of $b_q, b^\dagger_q$. The fermion vertex supplies this representation; the interacting LL ground state is then determined later. A useful counting check is available at finite size. In a fixed branch-number sector, moving occupied fermion levels upward by a total of n momentum spacings labels an integer partition of n. Bosonic occupations $m_j \geq 0$ obeying $\sum_{j\geq 1} j m_j = n$ label the same partitions. Thus each chiral fermionic excitation sector and its bosonic oscillator sector have the same degeneracies. The zero-mode labels must still be retained separately.

3.7 From a density commutator to a fermion vertex

$$H = H_0 + H_{\mathrm{int}} + H_T. \tag{3.55}$$

A contact joining two parts of a wire is illustrated schematically in Fig. 3.4. The density terms are already quadratic in bosons; a fermionic contact term still has to be represented.

$$H_T = \gamma\, \psi^\dagger_L(0)\psi_R(0) + \gamma^* \psi^\dagger_R(0)\psi_L(0). \tag{3.56}$$

The contact amplitude γ is unrelated to a quasiparticle linewidth. At this stage the sketch motivates the operator calculation; a bulk impurity and two reflecting endpoints will require different boundary treatments.

$$\psi_r(x) = \frac{1}{\sqrt{L}} \sum_k e^{ikx} c_{r,k}. \tag{3.57}$$

$$[\rho_r(p), \psi_r(x)] = \frac{1}{\sqrt{L}} \sum_{k,k_1} e^{ik_1 x} [c^\dagger_{r,k+p} c_{r,k}, c_{r,k_1}]. \tag{3.58}$$

$$= -\frac{1}{\sqrt{L}} \sum_k e^{i(k+p)x} c_{r,k} = -e^{ipx} \psi_r(x). \tag{3.59}$$

$$[A,B]=c\,\mathbf{1}\quad\Longrightarrow\quad[A,e^{\alpha B}]=\sum_{n=0}^{\infty}\frac{\alpha^n}{n!}(AB^n-B^nA). \tag{3.60}$$

$$[A,e^{\alpha B}]=\sum_{n=1}^{\infty}\frac{\alpha^n}{n!}nB^{n-1}c=\alpha c\,e^{\alpha B}. \tag{3.61}$$

$$\psi_r(x)\propto e^{S_r(x)},\qquad S_r(x)=\sum_{p'\neq0}\frac{2\pi r}{Lp'}e^{ip'x}\rho_r(-p'). \tag{3.62}$$

A factor commuting with all nonzero density modes is not fixed by this equation.

$$A=\rho_r(p),\quad B_{p'}=\rho_r(-p'),\quad \alpha_{p'}=\frac{2\pi r}{Lp'}e^{ip'x},\quad [A,B_{p'}]=-\delta_{p,p'}\frac{rpL}{2\pi}. \tag{3.63}$$

$$[\rho_r(p),e^{S_r(x)}]=-\sum_{p'\neq0}\delta_{p,p'}\frac{rpL}{2\pi}\frac{2\pi r}{Lp'}e^{ip'x}e^{S_r(x)}. \tag{3.64}$$

$$[\rho_r(p),e^{S_r(x)}]=-e^{ipx}e^{S_r(x)},\qquad p\neq0. \tag{3.65}$$

Two ingredients remain unfixed: the normalization, determined by $\{\psi_r(x),\psi_r^\dagger(x')\}$, and an additional factor ensuring anticommutation between different branches, whose density operators commute.

$$[A,e^{\beta B}]=c\beta e^{\beta B},\qquad Ae^{\beta B}=e^{\beta B}(A+c\beta). \tag{3.66}$$

$$A^2e^{\beta B}=Ae^{\beta B}(A+c\beta)=e^{\beta B}(A+c\beta)^2,\qquad A^ne^{\beta B}=e^{\beta B}(A+c\beta)^n. \tag{3.67}$$

$$\begin{aligned}[e^{\alpha A},e^{\beta B}]&=\sum_{n=0}^{\infty}\frac{\alpha^n}{n!}[A^n,e^{\beta B}]\\&=e^{\beta B}\sum_{n=0}^{\infty}\frac{\alpha^n}{n!}[(A+c\beta)^n-A^n].\end{aligned} \tag{3.68}$$

$$=e^{\beta B}e^{\alpha(A+c\beta)}-e^{\beta B}e^{\alpha A}=e^{\beta B}e^{\alpha A}(e^{\alpha\beta c}-1). \tag{3.69}$$

$$e^{\alpha A}e^{\beta B}=e^{\beta B}e^{\alpha A}e^{\alpha\beta c}. \tag{3.70}$$

This reordering identity, valid when the commutator is a scalar, will generate the exchange phase of the vertex operators.

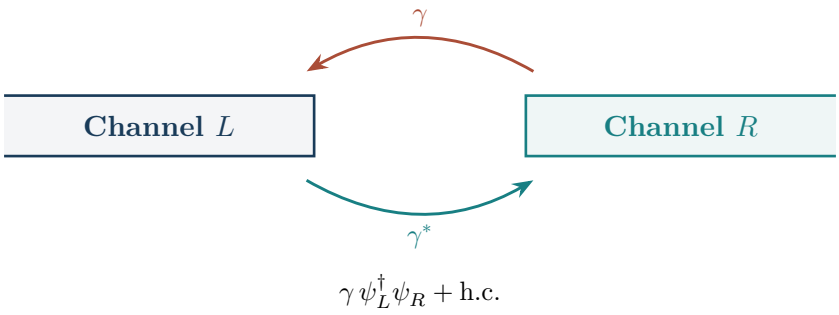


Figure 3.4: Schematic local coupling motivating bosonization of a fermionic transfer operator. The drawing specifies two coupled channel labels and a Hermitian-conjugate pair of transfer amplitudes. It does not yet select a bulk impurity or a reflecting-endpoint boundary condition; those geometries are established in the transport sections.

3.8 The local density and the Klein factors

$$c_{r,k} = \frac{1}{\sqrt{L}} \int_0^L dx\, e^{-ikx} \psi_r(x), \qquad c_{r,k}^\dagger = \frac{1}{\sqrt{L}} \int_0^L dx\, e^{ikx} \psi_r^\dagger(x). \tag{3.71}$$

$$\rho_r(q) = \frac{1}{L} \sum_k \int_0^L dx_1 \int_0^L dx_2\, e^{i(k+q)x_1 - ikx_2} {:}\psi_r^\dagger(x_1)\psi_r(x_2){:}. \tag{3.72}$$

$$= \int_0^L dx_1 \int_0^L dx_2\, \delta_L(x_1 - x_2) e^{iqx_1} {:}\psi_r^\dagger(x_1)\psi_r(x_2){:}. \tag{3.73}$$

$$\rho_r(q) = \int_0^L dx\, e^{iqx} {:}\psi_r^\dagger(x)\psi_r(x){:}. \tag{3.74}$$

$$[\rho_r(p), \psi_r(x)] = \int_0^L dx'\, e^{ipx'} [{:}\psi_r^\dagger(x')\psi_r(x'){:}, \psi_r(x)] = -e^{ipx}\psi_r(x). \tag{3.75}$$

$$\{\psi_r^\dagger(x'), \psi_r(x)\} = \delta_L(x - x'). \tag{3.76}$$

The local canonical anticommutator implies the density–fermion commutator. The reverse implication is only a consistency motivation; the vertex anticommutator itself will be checked explicitly below.

$$[\rho_R(p), \rho_L(q)] = 0 \quad \Longrightarrow \quad [e^{S_R(x)}, e^{S_L(x')}] = 0. \tag{3.77}$$

Oscillator exponentials from independent branches commute, whereas physical fermions from different branches must anticommute.

$$\psi_r(x) = \frac{F_r}{\sqrt{2\pi a}} {:}e^{-i\varphi_r(x)}{:}_a, \qquad \varphi_r = r\phi - \theta. \tag{3.78}$$

This is the infinite-line form with the finite-size factors specified below. The normal-ordering symbol is matched to the fixed reference sea and cutoff a.

$$F_r^\dagger F_r = F_r F_r^\dagger = 1, \qquad \{F_r^\dagger, F_{r'}\} = 2\delta_{rr'}. \tag{3.79}$$

A Klein factor is a unitary number-shift operator; it is not a single-level fermionic annihilation operator, and $F_r^2 \neq 0$.

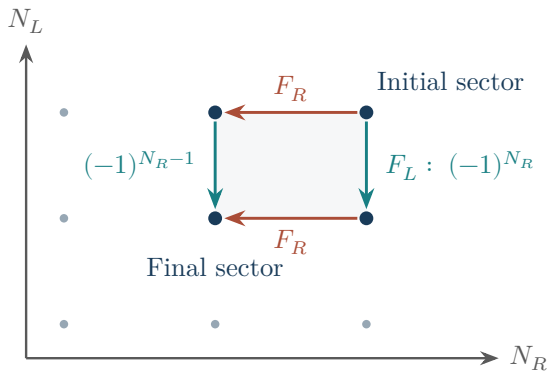
$$\{F_r, F_{r'}\} = \{F_r^\dagger, F_{r'}^\dagger\} = 0 \qquad (r \neq r'). \tag{3.80}$$

$$[\hat{N}_r, F_{r'}^\dagger] = \delta_{rr'} F_{r'}^\dagger, \qquad [\hat{N}_r, F_{r'}] = -\delta_{rr'} F_{r'}. \tag{3.81}$$

The factor F_r lowers branch particle number by one. The oscillator vertex supplies the particle–hole dynamics. These are separate roles: do not assign the same global number shift to both factors.

$$\begin{aligned} F_R|N_R, N_L\rangle &= |N_R - 1, N_L\rangle, \\ F_L|N_R, N_L\rangle &= (-1)^{N_R}|N_R, N_L - 1\rangle. \end{aligned} \tag{3.82}$$

This explicit cocycle realization satisfies the required interbranch signs. It acts on the integer zero-mode lattice and commutes with nonzero density oscillators. Figure 3.5 illustrates the number shifts.



Two orders of number shifts

$F_R F_L = -F_L F_R$

The same final number sector, with opposite amplitudes.

Figure 3.5: A Klein-factor realization on the integer branch-number lattice. F_R shifts N_R leftward; F_L shifts N_L downward and carries the parity $(-1)^{N_R}$. The two orders of these shifts differ by a minus sign, giving interbranch fermionic statistics. This diagram represents number sectors, not spatial trajectories.

3.9 Canonical fields and finite-size zero modes

$$S_r^{(a)}(x) = \sum_{p\neq 0} \frac{2\pi r}{Lp} e^{ipx - a|p|/2} \rho_r(-p), \qquad \psi_r(x) \propto F_r e^{S_r^{(a)}(x)}. \tag{3.83}$$

$$S_r^{(a)}(x) = -\frac{2\pi r}{L} \sum_{p\neq 0} \frac{e^{-ipx-a|p|/2}}{p} \rho_r(p). \tag{3.84}$$

$$S_r^{(a)}(x) = \frac{2\pi r}{L} \sum_{p>0} \frac{e^{-ap/2}}{p} \left[e^{ipx}\rho_r(-p) - e^{-ipx}\rho_r(p)\right]. \tag{3.85}$$

The mode sum is cumbersome for local calculations. Introduce position-dependent canonical fields so that the vertex and its anticommutators can be organized locally. The fields still contain the full mode sum; they are not classical phases. For the finite-size formulas in this subsection choose periodic branch fermions, $k = 2\pi n/L$, and occupy the zero-energy level in each reference sea. This convention fixes the otherwise unspecified occupation at the Fermi point. Write $N = \hat{N}_R + \hat{N}_L$ and $J = \hat{N}_R - \hat{N}_L$. We use winding fields without a separate global coordinate zero mode: the number-shifting operators F_r supply that part of the algebra.

$$\phi(x) = -\frac{\pi N x}{L} - \frac{i\pi}{L} \sum_{p\neq 0} \frac{e^{-ipx-a|p|/2}}{p} [\rho_R(p) + \rho_L(p)]. \tag{3.86}$$

$$\theta(x) = \frac{\pi J x}{L} + \frac{i\pi}{L} \sum_{p\neq 0} \frac{e^{-ipx-a|p|/2}}{p} [\rho_R(p) - \rho_L(p)]. \tag{3.87}$$

Two independent branches require two fields. Their sum and difference isolate the branches. The particular combinations have four advantages: they separate the physical branches, form a canonical pair in the local infinite-line limit, encode particle density, and simplify the fermionic anticommutator.

$$\phi(x) + \theta(x) = -\frac{2\pi \hat{N}_L x}{L} - \frac{2i\pi}{L} \sum_{p\neq 0} \frac{e^{-ipx-a|p|/2}}{p} \rho_L(p). \tag{3.88}$$

$$\phi(x) - \theta(x) = -\frac{2\pi \hat{N}_R x}{L} - \frac{2i\pi}{L} \sum_{p\neq 0} \frac{e^{-ipx-a|p|/2}}{p} \rho_R(p). \tag{3.89}$$

$$F_L e^{S_L^{(a)}(x)} = F_L \exp\left[\frac{2\pi}{L} \sum_{p\neq 0} \frac{e^{-ipx-a|p|/2}}{p} \rho_L(p)\right]. \tag{3.90}$$

$$= F_L \exp\left\{ i \left[\phi(x) + \theta(x) + \frac{2\pi \hat{N}_L x}{L} \right] \right\}. \tag{3.91}$$

$$e^{-ipx} = 1 - ipx + O(p^2), \qquad \frac{e^{-ipx}}{p}\rho_L(p) = \frac{\rho_L(p)}{p} - ix\rho_L(p) + O(p). \tag{3.92}$$

This formal expansion identifies an x-independent part and an x-linear part. On a finite ring, however, the nonzero momenta are discrete and do not determine $\rho_L(0) = \hat{N}_L$ by continuity. The number term must be inserted as independent zero-mode data; normal ordering cannot remove it.

$$F_L \exp\left[-\frac{2\pi i x \hat{N}_L}{L} + \frac{2\pi}{L} \sum_{p\neq 0} \frac{e^{-ipx - a|p|/2}}{p} \rho_L(p) \right]. \tag{3.93}$$

$$= F_L \exp\{ i[\phi(x) + \theta(x)] \} = F_L e^{-i\varphi_L(x)}. \tag{3.94}$$

The finite-size vertex also depends on operator ordering: F_r is written to the left of the number phase. For the periodic, filled-zero-level sector chosen here, the number phase is $e^{2\pi i r \hat{N}_r x/L}$. Another fermionic boundary phase changes this factor by a stated spin-structure phase; the oscillator commutators are unchanged. The normalization of the exponential is fixed after the anticommutator calculation.

3.10 The canonical commutator from the mode sum

$$F_R e^{S_R^{(a)}(x)} = F_R \exp\left[-\frac{2\pi}{L} \sum_{p\neq 0} \frac{e^{-ipx - a|p|/2}}{p} \rho_R(p) \right]. \tag{3.95}$$

$$-\frac{2\pi}{L}\frac{e^{-ipx}}{p}\rho_R(p) = -\frac{2\pi}{L}\frac{\rho_R(p)}{p} + \frac{2\pi i x}{L}\rho_R(p) + O(p). \tag{3.96}$$

As for the left branch, the formal singular constant is not a definition of the zero mode. The separately specified $\hat{N}_R$ supplies the finite number phase.

$$F_R \exp\left[\frac{2\pi i x \hat{N}_R}{L} + S_R^{(a)}(x) \right] = F_R \exp\{ -i[\phi(x) - \theta(x)] \}. \tag{3.97}$$

$$\psi_r(x) = \frac{F_r}{\sqrt{2\pi a}} :e^{-i[r\phi(x) - \theta(x)]}:_a, \qquad \Psi(x) \simeq \sum_r e^{irk_F x}\psi_r(x). \tag{3.98}$$

The cutoff is retained until distributions have been evaluated. The prefactor gives ψ_r the dimension length$^{-1/2}$. The rapid factors belong to the microscopic field and identify transport near the two Fermi points.

$$\hat{N}_r = \rho_r(0), \qquad [\hat{N}_R, \hat{N}_L] = 0, \qquad [\hat{N}_r, \rho_{r'}(p \neq 0)] = 0. \tag{3.99}$$

$$[\phi(x), \theta(x')] = \frac{\pi^2}{L^2} \sum_{p,p'\neq 0} \frac{e^{-i(px + p'x') - a(|p| + |p'|)/2}}{pp'} \times ([\rho_R(p), \rho_R(p')] - [\rho_L(p), \rho_L(p')]). \tag{3.100}$$

$$= \frac{\pi^2}{L^2} \sum_{p,p'\neq 0} \frac{e^{-i(px + p'x') - a(|p| + |p'|)/2}}{pp'} \frac{pL}{2\pi} \delta_{p,-p'}(-1 - 1). \tag{3.101}$$

$$= \frac{\pi^2}{L^2} \sum_{p\neq 0} \frac{1}{p^2} \frac{pL}{\pi} e^{-ip(x-x') - a|p|} = \frac{\pi}{L} \sum_{p\neq 0} \frac{e^{-ip(x-x') - a|p|}}{p}. \tag{3.102}$$

$$[\phi(x),\theta(x')] = \frac{1}{2}\log\frac{1-e^{-2\pi a/L+2\pi i(x-x')/L}}{1-e^{-2\pi a/L-2\pi i(x-x')/L}}. \tag{3.103}$$

The logarithms are chosen by continuous continuation from $a > 0$. On the ring, the $a \to 0^+$ result is a periodic sawtooth, not an unrestricted sign function. For $0 < x - x' < L$ it is $i\pi[(x-x')/L - 1/2]$. Thus its derivative gives $\delta_L - 1/L$ for the oscillator algebra.

$$\lim_{L\to\infty}[\phi(x),\theta(x')] = \frac{1}{2}\left(\int_{-\infty}^{0} dp + \int_{0}^{\infty} dp\right)\frac{e^{-ip(x-x')-a|p|}}{p}. \tag{3.104}$$

$$= \frac{1}{2}\int_0^\infty \frac{dp}{p} e^{-ap}\left[e^{-ip(x-x')} - e^{ip(x-x')}\right]. \tag{3.105}$$

$$= i\int_0^\infty \frac{dp}{p} e^{-ap}\sin[p(x'-x)] = i\arctan\frac{x'-x}{a}. \tag{3.106}$$

$$\frac{d}{dy}\int_0^\infty \frac{dp}{p} e^{-ap}\sin(py) = \int_0^\infty dp\, e^{-ap}\cos(py) = \frac{a}{a^2+y^2}. \tag{3.107}$$

The integral vanishes at $y = 0$, so integrating the last expression gives $\arctan(y/a)$. This also fixes the sign of the commutator.

$$\lim_{a\to 0^+}\arctan\frac{x'-x}{a} = \begin{cases} \frac{\pi}{2}\operatorname{sgn}(x'-x), & x' \neq x, \\ 0, & x' = x. \end{cases} \tag{3.108}$$

$$[\phi(x),\theta(x')] = \frac{i\pi}{2}\operatorname{sgn}(x'-x) \qquad \text{on the infinite line.} \tag{3.109}$$

The difference between the two limits is plotted in Fig. 3.6.

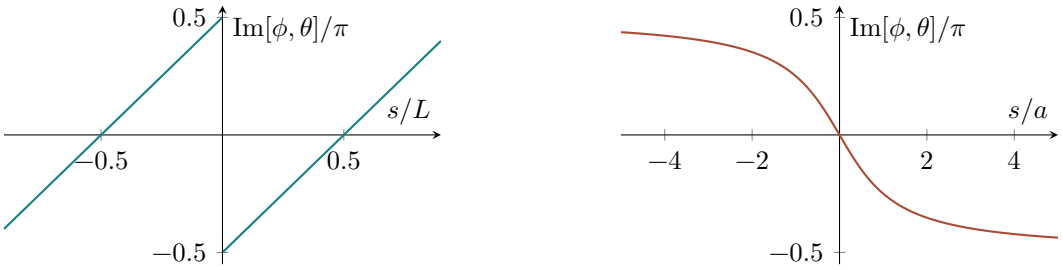


Figure 3.6: Two different limiting forms of the canonical commutator, with $s = x - x'$. Left: the periodic oscillator result at $a \to 0^+$, $\mathrm{Im}[\phi,\theta]/\pi = s/L - \frac{1}{2}\operatorname{sgn} s$ for $|s| < L$. Right: take $L \to \infty$ first at finite cutoff, giving $-\arctan(s/a)/\pi$. Only the subsequent $a \to 0^+$ limit gives an infinite-line sign function. These are formula-based curves, not numerical or experimental data.

3.11 The physical meaning of the canonical fields

$$[\phi(x),\theta(x')] = \frac{i\pi}{2}\operatorname{sgn}(x'-x). \tag{3.110}$$

The two fields do not commute: the density field and its dual phase are conjugate variables.

$$[\phi(x),\partial_{x'}\theta(x')] = i\pi\delta(x-x'), \qquad \Pi(x) \equiv \frac{1}{\pi}\partial_x\theta(x), \qquad [\phi(x),\Pi(x')] = i\delta(x-x'). \tag{3.111}$$

The density embedding supplies the fields with a directly observable meaning.

$$\phi(x) = -\frac{\pi N x}{L} - \frac{i\pi}{L}\sum_{p\neq 0}\frac{e^{-ipx-a|p|/2}}{p}[\rho_R(p)+\rho_L(p)]. \tag{3.112}$$

$$= -\frac{\pi N x}{L} - \frac{i\pi}{L}\sum_{p\neq 0}\frac{e^{-ipx-a|p|/2}}{p}[\rho_R(-p)^\dagger + \rho_L(-p)^\dagger]. \tag{3.113}$$

Hermitian conjugation reverses the density momentum; it does not create a second independent full density field.

$$= -\frac{\pi N x}{L} + \frac{i\pi}{L}\sum_{p\neq 0}\frac{e^{ipx-a|p|/2}}{p}[\rho_R(p)^\dagger + \rho_L(p)^\dagger]. \tag{3.114}$$

This is the same Hermitian field after $p \mapsto -p$. The regulator will be taken to zero only after differentiation or distributional evaluation.

$$\partial_x\phi(x) = -\frac{\pi N}{L} - \frac{\pi}{L}\sum_{p\neq 0} e^{ipx-a|p|/2}[\rho_R(p)^\dagger + \rho_L(p)^\dagger]. \tag{3.115}$$

$$\partial_x\phi(x) = -\frac{\pi}{L}\sum_p e^{-ipx}[\rho_R(p) + \rho_L(p)] = -\pi[\rho_R(x) + \rho_L(x)] \qquad (a \to 0^+). \tag{3.116}$$

$$\partial_x\theta(x) = \pi[\rho_R(x) - \rho_L(x)]. \tag{3.117}$$

The total density is a gradient of ϕ, while the branch-density imbalance is a gradient of θ. Density interactions consequently become quadratic gradient terms in the Hamiltonian.

$$\rho_L(x) = -\frac{1}{2\pi}[\partial_x\phi(x) + \partial_x\theta(x)]. \tag{3.118}$$

$$\rho_R(x) = -\frac{1}{2\pi}[\partial_x\phi(x) - \partial_x\theta(x)]. \tag{3.119}$$

At a boundary where the stated condition $\rho_R + \rho_L = 0$ holds, $\partial_x\phi = 0$ there. A reflecting boundary can pin the boundary value of ϕ; this does not mean that ϕ lacks bulk dynamics. The boundary condition and the bulk equation of motion are distinct statements.

$$[\rho_L(x), \rho_R(x')] = \frac{1}{4\pi^2}\partial_x\left\{-[\phi(x), \partial_{x'}\theta(x')] + [\theta(x), \partial_{x'}\phi(x')]\right\}. \tag{3.120}$$

$$= \frac{1}{4\pi^2}\partial_x[-i\pi\delta(x-x') + i\pi\delta(x-x')] = 0. \tag{3.121}$$

The two cross terms cancel. This reproduces the independence of the physical branch densities. The remaining local checks are $[\rho_L(x), \psi_R(x')]$ and $[\rho_L(x), \psi_L(x')]$. They distinguish the branch that is unchanged from the branch whose particle number is lowered.

3.12 Checking the density–vertex commutator

$$[\rho_L(x), \psi_r(x')] = -\frac{1}{2\pi}\frac{F_r}{\sqrt{2\pi a}}[\partial_x\phi(x) + \partial_x\theta(x), :e^{-i[r\phi(x')-\theta(x')]}:_a]. \tag{3.122}$$

This local infinite-line calculation suppresses the separated global zero-mode factors. Their finite-ring contribution restores the uniform $1/L$ part of the periodic density commutator.

$$\begin{aligned} A &= \partial_x\phi(x) + \partial_x\theta(x), \qquad B = -i[r\phi(x') - \theta(x')], \\ [A, B] &= -[B, A] = i[r\phi(x') - \theta(x'), \partial_x\phi(x) + \partial_x\theta(x)]. \end{aligned} \tag{3.123}$$

$$[A, B] = ir[\phi(x'), \partial_x\theta(x)] - i[\theta(x'), \partial_x\phi(x)]. \tag{3.124}$$

$$= -\pi r\delta(x-x') + \pi\delta(x-x') = -\pi(r-1)\delta(x-x'). \tag{3.125}$$

$$[A, B] = \begin{cases} 0, & r = +1\ (R), \\ 2\pi\delta(x-x'), & r = -1\ (L). \end{cases} \tag{3.126}$$

$$[A,B]=c\,\mathbf{1}\quad\Longrightarrow\quad AB=BA+c. \tag{3.127}$$

$$AB^2=B(AB)+cB=B^2A+2cB. \tag{3.128}$$

$$AB^n=B^nA+ncB^{n-1}. \tag{3.129}$$

$$[A,e^B]=\sum_{n=0}^{\infty}\frac{1}{n!}[A,B^n]=\sum_{n=1}^{\infty}\frac{nc}{n!}B^{n-1}=ce^B. \tag{3.130}$$

$$[\rho_L(x),\psi_r(x')]=-\frac{1}{2\pi}\frac{F_r}{\sqrt{2\pi a}}\delta_{r,L}\,2\pi\delta(x-x'):e^{-i[r\phi(x')-\theta(x')]}:_a. \tag{3.131}$$

$$[\rho_L(x),\psi_r(x')]=-\delta_{r,L}\delta(x-x')\psi_r(x'). \tag{3.132}$$

$$[\rho_s(x),\psi_r(x')]=-\delta_{sr}\delta(x-x')\psi_r(x'),\qquad[\hat{N}_s,F_r]=-\delta_{sr}F_r. \tag{3.133}$$

The local and global statements agree: annihilating a fermion lowers the particle number in its own branch by one and leaves the other branch unchanged. The zero-mode contribution must be counted once, through F_r, when working on a finite ring.

3.13 Vertex reordering and equal-branch statistics

To verify fermionic statistics, introduce the annihilation part of the right-branch oscillator phase,

$$d_R(x)=i\sqrt{\frac{2\pi}{L}}\sum_{p>0}\frac{e^{ipx-ap/2}}{\sqrt{p}}\,b_p,\qquad\varphi_{R,\mathrm{osc}}(x)=d_R(x)+d_R^\dagger(x).$$

The next subsection derives this decomposition and its finite-size sum. Define

$$D_a(s)=[d_R(x),d_R^\dagger(x')],\qquad s=x-x',\qquad z_a=e^{-2\pi a/L}.$$

We distinguish the oscillator phase from the physical fermion throughout.

$$D_a(s)=-\ln\left[1-e^{(2\pi/L)(-a+is)}\right]. \tag{3.134}$$

$$D_a(s)\simeq-\ln\left[\frac{2\pi}{L}(a-is)\right],\qquad L\gg a,|s|. \tag{3.135}$$

$$\psi_R(x)=\mathcal{A}_{a,L}\,F_R\,e^{-i[d_R(x)+d_R^\dagger(x)]}\quad\text{(oscillator part; number phase suppressed).} \tag{3.136}$$

The two exponentials cannot be split without a scalar factor because $[d_R,d_R^\dagger]\neq0$. The sign of the physical annihilation vertex is fixed by (3.78).

$$D_a(0)=-\ln(1-z_a)\simeq-\ln\frac{2\pi a}{L}. \tag{3.137}$$

$$\begin{aligned}e^{-i(d_R+d_R^\dagger)}&=e^{-id_R^\dagger}e^{-id_R}e^{-D_a(0)/2}\\&=e^{-id_R^\dagger}e^{-id_R}\sqrt{1-z_a}\simeq e^{-id_R^\dagger}e^{-id_R}\sqrt{\frac{2\pi a}{L}}.\end{aligned} \tag{3.138}$$

$$\psi_R(x)=\mathcal{A}_{a,L}F_R\sqrt{1-z_a}\,e^{-id_R^\dagger(x)}e^{-id_R(x)}. \tag{3.139}$$

$$\psi_R^\dagger(x)=\mathcal{A}_{a,L}^*\sqrt{1-z_a}\,e^{id_R^\dagger(x)}e^{id_R(x)}F_R^\dagger. \tag{3.140}$$

In a neutral product on one branch, the number shifts cancel after their phases have been tracked. In a product of two annihilation operators, an overall F_R^2 remains; it does not supply the local same-branch exchange sign.

$$\begin{aligned}\psi_R(x)\psi_R(x')=\mathcal{A}_{a,L}^2(1-z_a)F_R^2\\\times e^{-id_R^\dagger(x)}e^{-id_R(x)}e^{-id_R^\dagger(x')}e^{-id_R(x')},\end{aligned} \tag{3.141}$$

where the explicitly separated number phases are temporarily suppressed in this local oscillator calculation.

$$= \mathcal{A}_{a,L}^2(1-z_a)F_R^2\,\mathcal{O}_{--}(x,x')\,e^{-D_a(s)}, \tag{3.142}$$

with

$$\mathcal{O}_{--}(x,x') = e^{-id_R^\dagger(x)}e^{-id_R^\dagger(x')}e^{-id_R(x)}e^{-id_R(x')}.$$

Creation parts commute with one another, as do annihilation parts.

$$\psi_R(x)\psi_R(x') \simeq \mathcal{A}_{a,L}^2\frac{2\pi a}{L}\frac{2\pi}{L}(a-is)F_R^2\mathcal{O}_{--}(x,x'). \tag{3.143}$$

$$= \frac{4\pi^2\mathcal{A}_{a,L}^2 a}{L^2}(a-is)F_R^2\mathcal{O}_{--}(x,x'). \tag{3.144}$$

$$\{\psi_R(x),\psi_R(x')\} \simeq \frac{4\pi^2\mathcal{A}_{a,L}^2 a}{L^2}F_R^2\mathcal{O}_{--}(x,x')\big[(a-is)+(a+is)\big]. \tag{3.145}$$

$$= \frac{8\pi^2\mathcal{A}_{a,L}^2 a^2}{L^2}F_R^2\mathcal{O}_{--}(x,x') \longrightarrow 0. \tag{3.146}$$

After normalization gives $\mathcal{A}_{a,L}^2 \sim 1/(2\pi a)$, the coefficient tends to zero in the local continuum limit. This calculation retains the regulator instead of declaring an unregulated product to vanish by inspection. The exact finite-ring exchange also requires the number phases. With $h = 2\pi/L$, their ratio under $x \leftrightarrow x'$ is e^{-ihs}, whereas the oscillator ratio is

$$\lim_{a\to 0^+}\frac{1-z_a e^{ihs}}{1-z_a e^{-ihs}} = -e^{ihs}, \qquad s \notin L\mathbb{Z}.$$

Their product is -1. Thus the finite-size zero mode restores exact fermionic exchange around the ring; it cannot be discarded before taking that limit.

3.14 Splitting the phase and summing the finite-size contraction

$$\begin{aligned}\varphi_R(x) = \phi(x) - \theta(x) = &-\frac{2\pi\hat{N}_R x}{L}\\ &-\frac{2i\pi}{L}\sum_{p>0}\sqrt{\frac{Lp}{2\pi}}\frac{e^{-ap/2-ipx}}{p}b_p^\dagger\\ &-\frac{2i\pi}{L}\sum_{p<0}\sqrt{\frac{L|p|}{2\pi}}\frac{e^{-a|p|/2-ipx}}{p}b_{-p}.\end{aligned} \tag{3.147}$$

This is the physical right-branch phase in the free-oscillator basis. The winding term commutes with the oscillator fields and therefore does not enter their contraction; its number-shift phase is nevertheless needed in the complete finite-ring fermion. Change $p \mapsto -p$ in the negative-momentum sum.

$$\varphi_{R,\mathrm{osc}}(x) = -\frac{2i\pi}{L}\sum_{p>0}\sqrt{\frac{Lp}{2\pi}}\frac{e^{-ap/2}}{p}(e^{-ipx}b_p^\dagger - e^{ipx}b_p). \tag{3.148}$$

$$= -i\sqrt{\frac{2\pi}{L}}\sum_{p>0}\frac{e^{-ap/2}}{\sqrt{p}}(e^{-ipx}b_p^\dagger - e^{ipx}b_p). \tag{3.149}$$

The two terms are Hermitian conjugates after multiplication by $-i$, so the phase is Hermitian.

$$\varphi_{R,\mathrm{osc}}(x) = d_R^\dagger(x) + d_R(x). \tag{3.150}$$

$$d_R(x) = i\sqrt{\frac{2\pi}{L}} \sum_{p>0} \frac{e^{-ap/2+ipx}}{\sqrt{p}}\, b_p. \tag{3.151}$$

$$[d_R(x), d_R(x')] = 0, \qquad [d_R^\dagger(x), d_R^\dagger(x')] = 0. \tag{3.152}$$

$$D_a(s) = \frac{2\pi}{L} \sum_{p_1,p_2>0} \frac{e^{-a(p_1+p_2)/2+i(p_1x-p_2x')}}{\sqrt{p_1p_2}} [b_{p_1}, b_{p_2}^\dagger]. \tag{3.153}$$

$$D_a(s) = \frac{2\pi}{L} \sum_{p>0} \frac{e^{-ap+ips}}{p}. \tag{3.154}$$

A direct continuum integral $\int_0^\infty dp\, e^{-ap+ips}/p$ has an infrared divergence. Retain the finite ring until the sum has been evaluated.

$$D_a(s) = \frac{2\pi}{L} \sum_{n=1}^{\infty} \frac{1}{2\pi n/L} \exp\left[(-a+is)\frac{2\pi n}{L}\right]. \tag{3.155}$$

$$D_a(s) = \sum_{n=1}^{\infty} \frac{z^n}{n} \equiv \mathcal{S}_0(z), \qquad z = \exp\left[(-a+is)\frac{2\pi}{L}\right]. \tag{3.156}$$

$$|z| = e^{-2\pi a/L} < 1, \qquad \mathcal{S}_0(0) = 0. \tag{3.157}$$

$$\mathcal{S}_0'(z) = \sum_{n=1}^{\infty} z^{n-1} = \sum_{n=0}^{\infty} z^n = \frac{1}{1-z}. \tag{3.158}$$

$$\mathcal{S}_0(z) = \int_0^z \frac{d\zeta}{1-\zeta} = -\ln(1-z). \tag{3.159}$$

This proves (3.134); $a > 0$ fixes the analytic branch and makes the differentiation of the series legitimate.

3.15 The mixed anticommutator and the ultraviolet normalization

$$\begin{aligned}\psi_R^\dagger(x)\psi_R(x') \simeq |\mathcal{A}_{a,L}|^2 \frac{2\pi a}{L} \\ \times e^{id_R^\dagger(x)} e^{id_R(x)} e^{-id_R^\dagger(x')} e^{-id_R(x')}.\end{aligned} \tag{3.160}$$

We keep the same local large-L limit as in the preceding calculation; the finite-ring zero-mode check follows below.

$$= |\mathcal{A}_{a,L}|^2 \frac{2\pi a}{L}\, \mathcal{O}_{+-}(x,x')\, e^{D_a(s)}, \tag{3.161}$$

where

$$\mathcal{O}_{+-}(x,x') = e^{id_R^\dagger(x)} e^{-id_R^\dagger(x')} e^{id_R(x)} e^{-id_R(x')}.$$

$$\psi_R^\dagger(x)\psi_R(x') \simeq |\mathcal{A}_{a,L}|^2 \frac{2\pi a}{L} \mathcal{O}_{+-}(x,x') \left[\frac{2\pi}{L}(a-is)\right]^{-1}. \tag{3.162}$$

$$\begin{aligned}\psi_R(x')\psi_R^\dagger(x) \simeq |\mathcal{A}_{a,L}|^2 \frac{2\pi a}{L} \\ \times e^{-id_R^\dagger(x')} e^{-id_R(x')} e^{id_R^\dagger(x)} e^{id_R(x)}.\end{aligned} \tag{3.163}$$

$$= |\mathcal{A}_{a,L}|^2 \frac{2\pi a}{L} \mathcal{O}_{+-}(x,x')\, e^{D_a(-s)}. \tag{3.164}$$

Splitting creation and annihilation parts makes all reordering factors explicit. An unsplit calculation with ϕ, θ must reproduce the same scalar commutators.

$$\psi_R(x')\psi_R^\dagger(x) \simeq |\mathcal{A}_{a,L}|^2 \frac{2\pi a}{L} \mathcal{O}_{+-}(x,x') \left[\frac{2\pi}{L}(a+is)\right]^{-1}. \tag{3.165}$$

$$\{\psi_R^\dagger(x),\psi_R(x')\} \simeq |\mathcal{A}_{a,L}|^2 \frac{2\pi a}{L}\mathcal{O}_{+-}(x,x') \times \frac{L}{2\pi}\left(\frac{1}{a-is}+\frac{1}{a+is}\right). \tag{3.166}$$

$$= |\mathcal{A}_{a,L}|^2 \mathcal{O}_{+-}(x,x')\frac{2a^2}{a^2+s^2}. \tag{3.167}$$

Taking $a=0$ at the start only tests separated points and loses the contact term. For $|s| \gg a$ the regulated kernel is small; for $|s| \lesssim a$ it contains the peak that becomes the delta function. Both regimes are necessary for canonical normalization. Figure 3.7 shows why the cutoff must be retained through normalization.

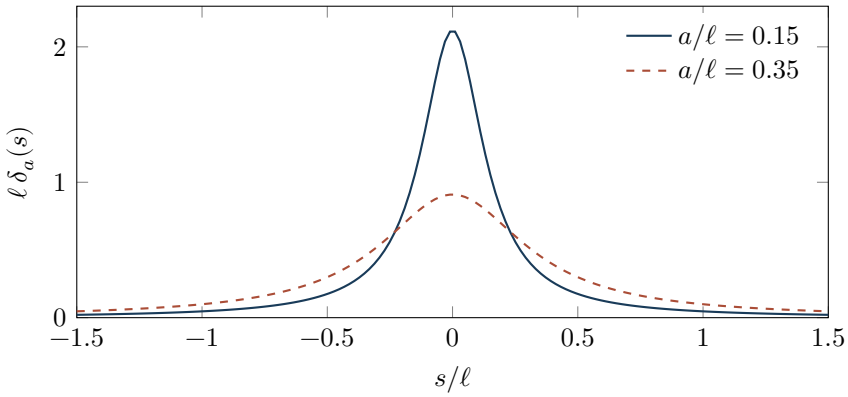


Figure 3.7: The regulated contact kernel $\delta_a(s) = a/[\pi(a^2+s^2)]$ in units of an arbitrary fixed length ℓ. Each curve has unit area on the entire real line. As a decreases the peak narrows and grows; evaluating only separated points would miss its distributional limit. The curves use exactly the formula entering the fermion normalization.

$$\int_{-\infty}^{\infty} ds\,\frac{a}{a^2+s^2} = \pi. \tag{3.168}$$

$$\frac{a}{\pi(a^2+s^2)} \xrightarrow[a\to 0^+]{\mathcal{D}'} \delta(s), \qquad \frac{2a^2}{a^2+s^2} = 2\pi a\,\frac{a}{\pi(a^2+s^2)}. \tag{3.169}$$

The second expression carries an extra factor of a. It must be combined with the diverging normalization $|\mathcal{A}_{a,L}|^2$ before the limit is taken.

$$\{\psi_R^\dagger(x),\psi_R(x')\} = \lim_{a\to 0^+} 2\pi a|\mathcal{A}_{a,L}|^2\delta(x-x')\mathcal{O}_{+-}(x,x'). \tag{3.170}$$

At coincident positions, $\mathcal{O}_{+-}(x,x)=1$; the delta distribution selects precisely that value.

$$\{\psi_R^\dagger(x),\psi_R(x')\} = \lim_{a\to 0^+} 2\pi a|\mathcal{A}_{a,L}|^2\delta(x-x'). \tag{3.171}$$

$$\mathcal{A}_{a,\infty} = \frac{1}{\sqrt{2\pi a}}, \qquad \{\psi_R^\dagger(x),\psi_R(x')\} = \delta(x-x'). \tag{3.172}$$

The short-distance cutoff is essential while the operator products are manipulated. Removing it prematurely produces an undefined subtraction of singular terms and obscures the finite contact contribution. For completeness, finite-size matching can be done before the large-L approximation. With $z_a = e^{-2\pi a/L}$, choose

$$\mathcal{A}_{a,L} = \frac{1}{\sqrt{L(1-z_a)}}, \qquad \mathcal{A}_{a,L}\sqrt{1-z_a} = \frac{1}{\sqrt{L}}.$$

The periodic finite-ring vertex is therefore

$$\psi_r(x) = \frac{F_r}{\sqrt{L}} e^{2\pi i r \hat{N}_r x/L} e^{-i\varphi^{\rm cr}_{r,\rm osc}(x)} e^{-i\varphi^{\rm an}_{r,\rm osc}(x)},$$

where the superscripts denote the creation and annihilation parts and the number phase acts before the number-shift operator on a number eigenstate. The notation $:e^{-i\varphi_r}:_a$ used in the main bosonization identity is the cutoff-matched vertex: relative to the raw oscillator Wick-ordered exponential it includes the factor $\sqrt{2\pi a/L}$. Equivalently, relative to the unsplit exponential it includes $\sqrt{2\pi a/[L(1-z_a)]}$. This convention preserves the stated prefactor $(2\pi a)^{-1/2}$ without absorbing any amplitude into F_r. The complete periodic-ring mixed anticommutator can be checked with the number phases retained. Put $h = 2\pi/L$. Up to a common ordered oscillator product, its scalar kernel is

$$\frac{e^{-ih\hat{N}_R s}}{L}\left[\frac{1}{1-z_a e^{ihs}} + \frac{e^{-ihs}}{1-z_a e^{-ihs}}\right].$$

The first geometric series contains modes $n \geq 0$, and the second modes $n \leq -1$. Hence, as $a \to 0^+$,

$$\frac{1}{L}\sum_{n\in\mathbb{Z}} e^{2\pi i n s/L} = \delta_L(s).$$

At its support the common number phase and ordered oscillator product reduce to one. This supplies the periodic delta function. Taking $L \to \infty$ first instead gives the local Dirac delta derived above. Distinct branches anticommute because of $F_R F_L = -F_L F_R$.

4 The interacting Luttinger Hamiltonian

Having fixed the operator dictionary, we translate the free kinetic energy and the forward density interactions into the canonical fields. The coefficients define the physical velocity and Luttinger parameter; all stability conditions are kept explicit.

4.1 The free Hamiltonian in canonical fields

The density oscillators make the free fermion problem diagonal. Our next aim is to express its energy in the local canonical fields, because a density interaction will then become a quadratic field Hamiltonian. Throughout this section the fermions are spinless. The reference state is the linearized filled sea $|0_{\rm FS}\rangle$, annihilated by every b_q. Normal ordering is relative to this sea. On the ring $q = 2\pi n/L$, with $n \in \mathbb{Z}\setminus\{0\}$. For manipulations that invert the exponential regulator, first keep a finite symmetric set $0 < |q| \leq \Lambda$; take $a \to 0$ at fixed Λ, and only then remove this auxiliary mode bound within the low-energy theory.

The finite-ring fermion convention used earlier fills the zero-energy level of each branch and uses integer momenta. Adding $N_r \geq 0$ particles to that reference therefore costs

$$\sum_{j=1}^{N_r} \frac{2\pi v_F j}{L} = \frac{2\pi v_F}{L}\frac{N_r(N_r+1)}{2} = \frac{\pi v_F}{L} N_r(N_r+1). \tag{4.1}$$

For $N_r = -M < 0$, removing the occupied levels of energies $0, -2\pi v_F/L, \ldots, -2\pi v_F(M-1)/L$ costs $\pi v_F M(M-1)/L$, the same polynomial. Let $\mathcal{H}_{0,\rm ref}$ denote this original generator. To

use symmetric quadratic zero-mode energies below, we explicitly shift the finite-size chemical potential:

$$\mathcal{H}_{0,\mathrm{ref}} = H_{0,\mathrm{osc}} + \frac{\pi v_F}{L} \sum_{r=R,L} \hat{N}_r(\hat{N}_r + 1), \tag{4.2}$$

$$H_0 \equiv \mathcal{H}_{0,\mathrm{ref}} - \delta\mu_L \hat{N}, \qquad \delta\mu_L = \frac{\pi v_F}{L}, \qquad \hat{N} = \hat{N}_R + \hat{N}_L. \tag{4.3}$$

This shift does not change the periodic spin structure or any fixed-number oscillator state. All Hamiltonians denoted H in the present Hamiltonian and ground-state sections use this shifted generator. A finite-size annihilation operator evolved with the unshifted generator instead has the extra phase $\psi_{r,\mathrm{ref}}(t) = e^{-i\delta\mu_L t}\psi_r(t)$; the generator must be matched when comparing correlators.

$$H_0 = \sum_{q\neq 0} v_F|q| b_q^\dagger b_q + \frac{\pi v_F}{L}(\hat{N}_R^2 + \hat{N}_L^2). \tag{4.4}$$

The last term is constant only within a fixed branch-number sector. We retain it when comparing sectors, and suppress it in oscillator manipulations. Define $\hat{J} = \hat{N}_R - \hat{N}_L$. For the following explicit winding representation, the uniform phase operators are held fixed; number-changing vertices still require their Klein and zero-mode operators.

$$b_q^\dagger = \sqrt{\frac{2\pi}{L|q|}} \sum_{r=\pm 1} \Theta(rq)\rho_r(q), \tag{4.5}$$

$$= \sqrt{\frac{2\pi}{L|q|}} [\Theta(q)\rho_R(q) + \Theta(-q)\rho_L(q)]. \tag{4.6}$$

Taking the adjoint with $\rho_r(q)^\dagger = \rho_r(-q)$ gives

$$b_q^\dagger + b_{-q} = \sqrt{\frac{2\pi}{L|q|}} [\Theta(q)\rho_R(q) + \Theta(-q)\rho_L(q) + \Theta(-q)\rho_R(q) + \Theta(q)\rho_L(q)], \tag{4.7}$$

$$= \sqrt{\frac{2\pi}{L|q|}} [\rho_R(q) + \rho_L(q)]. \tag{4.8}$$

Likewise,

$$b_q^\dagger - b_{-q} = \sqrt{\frac{2\pi}{L|q|}} [\Theta(q)\rho_R(q) + \Theta(-q)\rho_L(q) - \Theta(-q)\rho_R(q) - \Theta(q)\rho_L(q)], \tag{4.9}$$

$$= \sqrt{\frac{2\pi}{L|q|}} \begin{cases} \rho_R(q) - \rho_L(q), & q > 0, \\ -\rho_R(q) + \rho_L(q), & q < 0, \end{cases} \tag{4.10}$$

$$= \sqrt{\frac{2\pi}{L|q|}} \operatorname{sgn}(q)[\rho_R(q) - \rho_L(q)]. \tag{4.11}$$

Thus the sum of branch densities determines the density field ϕ, while their difference determines the dual field θ. The proportionality here includes the Fourier derivative and the chirality sign; it is not an equality between a field and a density.

$$\phi(x) = -\frac{\pi \hat{N} x}{L} - \frac{i\pi}{L} \sum_{q\neq 0} \sqrt{\frac{L|q|}{2\pi}} \frac{e^{-a|q|/2 - iqx}}{q} (b_q^\dagger + b_{-q}), \tag{4.12}$$

$$\theta(x) = \frac{\pi \hat{J} x}{L} + \frac{i\pi}{L} \sum_{q\neq 0} \sqrt{\frac{L|q|}{2\pi}} \frac{e^{-a|q|/2 - iqx}}{|q|} (b_q^\dagger - b_{-q}). \tag{4.13}$$

The distinction between $1/q$ and $1/|q|$ is the sign in (4.11). To recover $\sum_q |q| b_q^\dagger b_q$, differentiation removes these inverse momenta and inverse Fourier transformation isolates each oscillator.

$$\partial_x \phi(x) = -\frac{\pi \hat{N}}{L} - \frac{\pi}{L} \sum_{q\neq 0} \sqrt{\frac{L|q|}{2\pi}} e^{-a|q|/2 - iqx} (b_q^\dagger + b_{-q}), \tag{4.14}$$

$$\partial_x \theta(x) = \frac{\pi \hat{J}}{L} + \frac{\pi}{L} \sum_{q\neq 0} \sqrt{\frac{L|q|}{2\pi}} \operatorname{sgn}(q) e^{-a|q|/2 - iqx} (b_q^\dagger - b_{-q}). \tag{4.15}$$

The opposite winding signs reproduce $\rho_{\rm sm} = -\partial_x \phi/\pi$ and $\rho_R - \rho_L = \partial_x \theta/\pi$. Squaring these gradients will produce the energy.

4.1.1 Inverting the field expansion

For $q' = 2\pi n'/L$, Fourier orthogonality on the ring reads

$$\int_0^L dx\, e^{iq'x} = L\delta_{q',0}, \tag{4.16}$$

$$\int_0^L dx\, e^{i(q'-q)x} = L\delta_{q,q'}. \tag{4.17}$$

$$\begin{aligned} \int_0^L dx\, e^{iq'x} \partial_x \phi(x) = &-\frac{\pi \hat{N}}{L} \int_0^L dx\, e^{iq'x} \\ &- \frac{\pi}{L} \sum_{q\neq 0} \sqrt{\frac{L|q|}{2\pi}} e^{-a|q|/2} (b_q^\dagger + b_{-q}) \int_0^L dx\, e^{i(q'-q)x}, \end{aligned} \tag{4.18}$$

$$= -\pi \hat{N} \delta_{q',0} - \pi \sum_{q\neq 0} \sqrt{\frac{L|q|}{2\pi}} e^{-a|q|/2} (b_q^\dagger + b_{-q}) \delta_{q,q'}, \tag{4.19}$$

$$= -\pi \hat{N} \delta_{q',0} - \pi \sqrt{\frac{L|q'|}{2\pi}} e^{-a|q'|/2} (b_{q'}^\dagger + b_{-q'}). \tag{4.20}$$

The oscillator term in this last line is present only for $q' \neq 0$; there is no b_0. The $q' = 0$ contribution is the fixed winding sector. It drops out of the nonzero-mode inversion, but is not generally negligible as an operator.

$$\int_0^L dx\, e^{iq'x} \partial_x \theta(x) = \pi \hat{J} \delta_{q',0} + \pi \sqrt{\frac{L|q'|}{2\pi}} e^{-a|q'|/2} \operatorname{sgn}(q') (b_{q'}^\dagger - b_{-q'}). \tag{4.21}$$

For $q > 0$ the two combinations can now be isolated:

$$b_q^\dagger + b_{-q} = -\sqrt{\frac{2}{\pi L q}} e^{aq/2} \int_0^L dx\, e^{iqx} \partial_x \phi(x), \tag{4.22}$$

$$b_q^\dagger - b_{-q} = \sqrt{\frac{2}{\pi L q}} e^{aq/2} \int_0^L dx\, e^{iqx} \partial_x \theta(x). \tag{4.23}$$

Adding and subtracting the preceding two equations gives

$$b_q^\dagger = \frac{e^{aq/2}}{\sqrt{2\pi qL}} \int_0^L dx\, e^{iqx}[-\partial_x\phi(x) + \partial_x\theta(x)], \tag{4.24}$$

$$b_{-q} = -\frac{e^{aq/2}}{\sqrt{2\pi qL}} \int_0^L dx\, e^{iqx}[\partial_x\phi(x) + \partial_x\theta(x)]. \tag{4.25}$$

The second formula is an annihilator in the opposite physical branch. Replacing $-q$ by q without changing the chirality condition would give the wrong field combination. Taking the adjoint of $b_q^\dagger$ and multiplying yields

$$\begin{aligned}\sum_{q>0} q\, b_q^\dagger b_q = \frac{1}{2\pi L}\sum_{q>0} e^{aq} \int_0^L dx \int_0^L dx'\, e^{iq(x-x')} \\ \times :[-\partial_x\phi(x) + \partial_x\theta(x)][-\partial_{x'}\phi(x') + \partial_{x'}\theta(x')]:.\end{aligned} \tag{4.26}$$

The $a \to 0$ limit must be taken with the finite-mode prescription stated above. The inverse factor e^{aq} exactly compensates the two regulated gradients; it must not be pulled out of the momentum sum.

4.1.2 Combining the two propagation directions

For $q < 0$ the corresponding oscillator formulas are

$$b_q = -\frac{e^{a|q|/2}}{\sqrt{2\pi|q|L}} \int_0^L dx\, e^{-iqx}[\partial_x\phi(x) + \partial_x\theta(x)], \tag{4.27}$$

$$b_q^\dagger = -\frac{e^{a|q|/2}}{\sqrt{2\pi|q|L}} \int_0^L dx\, e^{iqx}[\partial_x\phi(x) + \partial_x\theta(x)]. \tag{4.28}$$

Positive and negative oscillator momenta label different propagation directions. Their opposite signs in the dual-field contribution are essential.

$$\begin{aligned}\sum_{q<0} |q| b_q^\dagger b_q = \frac{1}{2\pi L}\sum_{q<0} e^{a|q|} \int_0^L dx \int_0^L dx'\, e^{iq(x-x')} \\ \times :[\partial_x\phi(x) + \partial_x\theta(x)][\partial_{x'}\phi(x') + \partial_{x'}\theta(x')]:.\end{aligned} \tag{4.29}$$

To display the cancellation without hiding operator order, write $X(x) = \partial_x\phi(x)$ and $Y(x) = \partial_x\theta(x)$ within this calculation, and define

$$D(x,x') = :X(x)X(x') + Y(x)Y(x'):, \qquad M(x,x') = :X(x)Y(x') + Y(x)X(x'):. \tag{4.30}$$

These are only abbreviations for derivatives of the same canonical fields. All products below retain the reference-sea normal ordering.

$$\sum_{q\neq 0} |q| b_q^\dagger b_q = \frac{1}{2\pi L}\sum_{q\neq 0} e^{a|q|} \int_0^L dx \int_0^L dx'\, e^{iq(x-x')} D(x,x') + \mathcal{X}, \tag{4.31}$$

$$\mathcal{X} = -\frac{1}{2\pi L}\sum_{q\neq 0} \mathrm{sgn}(q) e^{a|q|} \int_0^L dx \int_0^L dx'\, e^{iq(x-x')} M(x,x'). \tag{4.32}$$

Pairing q with $-q$ gives

$$\mathcal{X} = -\frac{1}{2\pi L}\sum_{q>0} e^{aq}\int_0^L dx\int_0^L dx'\ \left[e^{iq(x-x')}M(x,x') - e^{-iq(x-x')}M(x,x')\right], \tag{4.33}$$

$$= -\frac{1}{2\pi L}\sum_{q>0} e^{aq}\int_0^L dx\int_0^L dx'\, e^{iq(x-x')}\left[M(x,x') - M(x',x)\right], \tag{4.34}$$

$$= 0. \tag{4.35}$$

In the second line we interchanged the dummy integration variables. In the third line, normal-ordered products of linear boson fields are symmetric under exchanging their factors. Without normal ordering the same reordering generates a state-independent commutator contribution, which belongs to the vacuum energy; it cannot simply be discarded in an unnormal-ordered identity. Consequently, in the oscillator sector,

$$H_{0,\mathrm{osc}} = \frac{v_F}{2\pi L}\sum_{q\neq 0} e^{a|q|}\int_0^L dx\int_0^L dx'\, e^{iq(x-x')}D(x,x'). \tag{4.36}$$

At $a = 0$ and finite Λ, the Fourier kernel is the projector onto the retained nonzero modes:

$$\frac{1}{L}\sum_{0<|q|\leq\Lambda} e^{iq(x-x')} = \delta_{L,\Lambda}(x-x') - \frac{1}{L}. \tag{4.37}$$

This kernel acts as the identity on the retained oscillator gradients, whose spatial integrals vanish. In the ring limit it is $\delta_L(x-x') - 1/L$, rather than the infinite-line delta function. The winding contribution is then restored separately.

$$H_0 = \frac{v_F}{2\pi}\int_0^L dx :\left[(\partial_x\phi)^2 + (\partial_x\theta)^2\right]:. \tag{4.38}$$

Use the canonical momentum specified by the project normalization:

$$\Pi(x) = \frac{1}{\pi}\partial_x\theta(x), \qquad [\phi(x),\Pi(x')] = i\delta(x-x') \tag{4.39}$$

on the infinite line; the oscillator-only ring version has $\delta_L - 1/L$. Thus

$$H_0 = \frac{v_F}{2\pi}\int dx :\left[(\pi\Pi)^2 + (\partial_x\phi)^2\right]:. \tag{4.40}$$

The Hamiltonian is quadratic in a canonical coordinate and its momentum. Its momentum-squared term is the kinetic contribution, and its field-gradient term is an elastic potential energy. In Fourier space each independent mode is a harmonic oscillator. A noninteracting linearized fermion lead therefore also has a noninteracting bosonic description. The bosons are collective density excitations, and their free character does not turn a bosonic oscillator into an individual electron.

4.2 Forward interactions and the Luttinger parameters

We now ask how interactions change the energy cost of smooth density and current fluctuations. The model contains two physical branches of one spinless species, a linearized dispersion, and homogeneous short-range forward interactions. The coefficients g_2 and g_4 are defined by the density Hamiltonian below and have dimensions of velocity in units $\hbar = 1$. Products of densities are point-split or cutoff-regularized and normal-ordered relative to the same reference sea. In

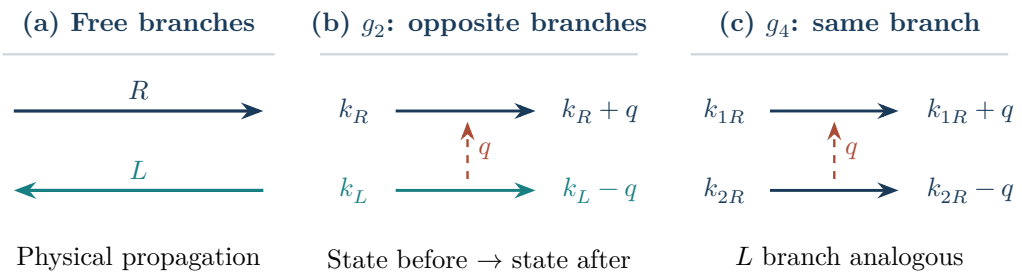


Figure 4.1: Forward-scattering geometry (schematic). The free branch arrows specify physical propagation directions. In a g_2 event, $k_R \to k_R + q$ and $k_L \to k_L - q$ transfer a small momentum between opposite branches. A g_4 event transfers q between two momenta in the same branch. Both preserve total momentum and branch particle number; neither converts a right mover into a left mover.

particular, an intrabranch density interaction is not an unregularized coincident product of four identical spinless fermion fields. Without interbranch coupling, a right-branch disturbance travels to the right and a left-branch disturbance travels to the left. Figure 4.1 retains this branch geometry and distinguishes the two forward processes. The g_2 interaction couples opposite physical branches, whereas g_4 couples density fluctuations within the same branch.

$$H_{\text{int}} = H_{\text{inter}} + H_{\text{intra}}. \tag{4.41}$$

The branch densities are

$$\rho_L(x) = -\frac{1}{2\pi}[\partial_x\phi(x) + \partial_x\theta(x)], \tag{4.42}$$

$$\rho_R(x) = -\frac{1}{2\pi}[\partial_x\phi(x) - \partial_x\theta(x)]. \tag{4.43}$$

Their interbranch interaction becomes

$$H_{\text{inter}} = g_2 \int dx\, {:}\psi_R^\dagger(x)\psi_R(x)\psi_L^\dagger(x)\psi_L(x){:}, \tag{4.44}$$

$$= g_2 \int dx\, {:}\rho_R(x)\rho_L(x){:}, \tag{4.45}$$

$$= \frac{g_2}{(2\pi)^2} \int dx\, {:}(\partial_x\phi - \partial_x\theta)(\partial_x\phi + \partial_x\theta){:}, \tag{4.46}$$

$$= \frac{g_2}{4\pi^2} \int dx\, {:}\left[(\partial_x\phi)^2 - (\partial_x\theta)^2\right]{:}. \tag{4.47}$$

A positive g_2 raises the energy of coincident positive branch-density fluctuations, while a negative g_2 lowers it. This is the sense in which the forward coupling is repulsive or attractive; excess densities can individually have either sign. In either case the field Hamiltonian remains quadratic. Similarly, retaining the factor of $1/2$ that prevents double counting within a branch,

$$H_{\text{intra}} = \frac{g_4}{2} \int dx\, {:}\left[\psi_R^\dagger\psi_R\psi_R^\dagger\psi_R + \psi_L^\dagger\psi_L\psi_L^\dagger\psi_L\right]{:}, \tag{4.48}$$

$$= \frac{g_4}{2} \int dx\, {:}[\rho_R^2(x) + \rho_L^2(x)]{:}, \tag{4.49}$$

$$= \frac{g_4}{2(2\pi)^2} \int dx\, {:}\left[(\partial_x\phi - \partial_x\theta)^2 + (\partial_x\phi + \partial_x\theta)^2\right]{:}, \tag{4.50}$$

$$= \frac{g_4}{8\pi^2} \int dx\, {:}[(\partial_x\phi)^2 - 2\partial_x\phi\,\partial_x\theta + (\partial_x\theta)^2 + (\partial_x\phi)^2 + 2\partial_x\phi\,\partial_x\theta + (\partial_x\theta)^2]{:}, \tag{4.51}$$

$$= \frac{g_4}{4\pi^2} \int dx\, {:}\left[(\partial_x\phi)^2 + (\partial_x\theta)^2\right]{:}. \tag{4.52}$$

Combining the free and interacting parts gives

$$\begin{aligned} H &= H_0 + H_{\text{inter}} + H_{\text{intra}} \\ &= \frac{v_F}{2\pi}\int dx\,{:}[(\partial_x\phi)^2 + (\partial_x\theta)^2]{:} \\ &\quad + \frac{g_2}{4\pi^2}\int dx\,{:}[(\partial_x\phi)^2 - (\partial_x\theta)^2]{:} + \frac{g_4}{4\pi^2}\int dx\,{:}[(\partial_x\phi)^2 + (\partial_x\theta)^2]{:}, \qquad (4.53) \\ &= \frac{1}{2\pi}\int dx\,{:}\left[\left(v_F + \frac{g_4+g_2}{2\pi}\right)(\partial_x\phi)^2 + \left(v_F + \frac{g_4-g_2}{2\pi}\right)(\partial_x\theta)^2\right]{:}, \qquad (4.54) \\ &= \frac{u}{2\pi}\int dx\,{:}\left[\frac{1}{K}(\partial_x\phi)^2 + K(\partial_x\theta)^2\right]{:}. \qquad (4.55) \end{aligned}$$

The common velocity scale is the geometric mean of the two stiffnesses:

$$u^2 = v_F^2\left(1 + \frac{g_2+g_4}{2\pi v_F}\right)\left(1 + \frac{g_4-g_2}{2\pi v_F}\right). \qquad (4.56)$$

Their relative size is encoded in K:

$$uK = v_F\left(1 + \frac{g_4-g_2}{2\pi v_F}\right), \qquad \frac{u}{K} = v_F\left(1 + \frac{g_4+g_2}{2\pi v_F}\right). \qquad (4.57)$$

Both stiffnesses must be positive. Defining the velocity-valued coefficients $A = v_F + g_4/(2\pi)$ and $B = g_2/(2\pi)$, stability means $A > |B|$. The winding energy is

$$H_{\text{zm}} = \frac{\pi u}{2L}\left(\frac{\hat{N}^2}{K} + K\hat{J}^2\right). \qquad (4.58)$$

It follows by inserting the winding gradients in (4.55); oscillator–winding cross terms integrate to zero. The number and current stiffnesses are u/K and uK. From the canonical equation $\partial_t\phi = uK\partial_x\theta$, the particle current is $j_N = uK\partial_x\theta/\pi$, and the electric current is $j_Q = -ej_N$. These are currents of physical particles, not counts of diagonal bosons. With the earlier integer-momentum filled-zero reference and its unshifted chemical potential, restore the linear contribution explicitly:

$$\mathcal{H}_{\text{zm,ref}} = \frac{\pi u}{2L}\left(\frac{\hat{N}^2}{K} + K\hat{J}^2\right) + \delta\mu_L\hat{N}, \qquad \delta\mu_L = \frac{\pi v_F}{L}. \qquad (4.59)$$

The forward density interaction is defined in excess densities and adds the displayed quadratic coefficients; it does not remove this declared finite-size reference shift.

4.2.1 What u and K measure

The homogeneous Gaussian model has the canonical form

$$H_{\text{LL}} = \frac{u}{2\pi}\int dx\,{:}\,[K^{-1}(\partial_x\phi)^2 + K(\partial_x\theta)^2]\,{:}. \qquad (4.60)$$

$$u = v_F\sqrt{\left(1 + \frac{g_4}{2\pi v_F}\right)^2 - \left(\frac{g_2}{2\pi v_F}\right)^2}, \qquad (4.61)$$

$$K = \sqrt{\frac{1 + (g_4-g_2)/(2\pi v_F)}{1 + (g_4+g_2)/(2\pi v_F)}}. \qquad (4.62)$$

The noninteracting reference has $g_2 = g_4 = 0$, hence $K = 1$ and $u = v_F$. A value $K = 1$ alone, however, does not force every interaction coefficient to vanish.

$$g_2 = 0, \quad g_4 \neq 0 \quad \Longrightarrow \quad K = 1, \qquad u = v_F + \frac{g_4}{2\pi}. \tag{4.63}$$

Within this linearized effective model the g_4-only case has the free-fermion exponents and may be refermionized with the shifted velocity u. This is a statement about this model's low-energy structure, not a claim that its microscopic interaction has disappeared. For a stable model, $g_2 \neq 0$ implies $K \neq 1$. For $g_2 > 0$, a positive overlap of the two excess densities costs energy: $u/K > uK$ and $K < 1$. Density fluctuations are relatively stiff. For $g_2 < 0$, the relative stiffness is reversed and $K > 1$. Figure 4.2 displays these relations directly from (4.62).

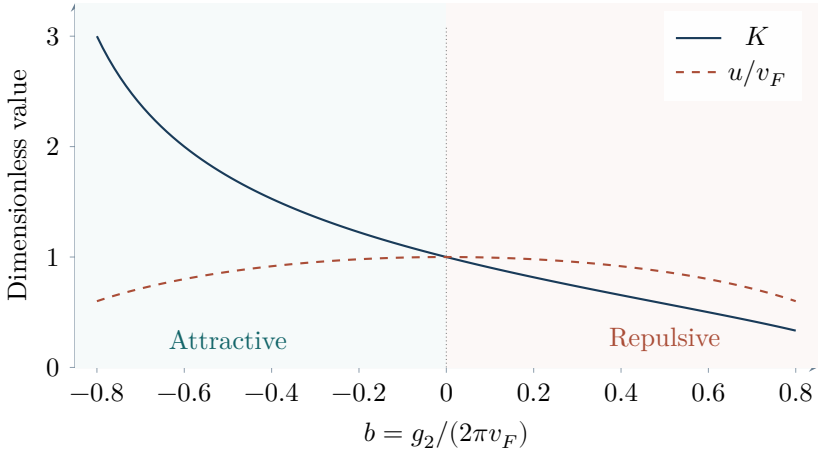


Figure 4.2: Parameters of the stable forward-scattering model. Here $g_4 = 0$, $b = g_2/(2\pi v_F)$, $K = \sqrt{(1-b)/(1+b)}$, and $u/v_F = \sqrt{1-b^2}$. The displayed interval is $-0.8 \leq b \leq 0.8$, strictly inside the stability domain $|b| < 1$. The curves are calculations from (4.62), not data. Increasing repulsion lowers K because it raises the density stiffness relative to the current stiffness.

Backscattering is not included in this forward Hamiltonian. A branch-converting perturbation has the form

$$H_{\mathrm{bs}} = \int dx \, [\gamma_b(x)\psi_L^\dagger(x)\psi_R(x) + \mathrm{h.c.}], \tag{4.64}$$

where $\gamma_b(x)$ contains the physical momentum-transfer envelope; a local impurity is one important case. Such a term is not diagonalized by the following Gaussian transformation. Its importance will instead be determined by a boundary or renormalization-group analysis.

4.3 From the interaction Hamiltonian to the ground state

The Hamiltonian is quadratic in fields but need not be diagonal in the original density oscillators. We now preserve the field-to-oscillator route and determine which linear combinations annihilate the interacting ground state. The states $|0_{\mathrm{FS}}\rangle$ and $|\Omega_{\mathrm{LL}}\rangle$ are compared in the same number and winding sector. The Fourier gradients, repeated here because they are the input to the interacting calculation, are

$$\int_0^L dx \, e^{iqx}\partial_x\phi(x) = -\pi\hat{N}\delta_{q,0} - \pi\sqrt{\frac{L|q|}{2\pi}}e^{-a|q|/2}(b_q^\dagger + b_{-q}), \tag{4.65}$$

$$\int_0^L dx \, e^{iqx}\partial_x\theta(x) = \pi\hat{J}\delta_{q,0} + \pi\sqrt{\frac{L|q|}{2\pi}}e^{-a|q|/2}\,\mathrm{sgn}(q)(b_q^\dagger - b_{-q}). \tag{4.66}$$

The inverse step uses

$$\partial_x\phi(x) = \frac{1}{L}\sum_q e^{-iqx}\left[\int_0^L dx'\, e^{iqx'}\partial_{x'}\phi(x')\right], \tag{4.67}$$

with $\int_0^L dx'\, e^{i(q-q')x'} = L\delta_{q,q'}$.

$$\partial_x\phi(x) = \frac{\pi}{L}\sum_q e^{-iqx}\left[-\hat{N}\delta_{q,0} - \sqrt{\frac{L|q|}{2\pi}}e^{-a|q|/2}(b_q^\dagger + b_{-q})\right], \tag{4.68}$$

$$= -\frac{\pi\hat{N}}{L} - \frac{\pi}{L}\sum_{q\neq 0}\sqrt{\frac{L|q|}{2\pi}}e^{-iqx-a|q|/2}(b_q^\dagger + b_{-q}), \tag{4.69}$$

$$\partial_x\theta(x) = \frac{\pi\hat{J}}{L} + \frac{\pi}{L}\sum_{q\neq 0}\sqrt{\frac{L|q|}{2\pi}}e^{-iqx-a|q|/2}\,\mathrm{sgn}(q)(b_q^\dagger - b_{-q}). \tag{4.70}$$

The $q = 0$ oscillator term is absent in every line. The factor π/L multiplies the oscillator sums as well as the winding term.

4.3.1 The anomalous terms pair opposite momenta

Consider the difference of gradient energies

$$\mathcal{D}_- \equiv \int_0^L dx : \left[(\partial_x\phi)^2 - (\partial_x\theta)^2\right] :. \tag{4.71}$$

The winding energy is set aside in a fixed sector. For fixed excess numbers it is of order $1/L$; if the numbers grow with L, this estimate no longer makes it negligible. Linear oscillator–winding terms vanish after spatial integration. Take the regulator limit at fixed retained modes before using the expressions below.

$$\int_0^L dx\, e^{-iqx} = L\delta_{q,0}. \tag{4.72}$$

The oscillator part of $\mathcal{D}_-$ is

$$\mathcal{D}_{-,\mathrm{osc}} = \frac{\pi}{2L}\int_0^L dx \sum_{q_1,q_2\neq 0}\sqrt{|q_1 q_2|}\, e^{-i(q_1+q_2)x}\Big[:(b_{q_1}^\dagger + b_{-q_1})(b_{q_2}^\dagger + b_{-q_2}):$$
$$- \mathrm{sgn}(q_1)\,\mathrm{sgn}(q_2):(b_{q_1}^\dagger - b_{-q_1})(b_{q_2}^\dagger - b_{-q_2}):\Big]. \tag{4.73}$$

The integral imposes $q_2 = -q_1$, hence $\mathrm{sgn}(q_1)\,\mathrm{sgn}(q_2) = -1$.

$$\mathcal{D}_{-,\mathrm{osc}} = \frac{\pi}{2}\sum_{q\neq 0}|q| :\Big[(b_q^\dagger + b_{-q})(b_{-q}^\dagger + b_q) + (b_q^\dagger - b_{-q})(b_{-q}^\dagger - b_q)\Big]:, \tag{4.74}$$

$$= \frac{\pi}{2}\sum_{q\neq 0}|q| :\Big[b_q^\dagger b_{-q}^\dagger + b_q^\dagger b_q + b_{-q}b_{-q}^\dagger + b_{-q}b_q$$
$$+ b_q^\dagger b_{-q}^\dagger - b_q^\dagger b_q - b_{-q}b_{-q}^\dagger + b_{-q}b_q\Big]:, \tag{4.75}$$

$$= \pi\sum_{q\neq 0}|q|\left(b_q^\dagger b_{-q}^\dagger + b_{-q}b_q\right). \tag{4.76}$$

For $q > 0$, $b_q^\dagger \propto \rho_R(q)$ and $b_{-q}^\dagger \propto \rho_L(-q)$. These create a pair of density excitations with total momentum zero. The anomalous quadratic structure resembles a superconducting pairing

Hamiltonian, but the paired objects here are neutral particle–hole density modes, not charged Cooper pairs. Using the sum and difference of gradient squares,

$$H_{\rm osc} = \frac{u}{2\pi}\int dx : \left[K^{-1}(\partial_x\phi)^2 + K(\partial_x\theta)^2\right] :_{\rm osc}, \tag{4.77}$$

$$= \frac{u}{4\pi}\int dx : \Big[(K^{-1}+K)((\partial_x\phi)^2+(\partial_x\theta)^2) + (K^{-1}-K)((\partial_x\phi)^2-(\partial_x\theta)^2)\Big] :_{\rm osc}, \tag{4.78}$$

$$= \frac{u}{4\pi}\,2\pi\left[(K^{-1}+K)\sum_{q\neq 0}|q| b_q^\dagger b_q + \frac{K^{-1}-K}{2}\sum_{q\neq 0}|q|(b_q^\dagger b_{-q}^\dagger + b_{-q}b_q)\right], \tag{4.79}$$

$$= \frac{u}{2}\sum_{q\neq 0}|q|\left[(K^{-1}+K)b_q^\dagger b_q + \frac{K^{-1}-K}{2}(b_q^\dagger b_{-q}^\dagger + b_{-q}b_q)\right]. \tag{4.80}$$

The anomalous term vanishes when $K = 1$, including the g_4-only model. Both the normal and anomalous coefficients multiply $|q|$. The stable diagonal spectrum therefore remains gapless as $q \to 0$. Grouping each independent pair once,

$$H_{\rm osc} = \sum_{q>0} q\left[A(b_q^\dagger b_q + b_{-q}^\dagger b_{-q}) + B(b_q^\dagger b_{-q}^\dagger + b_q b_{-q})\right]. \tag{4.81}$$

$$A = \frac{u}{2}(K^{-1}+K) = v_F + \frac{g_4}{2\pi}, \qquad B = \frac{u}{2}(K^{-1}-K) = \frac{g_2}{2\pi}. \tag{4.82}$$

A bosonic Bogoliubov transformation has the form

$$\beta_q = \cosh\xi\, b_q - \sinh\xi\, b_{-q}^\dagger, \tag{4.83}$$

$$\beta_q^\dagger = \cosh\xi\, b_q^\dagger - \sinh\xi\, b_{-q}. \tag{4.84}$$

Its canonical algebra follows directly:

$$[\beta_q, \beta_{q'}^\dagger] = \cosh^2\xi\,[b_q, b_{q'}^\dagger] + \sinh^2\xi\,[b_{-q}^\dagger, b_{-q'}], \tag{4.85}$$

$$= (\cosh^2\xi - \sinh^2\xi)\delta_{q,q'} = \delta_{q,q'}. \tag{4.86}$$

Preserving the bosonic commutator does not yet fix ξ. Its value follows by cancelling the anomalous terms in the Hamiltonian, as we now calculate.

5 Bogoliubov diagonalization and the ground state

The interaction mixes opposite density modes. A canonical Bogoliubov transformation removes anomalous terms; its vacuum conditions determine the full coefficient recurrence and normalization of the interacting ground state.

5.1 Explicit Bogoliubov diagonalization

Write $c = \cosh\xi$ and $s = \sinh\xi$, so $c^2 - s^2 = 1$. The squeezing parameter ξ is determined below; it must not be confused with the canonical field $\theta(x)$.

$$b_q = c\beta_q + s\beta_{-q}^\dagger, \tag{5.1}$$

$$b_{-q}^\dagger = s\beta_q + c\beta_{-q}^\dagger. \tag{5.2}$$

The inverse relations are

$$\beta_q = cb_q - sb^\dagger_{-q}, \tag{5.3}$$
$$\beta^\dagger_{-q} = -sb_q + cb^\dagger_{-q}. \tag{5.4}$$

As a check on the inverse and its minus sign,

$$[\beta_p, \beta^\dagger_q] = c^2[b_p, b^\dagger_q] + s^2[b^\dagger_{-p}, b_{-q}], \tag{5.5}$$
$$= (c^2 - s^2)\delta_{p,q} = \delta_{p,q}. \tag{5.6}$$

For the stable model $A > |B|$, and

$$-\frac{B}{A} = \frac{K^2-1}{K^2+1}. \tag{5.7}$$

For one positive q, remove the common factor q and define

$$h_q = A(b^\dagger_q b_q + b^\dagger_{-q} b_{-q}) + B(b^\dagger_q b^\dagger_{-q} + b_q b_{-q}). \tag{5.8}$$

Substitution retains each of the four products:

$$\begin{aligned} h_q = A\Big[&(c\beta^\dagger_q + s\beta_{-q})(c\beta_q + s\beta^\dagger_{-q}) \\ &+ (s\beta_q + c\beta^\dagger_{-q})(s\beta^\dagger_q + c\beta_{-q})\Big] \\ + B\Big[&(c\beta^\dagger_q + s\beta_{-q})(s\beta_q + c\beta^\dagger_{-q}) \\ &+ (c\beta_q + s\beta^\dagger_{-q})(s\beta^\dagger_q + c\beta_{-q})\Big]. \end{aligned} \tag{5.9}$$

The useful double-angle identities can be checked before collecting terms:

$$sc = \frac{e^\xi - e^{-\xi}}{2}\frac{e^\xi + e^{-\xi}}{2} = \frac{1}{4}(e^{2\xi} - e^{-2\xi}) = \frac{1}{2}\sinh(2\xi), \tag{5.10}$$
$$c^2 + s^2 = \frac{1}{4}(e^{2\xi} + e^{-2\xi} + 2) + \frac{1}{4}(e^{2\xi} + e^{-2\xi} - 2) = \cosh(2\xi). \tag{5.11}$$

After reordering the diagonal products,

$$\begin{aligned} h_q = &[2Asc + B(c^2 + s^2)]\beta^\dagger_q \beta^\dagger_{-q} \\ &+ [2Asc + B(s^2 + c^2)]\beta_q \beta_{-q} \\ &+ [Ac^2 + As^2 + 2Bsc]\beta^\dagger_q \beta_q \\ &+ [As^2 + Ac^2 + 2Bsc]\beta^\dagger_{-q}\beta_{-q} + 2As^2 + 2Bsc. \end{aligned} \tag{5.12}$$

The last two terms come from $\beta\beta^\dagger = \beta^\dagger\beta + 1$ and are a scalar shift of the ground-state energy. They matter when the energy zero remains that of the reference sea, even though they do not affect the eigenvectors. Both anomalous coefficients vanish if

$$A\sinh(2\xi) + B\cosh(2\xi) = 0. \tag{5.13}$$

Thus

$$\tanh(2\xi) = -\frac{B}{A} = \frac{K^2-1}{K^2+1}, \qquad \xi = \frac{1}{2}\ln K. \tag{5.14}$$

$$\cosh(2\xi) = \frac{A}{\sqrt{A^2 - B^2}}, \tag{5.15}$$
$$\sinh(2\xi) = -\frac{B}{\sqrt{A^2 - B^2}}. \tag{5.16}$$

The number-operator contribution is then

$$[A\cosh(2\xi) + B\sinh(2\xi)](\beta_q^\dagger\beta_q + \beta_{-q}^\dagger\beta_{-q}) \tag{5.17}$$

$$= \frac{A^2 - B^2}{\sqrt{A^2 - B^2}}(\beta_q^\dagger\beta_q + \beta_{-q}^\dagger\beta_{-q}) \tag{5.18}$$

$$= \sqrt{A^2 - B^2}(\beta_q^\dagger\beta_q + \beta_{-q}^\dagger\beta_{-q}). \tag{5.19}$$

For constant g_2 and g_4, ξ is momentum independent. Figure 5.1 shows what this transformation mixes: an annihilator in one density sector and a creator in the opposite sector. This is a canonical transformation in oscillator space, rather than a rotation between individual electrons.

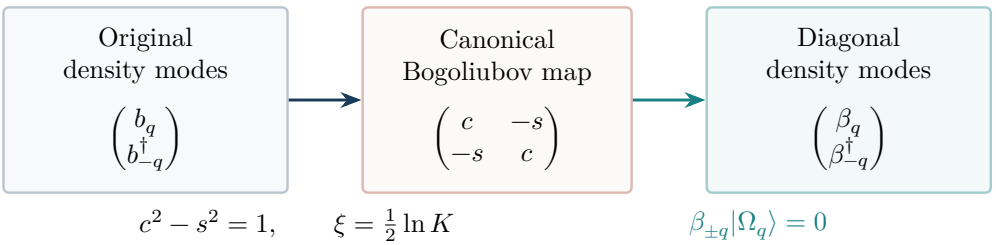


Figure 5.1: Bogoliubov mixing (schematic). The pair $(b_q, b_{-q}^\dagger)$ is transformed by a real matrix with entries $c = \cosh\xi$ and $s = \sinh\xi$. The indefinite metric $\mathrm{diag}(1, -1)$ is preserved because $c^2 - s^2 = 1$. At $K = 1$, $s = 0$; away from this point, each diagonal annihilator contains both original density sectors.

5.2 The vacuum conditions in the original oscillator basis

The diagonal velocity is

$$\sqrt{A^2 - B^2} = \frac{u}{2}\sqrt{(K^{-1} + K)^2 - (K^{-1} - K)^2} = u. \tag{5.20}$$

This is momentum independent for the local model. A nonlocal interaction instead requires starting with $A(q)$ and $B(q)$ and obtaining $u(q)$ and $\xi(q)$ separately. The diagonal Hamiltonian is

$$H = H_{\mathrm{zm}} + E_{\mathrm{vac}} + u\sum_{q\neq 0}|q|\beta_q^\dagger\beta_q. \tag{5.21}$$

Relative to the reference-sea zero, the oscillator vacuum shift is

$$E_{\mathrm{vac}} = \sum_{q>0} q\,[2As^2 + 2Bsc] = \sum_{q>0} q\,[A(\cosh 2\xi - 1) + B\sinh 2\xi] = \sum_{q>0} q(u - A). \tag{5.22}$$

The mode bound or microscopic cutoff is required in this constant. Setting the interacting vacuum energy to zero subtracts it; it does not change the state. The interacting ground state is characterized by the absence of diagonal bosons:

$$\beta_q|\Omega_{\mathrm{LL}}\rangle = 0 \qquad (q \neq 0). \tag{5.23}$$

$$\beta_q = cb_q - sb_{-q}^\dagger. \tag{5.24}$$

Recall

$$\tanh(2\xi) = -\frac{B}{A} = \frac{K^2 - 1}{K^2 + 1}, \qquad |\tanh(2\xi)| < 1. \tag{5.25}$$

For comparison, a spinful s-wave BCS mean-field Hamiltonian has the quadratic form

$$H_{\mathrm{BCS}} = \sum_{k,\sigma}\xi_k c_{k\sigma}^\dagger c_{k\sigma} + \sum_k \left(\Delta\, c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger + \mathrm{h.c.}\right), \tag{5.26}$$

where ξ_k is measured from the chemical potential. This is a separate comparison model; the LL continues to be spinless. Within each paired block, BCS diagonalization gives

$$E_k(\gamma_{k\uparrow}^\dagger\gamma_{k\uparrow} + \gamma_{-k\downarrow}^\dagger\gamma_{-k\downarrow}) + \text{constant}, \qquad E_k = \sqrt{\xi_k^2 + |\Delta|^2}. \tag{5.27}$$

For a nonzero uniform s-wave gap, the fermionic quasiparticle spectrum has a hard gap, unlike $u|q|$ for the LL density modes. Our question is now concrete: how is $|\Omega_{\mathrm{LL}}\rangle$ built out of particle–hole excitations of $|0_{\mathrm{FS}}\rangle$?

$$0 = \beta_q|\Omega_{\mathrm{LL}}\rangle = \left(cb_q - sb_{-q}^\dagger\right)|\Omega_{\mathrm{LL}}\rangle. \tag{5.28}$$

Since $c = (K+1)/(2\sqrt{K})$ and $s = (K-1)/(2\sqrt{K})$, this condition is

$$(K+1)b_q|\Omega_{\mathrm{LL}}\rangle = (K-1)b_{-q}^\dagger|\Omega_{\mathrm{LL}}\rangle. \tag{5.29}$$

The ratio relevant to the state is therefore the single-angle quantity

$$\lambda \equiv \frac{s}{c} = \tanh\xi = \frac{K-1}{K+1}, \qquad |\lambda| < 1. \tag{5.30}$$

It is $\tanh(2\xi)$, rather than λ, that appears in the Hamiltonian's cancellation condition (5.14). Distinguishing the two is necessary before deriving the coefficients. Momentum conservation splits the problem into independent pairs $(q, -q)$, with $q > 0$. The Hamiltonian does mix the two members of a pair, but does not couple different absolute momenta. We can solve for each pair separately. In the original, noninteracting oscillator basis, use the trial state

$$|\Omega_q\rangle = \sum_{n_1,n_2=0}^{\infty} C_{n_1,n_2}|n_1, n_2\rangle, \tag{5.31}$$

where n_1 and n_2 count the q and $-q$ density oscillators, respectively.

$$b_q|n_1, n_2\rangle = \sqrt{n_1}|n_1 - 1, n_2\rangle, \tag{5.32}$$

$$b_{-q}^\dagger|n_1, n_2\rangle = \sqrt{n_2+1}|n_1, n_2+1\rangle. \tag{5.33}$$

Inserting the ansatz and shifting the summation indices gives

$$\begin{aligned} 0 &= \left[(K+1)b_q - (K-1)b_{-q}^\dagger\right]|\Omega_q\rangle \\ &= \sum_{n_1,n_2\geq 0}\Big[(K+1)\sqrt{n_1+1}\,C_{n_1+1,n_2} - (K-1)\sqrt{n_2}\,C_{n_1,n_2-1}\Big]|n_1, n_2\rangle. \end{aligned} \tag{5.34}$$

Orthonormality of the two-mode Fock basis requires each bracket to vanish. The companion condition $\beta_{-q}|\Omega_q\rangle = 0$ must also hold. Together they give

$$\sqrt{n_1+1}\,C_{n_1+1,n_2} = \lambda\sqrt{n_2}\,C_{n_1,n_2-1}, \tag{5.35}$$

$$\sqrt{n_2+1}\,C_{n_1,n_2+1} = \lambda\sqrt{n_1}\,C_{n_1-1,n_2}. \tag{5.36}$$

Coefficients with a negative index are zero. A single annihilation condition alone would leave part of the coefficient array unconstrained; both are needed for the unique normalized two-mode vacuum. Zero total momentum suggests $n_1 = n_2$. We can establish the stronger statement directly from the recurrences:

$$C_{n_1,n_2} = C_{n_1}\delta_{n_1,n_2}. \tag{5.37}$$

First set $n_2 = 0$ in (5.35). Then

$$\sqrt{n_1+1}\,C_{n_1+1,0} = 0 \quad\Longrightarrow\quad C_{m,0} = 0 \quad (m \geq 1). \tag{5.38}$$

Likewise, setting $n_1 = 0$ in (5.36) yields $C_{0,m} = 0$ for $m \geq 1$. At $n_2 = 1$,

$$(K+1)\sqrt{n_1+1}\, C_{n_1+1,1} = (K-1)C_{n_1,0}. \tag{5.39}$$

The right side vanishes unless $n_1 = 0$; together with $C_{0,1} = 0$, this leaves only $C_{1,1}$ potentially nonzero in the row $n_2 = 1$. Assume the row $n_2 = n-1$ has support only at $n_1 = n-1$. Then the row $n_2 = n$ satisfies

$$(K+1)\sqrt{n_1+1}\, C_{n_1+1,n} = (K-1)\sqrt{n}\, C_{n_1,n-1}. \tag{5.40}$$

The right side can be nonzero only at $n_1 = n-1$, so the next row has support only at $C_{n,n}$, again using $C_{0,n} = 0$. Induction proves diagonal support. At $K = 1$, even these diagonal coefficients vanish for $n > 0$.

5.3 Solving and normalizing the paired state

With $C_{n_1,n_2} = \delta_{n_1,n_2} C_{n_1}$, choose $n_2 = n$ and $n_1 + 1 = n$ in the recurrence.

$$(K+1)C_n\sqrt{n} = (K-1)C_{n-1}\sqrt{n}, \qquad n \geq 1. \tag{5.41}$$

When $C_{n-1} \neq 0$, the ratio is

$$\frac{C_n}{C_{n-1}} = \frac{K-1}{K+1} = \lambda. \tag{5.42}$$

Equivalently, $C_n = \lambda C_{n-1}$ remains valid also at $K = 1$, where $C_{n>0} = 0$ and higher ratios would be $0/0$. The solution is a geometric sequence,

$$C_n = C_0 \lambda^n, \qquad |\lambda| < 1. \tag{5.43}$$

Restoring the mode label for later use,

$$C_{n_q} = C_{0q} \lambda^{n_q}. \tag{5.44}$$

$$|\Omega_q\rangle = \sum_{n_q=0}^{\infty} C_{n_q} |n_q, n_q\rangle, \tag{5.45}$$

$$= \sum_{n=0}^{\infty} C_{0q} \lambda^n \frac{(b_q^\dagger)^n}{\sqrt{n!}} \frac{(b_{-q}^\dagger)^n}{\sqrt{n!}} |0,0\rangle_q, \tag{5.46}$$

$$= C_{0q} \exp\left(\lambda b_q^\dagger b_{-q}^\dagger\right) |0,0\rangle_q. \tag{5.47}$$

The two factors $1/\sqrt{n!}$ multiply to $1/n!$, exactly the coefficient in the exponential series. The ket $|0,0\rangle_q$ is the reference-sea vacuum of the two original oscillators; it is not the interacting vacuum when $\lambda \neq 0$. Normalization gives

$$\langle \Omega_q | \Omega_q \rangle = \sum_{n=0}^{\infty} |C_n|^2, \tag{5.48}$$

$$= |C_{0q}|^2 \sum_{n=0}^{\infty} |\lambda|^{2n}, \tag{5.49}$$

$$= \frac{|C_{0q}|^2}{1 - |\lambda|^2} = 1. \tag{5.50}$$

Choosing the unobservable overall phase so $C_{0q} > 0$,

$$C_{0q} = \sqrt{1-\lambda^2} = \frac{1}{\cosh\xi} = \frac{2\sqrt{K}}{K+1} \equiv C_0. \tag{5.51}$$

It is momentum independent for the present local model. The full oscillator ground state is

$$|\Omega_{\mathrm{LL}}\rangle = \prod_{0<q\le\Lambda} \left[\frac{1}{\cosh\xi} \exp\left(\lambda b_q^\dagger b_{-q}^\dagger\right)\right] |0_{\mathrm{FS}}\rangle, \tag{5.52}$$

within the fixed number and winding sector. The product runs over $q > 0$, not over both signs: otherwise the same pair would be counted twice. Each factor has unit norm at finite L and cutoff. If M positive modes are retained, the overlap with the reference sea is $(\cosh\xi)^{-M}$. It tends to zero as $M \to \infty$ for $K \neq 1$, even though every finite regulated state remains normalized. One must not replace this cutoff-dependent product by a finite universal constant. For one pair, the probability of n density excitations in each direction is

$$P_n = (1-\lambda^2)\lambda^{2n}, \qquad \langle b_q^\dagger b_q\rangle = \frac{\lambda^2}{1-\lambda^2} = \sinh^2\xi, \qquad \langle b_q b_{-q}\rangle = \frac{\lambda}{1-\lambda^2} = sc. \tag{5.53}$$

The mean occupation follows from differentiating $\sum_{n\ge0} z^n = (1-z)^{-1}$ with respect to $z = \lambda^2$. The cross expectation follows by applying $b_q b_{-q}|n,n\rangle = n|n-1,n-1\rangle$ and summing $n\lambda^{2n-1}$. Figure 5.2 makes the pair content of the state explicit.

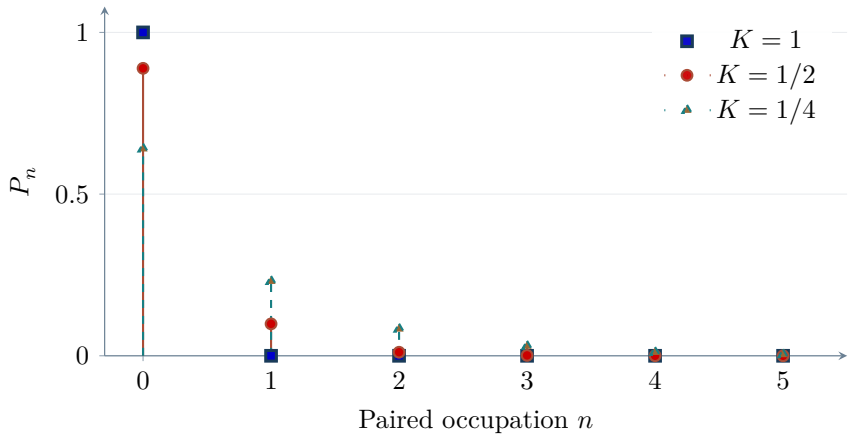


Figure 5.2: Occupation probabilities for a single $(q,-q)$ density-oscillator pair, calculated from (5.53). The shown choices are $K = 1$, $K = 1/2$, and $K = 1/4$; no experimental data are used. At $K = 1$ only $|0,0\rangle_q$ occurs. Increasing $|\xi|$ broadens the geometric distribution, while the two original mode occupations remain equal in each term of the ground-state superposition.

5.4 Physical content of the interacting ground state

The paired state can be written again as

$$|\Omega_{\mathrm{LL}}\rangle = \prod_{q>0} C_0 \exp(\lambda b_q^\dagger b_{-q}^\dagger)|0_{\mathrm{FS}}\rangle, \qquad \lambda = \frac{K-1}{K+1}, \tag{5.54}$$

with the finite cutoff understood. The reference ket is the filled noninteracting Fermi sea in the specified sector. The LL ground state contains paired bosonic density excitations relative to this reference sea. For nonzero g_2, every retained momentum pair is mixed in the local model. The g_4-only interaction is an exception: it changes the velocity without squeezing the ground state.

The many-pair structure signals the failure of a simple finite-overlap single-particle picture; the electron correlator below will make the absence of a Landau quasiparticle pole precise. For $K \neq 1$, the elementary eigenmodes are collective density excitations, rather than single-fermion quasiparticles. An injected physical fermion is described by a vertex operator that excites these collective sectors; its subsequent propagation is not that of a single long-lived electron pole. At $K = 1$, refermionization recovers the free-form propagator with velocity u. Fermion number is conserved because each density oscillator is built from a particle–hole bilinear:

$$[\hat{N}_r, b_q^\dagger] = [\hat{N}_r, b_q] = 0. \tag{5.55}$$

The number of original particle–hole density oscillators is not conserved when $B \neq 0$: the terms $b_q^\dagger b_{-q}^\dagger$ and $b_q b_{-q}$ change it by two. Measuring energy from the interacting ground state in a fixed sector,

$$H - E_0 = u \sum_{q \neq 0} |q| \beta_q^\dagger \beta_q. \tag{5.56}$$

$$\beta_q^\dagger = c b_q^\dagger - s b_{-q}, \qquad \beta_q |\Omega_{\mathrm{LL}}\rangle = 0. \tag{5.57}$$

A normalized basis of oscillator excited states is

$$|\{m_q\}\rangle_\beta = \prod_{q \neq 0} \frac{(\beta_q^\dagger)^{m_q}}{\sqrt{m_q!}} |\Omega_{\mathrm{LL}}\rangle, \qquad E - E_0 = u \sum_{q \neq 0} |q| m_q. \tag{5.58}$$

General states are linear combinations of these basis states. A physical electron creation operator also changes the number sector, through its Klein/zero-mode factor. It cannot be represented by $\beta_q^\dagger$ alone. The collective spectrum is gapless: the smallest oscillator energy is $2\pi u/L$ on a ring and tends to zero as $L \to \infty$. The ground state is therefore not separated from all excitations by a nonzero bulk spectral gap. This observation is a statement about spectral isolation, not a general theorem about every type of ground-state protection. An excitation can create an imbalance between opposite-momentum modes. More generally, equal excitations of both diagonal modes also cost energy; imbalance is one possible excitation pattern, not a necessary condition for excitation. Figure 5.3 separates the reference vacuum, its interacting paired superposition, and the diagonal excitations. The difference between original and diagonal occupations is the reason a state containing many b bosons can nevertheless be the β vacuum.

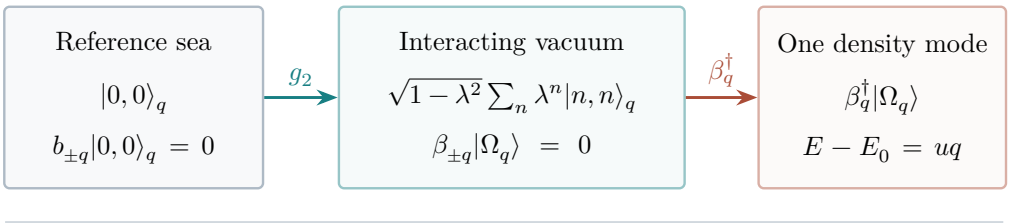


Figure 5.3: States for one independent momentum pair (schematic). The filled-sea oscillator vacuum is $|0,0\rangle_q$. Interbranch coupling produces the normalized superposition $\sqrt{1-\lambda^2}\sum_n \lambda^n |n,n\rangle_q$, which has no diagonal bosons. Acting with $\beta_q^\dagger$ creates a collective excitation of energy uq while preserving the total electron number. A physical electron insertion additionally changes a zero-mode number sector.

5.4.1 The useful and limited BCS analogy

For each bosonic pair, the exponential contains arbitrarily high powers,

$$\exp(\lambda b_q^\dagger b_{-q}^\dagger) = 1 + \lambda b_q^\dagger b_{-q}^\dagger + \frac{\lambda^2}{2!}(b_q^\dagger b_{-q}^\dagger)^2 + \cdots. \tag{5.59}$$

By contrast, the spinful BCS variational state is

$$|\mathrm{BCS}\rangle = \prod_k (U_k + V_k c^\dagger_{k\uparrow} c^\dagger_{-k\downarrow})|\mathrm{vac}\rangle, \qquad |U_k|^2 + |V_k|^2 = 1, \tag{5.60}$$

where each distinct paired orbital is included once. The notation U_k, V_k keeps these coherence factors distinct from the LL velocity u. Higher powers within one pair vanish by Pauli exclusion: $(c^\dagger_{k\uparrow} c^\dagger_{-k\downarrow})^2 = 0$. The LL pair consists of two particle–hole density modes; a BCS Cooper pair consists of two electrons in the paired spin and momentum orbitals. The LL excitations are gapless collective bosons, with $E_q = u|q|$ and possible opposite-direction occupation imbalance. The uniform s-wave BCS comparison has gapped fermionic quasiparticles, $E_k = \sqrt{\xi_k^2 + |\Delta|^2}$. The LL Hamiltonian and its squeezed state preserve the fixed electron-number sector. The displayed BCS mean-field Hamiltonian and variational state do not have definite particle number. The underlying number-conserving microscopic superconducting model does conserve it; the distinction concerns the mean-field description. In the Gaussian LL, the forward interaction sets u, K, and the squeezing. In BCS mean-field theory, a pairing channel produces the gap parameter Δ. The zero-temperature long-distance LL electron correlators decay as powers; their derivation follows next. A gapped BCS single-particle correlator is exponentially suppressed in long imaginary time, with a power prefactor possible. This does not mean that every real-time correlator or every superconducting correlation function decays exponentially.

Table 5.1 collects these distinctions.

Table 5.1: Scope of the LL–BCS analogy. The BCS column refers to a uniform spinful s-wave mean-field model.

Feature	Spinless Luttinger liquid	BCS comparison
Paired objects	Neutral density oscillators	Electron Cooper pairs
Pair occupation	Any nonnegative integer	Zero or one in a paired orbital
Elementary modes	Bosons, $E_q = u\|q\|$	Fermions, $E_k = \sqrt{\xi_k^2 + \|\Delta\|^2}$
Number sector	Fixed	Superposed in the mean-field state
Interaction effect	Stiffness and power exponents	Pairing gap

Figure 5.4 illustrates the spectral distinction behind this comparison.

6 Correlations, temperature, and propagating modes

Transport probes electron correlation functions. We first keep their operator order and cutoff phase, evaluate the Gaussian contractions in the interacting state, and then distinguish physical branch phases from the modes that propagate at the collective velocity.

6.1 Why transport requires electron correlation functions

We now study quantum transport in Luttinger liquids, with a single local barrier as the eventual example. Electrons tunnel between leads, whereas collective density waves carry the low-energy dynamics within each lead. Figure 6.1 distinguishes these processes. The calculation below keeps the leading nonzero tunnelling contribution. Linear response will be taken only after the bias-dependent current has been obtained. Consider two initially disconnected leads, labelled $\ell = 1, 2$; these labels do not denote chirality. Each contact electron d_ℓ has dimensions length$^{-1/2}$. Operators act on the fermionic product of the two lead spaces, with a fixed global fermion

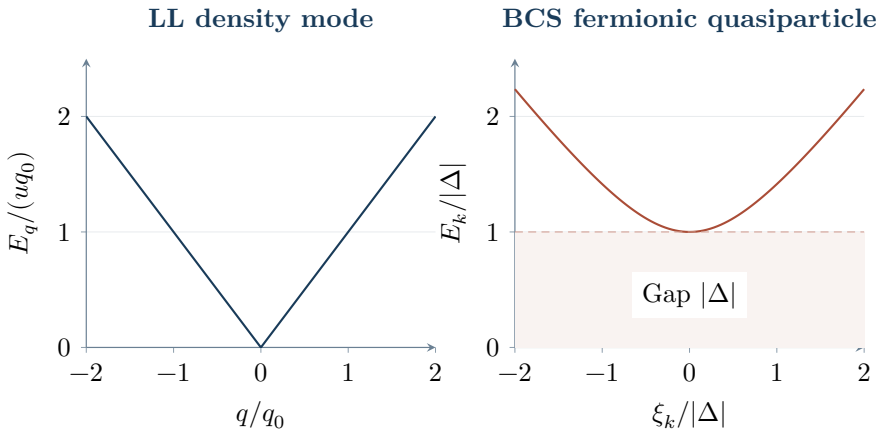


Figure 5.4: Spectral comparison calculated from the displayed model formulas. Left: the LL density mode has $E_q/(uq_0) = |q/q_0|$, where $q_0 > 0$ is an arbitrary momentum unit within the linear regime. Right: a uniform s-wave BCS fermionic quasiparticle has $E_k/|\Delta| = \sqrt{1 + (\xi_k/|\Delta|)^2}$. The horizontal axes represent different quantities. Pair mixing does not by itself imply a gap; the LL coefficients vanish with $|q|$, whereas the BCS gap stays finite at $\xi_k = 0$.

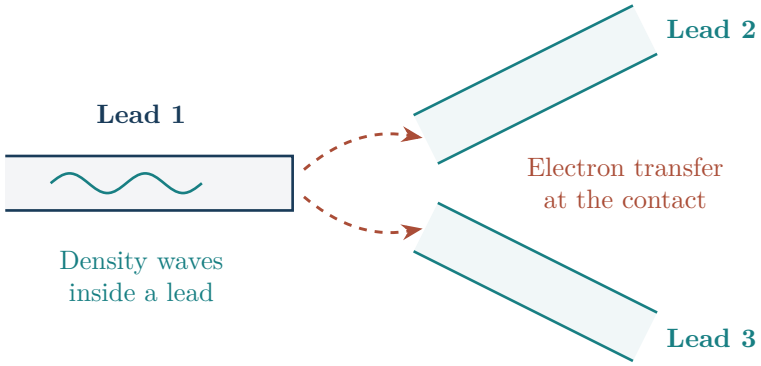


Figure 6.1: Schematic multi-terminal contact from the source. A particular tunnelling amplitude couples a chosen pair of leads; the subsequent formulas use leads 1 and 2. Electron transfer changes lead number by one, while the disturbance inside a lead is described by bosonic density modes.

ordering. Let $H_0 = H_1 + H_2$ and initially $\rho_0 \propto \exp[-\beta \sum_\ell (H_\ell - \mu_\ell N_\ell)]$. All operators in this subsection evolve with H_0, and tunnelling is switched on adiabatically from the remote past.

$$H_T = \gamma A + \gamma^* A^\dagger, \qquad A = d_1^\dagger d_2, \qquad \gamma_{21} = \gamma_{12}^*. \tag{6.1}$$

Here γ has dimensions of velocity. A sum over ordered lead pairs is equivalent, provided the Hermitian conjugate is not counted twice.

$$I \equiv e\dot{N}_2 = -e\dot{N}_1 = ie(\gamma A - \gamma^* A^\dagger). \tag{6.2}$$

Indeed $[N_1, A] = A$ and $[N_2, A] = -A$, so $\dot{N}_1 = i[H_T, N_1] = -i\gamma A + i\gamma^* A^\dagger$. Thus $I > 0$ denotes electron-number transfer from lead 1 to lead 2, multiplied by $e > 0$; equivalently it is electric charge entering lead 1. Restoring units inserts $1/\hbar$ into these rates. The mean current requires the interlead coherence $\langle d_1^\dagger(t) d_2(t)\rangle$, which vanishes in the disconnected state and begins at first order in γ.

$$\langle I(t)\rangle = \left\langle T_C I(t^+) \exp\left[-i \int_C dt' H_T(t')\right]\right\rangle_0 . \tag{6.3}$$

The contour C runs from $-\infty$ to $+\infty$ on $\kappa = +1$ and returns on $\kappa = -1$. Its integral is $\int_C dt' = \sum_{\kappa=\pm1} \kappa \int_{-\infty}^{\infty} dt'$. The average uses the uncoupled Hamiltonian, including each lead's interactions. The contour is shown in Fig. 6.2.

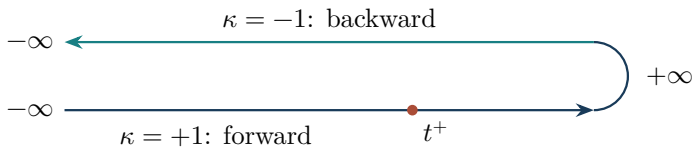


Figure 6.2: The closed Keldysh contour (schematic). Its backward branch carries the orientation sign $\kappa = -1$ in $\int_C dt = \sum_\kappa \kappa \int dt$. The two branches implement causal cancellation of perturbations later than the observation time.

$$\langle I(t)\rangle^{(2)} = e\gamma \int_C dt' \langle T_C A(t^+) H_T(t')\rangle_0 - e\gamma^* \int_C dt' \langle T_C A^\dagger(t^+) H_T(t')\rangle_0. \tag{6.4}$$

$$\langle I(t)\rangle^{(2)} = e\gamma \sum_\kappa \kappa \int dt' \left\langle T_C A(t^+) \left[\gamma A(t'^\kappa) + \gamma^* A^\dagger(t'^\kappa)\right]\right\rangle_0 \tag{6.5}$$

$$- e\gamma^* \sum_\kappa \kappa \int dt' \left\langle T_C A^\dagger(t^+) \left[\gamma A(t'^\kappa) + \gamma^* A^\dagger(t'^\kappa)\right]\right\rangle_0 . \tag{6.6}$$

Charge selection removes AA and $A^\dagger A^\dagger$ averages. Combining the two contour branches removes insertions later than the observation time. Cross-branch order is fixed by the contour; same-branch order depends on $t - t'$. This cancellation is a consequence of the closed contour and causality, rather than translation invariance alone.

$$\langle I(t)\rangle^{(2)} = -i \int_{-\infty}^{t} dt' \, \langle [I(t), H_T(t')]\rangle_0. \tag{6.7}$$

For stationary independent leads, the two transfer rates can therefore be expressed as full Fourier integrals:

$$R_{1\to 2} = |\gamma|^2 \int_{-\infty}^{\infty} d\tau \, \langle d_1^\dagger(\tau) d_1(0)\rangle_0 \langle d_2(\tau) d_2^\dagger(0)\rangle_0, \tag{6.8}$$

$$R_{2\to 1} = |\gamma|^2 \int_{-\infty}^{\infty} d\tau \, \langle d_1(\tau) d_1^\dagger(0)\rangle_0 \langle d_2^\dagger(\tau) d_2(0)\rangle_0, \qquad \langle I\rangle^{(2)} = e(R_{1\to 2} - R_{2\to 1}). \tag{6.9}$$

In the usual Green-function notation, $G_\ell^>(t) = -i\langle d_\ell(t) d_\ell^\dagger(0)\rangle_0$ and $G_\ell^<(t) = i\langle d_\ell^\dagger(0) d_\ell(t)\rangle_0$. The ordered correlators in (6.8) carry no extra i. Their complex phases are essential. The current is quadratic in γ because its operator is linear in γ and the contour exponential contributes one more power.

6.2 Gaussian reduction of a fermionic vertex correlator

The immediate target is the local ordered electron correlator in each participating channel, and ultimately its time integral.

$$\int dt \, \langle d_\ell^\dagger(t) d_\ell(0)\rangle_0 = \sum_{r=\pm 1} \int dt \, \langle \psi_{\ell r}^\dagger(t) \psi_{\ell r}(0)\rangle_0 \tag{6.10}$$

when the contact field is the sum of two independent physical branches. Interbranch anticommutation is necessary for fermion statistics. Vanishing cross averages additionally requires a state diagonal in the separate branch-number sectors. A reflecting boundary relates the branches and must be treated separately.

$$\langle \psi_R^\dagger(x,t) \psi_R(0,0)\rangle = \mathcal{A}_a \left\langle e^{i\varphi_R(x,t)} e^{-i\varphi_R(0,0)}\right\rangle , \qquad \varphi_R = \phi - \theta. \tag{6.11}$$

We need bosonic correlation functions with the correct boundary conditions for a weak or strong bond. The source attributes the Gaussian method below to Nazarov. The exponentials here mean cutoff-regulated Gaussian fields with their self-contraction matching included in $\mathcal{A}_a$. In a fixed short-distance matching convention one may choose $\mathcal{A}_a = 1/(2\pi a)$; re-normal-ordering in a different vacuum changes that amplitude. We work in the interacting oscillator ground state, with a fixed zero-mode sector, and take the thermodynamic limit before discarding its finite-size phases. An ordered Gaussian vertex average reduces to a two-point function. Operator ordering must be retained; an unsigned real variance alone does not determine the phase. Set $X = \varphi_R(x,t)$, $Y = \varphi_R(0,0)$, and define

$$f(\alpha) = \langle e^{i\alpha X} e^{-i\alpha Y}\rangle, \qquad f(0) = 1, \qquad \langle X\rangle = \langle Y\rangle = 0. \tag{6.12}$$

Translation invariance forces the one-point expectation of a nonzero-momentum oscillator to vanish. The spatially uniform and winding pieces are fixed separately; this is the zero-mean condition used here.

$$\frac{df}{d\alpha} = i\left(\langle X e^{i\alpha X} e^{-i\alpha Y}\rangle - \langle e^{i\alpha X} Y e^{-i\alpha Y}\rangle\right). \tag{6.13}$$

$$\langle X e^{i\alpha X} e^{-i\alpha Y}\rangle = \sum_{m,n\geq 0} \frac{i^m (-i)^n \alpha^{m+n}}{m!n!} \langle X X^m Y^n\rangle \tag{6.14}$$

$$= \sum_{m,n\geq 0} \frac{i^m (-i)^n \alpha^{m+n}}{m!n!} m\langle X^2\rangle\langle X^{m-1} Y^n\rangle \tag{6.15}$$

$$+ \sum_{m,n\geq 0} \frac{i^m (-i)^n \alpha^{m+n}}{m!n!} n\langle XY\rangle\langle X^m Y^{n-1}\rangle \tag{6.16}$$

$$= i\alpha\langle X^2\rangle f - i\alpha\langle XY\rangle f = i\alpha\langle X(X-Y)\rangle f. \tag{6.17}$$

Wick's theorem applies because X, Y are linear combinations of independent harmonic oscillators in a Gaussian state. The factor m counts possible contractions with the m copies of X, and n counts contractions with Y. Terms with a negative residual power are absent. Shifting the summation index in the second term produces $-i\alpha$, rather than $+i\alpha$.

$$\langle e^{i\alpha X} Y e^{-i\alpha Y}\rangle = \sum_{m,n\geq 0} \frac{i^m (-i)^n \alpha^{m+n}}{m!n!} m\langle XY\rangle\langle X^{m-1} Y^n\rangle \tag{6.18}$$

$$+ \sum_{m,n\geq 0} \frac{i^m (-i)^n \alpha^{m+n}}{m!n!} n\langle Y^2\rangle\langle X^m Y^{n-1}\rangle \tag{6.19}$$

$$= i\alpha\langle (X-Y)Y\rangle f. \tag{6.20}$$

Stationarity and spatial homogeneity give $\langle X^2\rangle = \langle Y^2\rangle$ with the same regulator.

$$\frac{df}{d\alpha} = 2\alpha\langle (X-Y)Y\rangle\langle e^{i\alpha X} e^{-i\alpha Y}\rangle = 2\alpha\langle (X-Y)Y\rangle f. \tag{6.21}$$

$$\frac{df}{f} = 2\alpha\langle (X-Y)Y\rangle\, d\alpha. \tag{6.22}$$

$$\ln f(\alpha)\big|_0^1 = \alpha^2\langle (X-Y)Y\rangle\big|_0^1. \tag{6.23}$$

$$\langle e^{iX} e^{-iY}\rangle = \exp\{\langle XY\rangle - \langle Y^2\rangle\}. \tag{6.24}$$

The problem has reduced to ordered two-point functions of the bosonic fields. The Baker–Campbell–Hausdorff formula provides an independent check. Since $[X,Y]$ is a scalar,

$$e^{iX} e^{-iY} = e^{i(X-Y)+[X,Y]/2}, \tag{6.25}$$

$$\langle e^{iX} e^{-iY}\rangle = e^{-\langle (X-Y)^2\rangle/2 + [X,Y]/2} = e^{\langle XY\rangle - \langle Y^2\rangle}. \tag{6.26}$$

Thus the commutator restores precisely the imaginary part lost by replacing the ordered answer with a real variance.

6.2.1 A direct oscillator route for the right branch

One can compute the right-branch phase directly before resolving the canonical fields separately. Keep the winding term on a ring and set the fixed sector $N_R = 0$ only when evaluating the oscillator correlator below.

$$\varphi_R(x) = -\frac{2\pi N_R x}{L} - \frac{2\pi i}{L}\sum_{p>0}\sqrt{\frac{L}{2\pi p}}e^{-ap/2-ipx}b_p^\dagger \tag{6.27}$$

$$+\frac{2\pi i}{L}\sum_{p<0}\sqrt{\frac{L}{2\pi|p|}}e^{-a|p|/2-ipx}b_{-p} \tag{6.28}$$

$$= -\frac{2\pi N_R x}{L} - i\sum_{p>0} f_p\left(e^{-ipx}b_p^\dagger - e^{ipx}b_p\right), \qquad f_p = \sqrt{\frac{2\pi}{Lp}}e^{-ap/2}. \tag{6.29}$$

$$\varphi_R(x) = -\frac{2\pi N_R x}{L} - i\sum_{p>0} f_p\left\{e^{-ipx}(s\beta_{-p} + c\beta_p^\dagger) - e^{ipx}(c\beta_p + s\beta_{-p}^\dagger)\right\}, \tag{6.30}$$

where $c = \cosh\xi$, $s = \sinh\xi$, and $\xi = \frac{1}{2}\ln K$. The common prefactor in a direct two-field contraction is

$$(-i)^2 f_p f_q = -\frac{4\pi^2}{L^2}\frac{L}{2\pi\sqrt{pq}}e^{-a(p+q)/2}. \tag{6.31}$$

$$\beta_p|\Omega_{\rm LL}\rangle = 0, \qquad \langle\beta_p\beta_q^\dagger\rangle = \delta_{pq}, \qquad \langle\beta_p^\dagger\beta_q\rangle = 0. \tag{6.32}$$

$$\varphi_R(x,t) = -i\sum_{p>0} f_p\Big\{e^{-ipx}\left(s\beta_{-p}e^{-iupt} + c\beta_p^\dagger e^{iupt}\right) \tag{6.33}$$

$$- e^{ipx}\left(c\beta_p e^{-iupt} + s\beta_{-p}^\dagger e^{iupt}\right)\Big\}. \tag{6.34}$$

$$\langle\varphi_R(x,t)\varphi_R(0,0)\rangle = -\sum_{p,q>0} f_p f_q\Big\langle[e^{-ipx}(s\beta_{-p}e^{-iupt} + c\beta_p^\dagger e^{iupt}) \tag{6.35}$$

$$- e^{ipx}(c\beta_p e^{-iupt} + s\beta_{-p}^\dagger e^{iupt})]\left[s\beta_{-q} + c\beta_q^\dagger - c\beta_q - s\beta_{-q}^\dagger\right]\Big\rangle \tag{6.36}$$

$$= -\frac{2\pi}{L}\sum_{p>0}\frac{e^{-ap}}{p}\left[-c^2e^{ipx-iupt} - s^2e^{-ipx-iupt}\right]. \tag{6.37}$$

Both eigenmode sectors appear unless $s = 0$, or $K = 1$. This direct calculation will agree with the separate ϕ, θ, and mixed contractions below; retaining both routes makes the origin of the two light-cone factors explicit.

6.3 Canonical-field correlation functions at zero temperature

Because ϕ and θ do not commute, reconstructing a fermion correlator requires $\phi\phi$, $\theta\theta$, and the mixed contractions. We first compute the density-field correlator. Throughout this calculation $p = 2\pi n/L$, $n \neq 0$, and the zero-mode sector is fixed.

$$\phi(x) = -\frac{\pi(N_R + N_L)x}{L} - \frac{i\pi}{L}\sum_{p\neq 0}\sqrt{\frac{L|p|}{2\pi}}\frac{e^{-a|p|/2-ipx}}{p}(b_p^\dagger + b_{-p}). \tag{6.38}$$

$$b_p^\dagger = s\beta_{-p} + c\beta_p^\dagger, \qquad b_{-p} = c\beta_{-p} + s\beta_p^\dagger. \tag{6.39}$$

$$\phi(x) = -\frac{\pi(N_R+N_L)x}{L} - \frac{i\pi(c+s)}{L}\sum_{p\neq 0}\sqrt{\frac{L|p|}{2\pi}}\frac{e^{-a|p|/2-ipx}}{p}(\beta_{-p}+\beta_p^\dagger). \tag{6.40}$$

$$\phi(x,t) = -\frac{i\pi(c+s)}{L}\sum_{p\neq 0}\sqrt{\frac{L|p|}{2\pi}}\frac{e^{-a|p|/2-ipx}}{p}(\beta_{-p}e^{-iu|p|t}+\beta_p^\dagger e^{iu|p|t}). \tag{6.41}$$

We have set the fixed excess-number sector to $N_R+N_L=0$. Oscillator–zero-mode cross averages vanish, and the remaining fixed-sector zero-mode phase is treated separately before the infinite-system limit. The diagonal oscillators annihilate $|\Omega_{\rm LL}\rangle$.

$$\langle\phi(x,t)\phi(0,0)\rangle = -\left(\frac{\pi}{L}\right)^2\frac{L}{2\pi}(c+s)^2\sum_{p,q\neq 0}\frac{\sqrt{|pq|}}{pq}e^{-a(|p|+|q|)/2-ipx} \tag{6.42}$$

$$\times\left\langle(\beta_{-p}e^{-iu|p|t}+\beta_p^\dagger e^{iu|p|t})(\beta_{-q}+\beta_q^\dagger)\right\rangle \tag{6.43}$$

$$= -\frac{\pi(c+s)^2}{2L}\sum_{p,q\neq 0}\frac{\sqrt{|pq|}}{pq}e^{-a(|p|+|q|)/2-ipx-iu|p|t}\delta_{q,-p} \tag{6.44}$$

$$= \frac{\pi(c+s)^2}{2L}\sum_{p\neq 0}\frac{e^{-a|p|-ipx-iu|p|t}}{|p|}. \tag{6.45}$$

$$(s+c)^2 = \frac{e^{2\xi}+e^{-2\xi}-2}{4}+\frac{e^{2\xi}+e^{-2\xi}+2}{4}+\frac{e^{2\xi}-e^{-2\xi}}{2} \tag{6.46}$$

$$= e^{2\xi} = \sinh(2\xi)+\cosh(2\xi) = \frac{A-B}{\sqrt{A^2-B^2}} = K. \tag{6.47}$$

$$A = \frac{u}{2}(K^{-1}+K),\quad B=\frac{u}{2}(K^{-1}-K),\quad \sqrt{A^2-B^2}=u,\quad A-B=uK,\quad \tanh(2\xi)=-\frac{B}{A}. \tag{6.48}$$

$$\langle\phi(x,t)\phi(0,0)\rangle = \frac{\pi K}{2L}\sum_{p\neq 0}\frac{e^{-a|p|-ipx-iu|p|t}}{|p|} \tag{6.49}$$

$$= \frac{\pi K}{2L}\sum_{p>0}\frac{e^{-ap-ipx-iupt}}{p} - \frac{\pi K}{2L}\sum_{p<0}\frac{e^{ap-ipx+iupt}}{p} \tag{6.50}$$

$$= \frac{\pi K}{2L}\sum_{p>0}\frac{e^{-ap-iupt}}{p}(e^{-ipx}+e^{ipx}) = Kg_L(x+ut)+Kg_R(x-ut). \tag{6.51}$$

The separate g_r contain an infrared constant. Only their subtracted differences enter neutral vertex correlators. Define

$$g_L(\ell) = \frac{\pi}{2L}\sum_{p>0}\frac{e^{-ap-ip\ell}}{p},\qquad g_R(\ell') = \frac{\pi}{2L}\sum_{p>0}\frac{e^{-ap+ip\ell'}}{p}. \tag{6.52}$$

$$\frac{d}{d\ell}Kg_L(\ell) = -\frac{i\pi K}{2L}\sum_{p>0}e^{-ap-ip\ell} \tag{6.53}$$

$$\longrightarrow -\frac{iK}{4}\int_0^\infty dp\, e^{-(a+i\ell)p} = -\frac{iK}{4}\left[\frac{e^{-(a+i\ell)p}}{-a-i\ell}\right]_0^\infty = -\frac{K}{4(\ell-ia)}. \tag{6.54}$$

$$\frac{d}{d\ell'}Kg_R(\ell') = \frac{i\pi K}{2L}\sum_{p>0}e^{-ap+ip\ell'} \tag{6.55}$$

$$\longrightarrow \frac{iK}{4}\int_0^\infty dp\, e^{-(a-i\ell')p} = \frac{iK}{4}\left[\frac{e^{-(a-i\ell')p}}{-a+i\ell'}\right]_0^\infty = -\frac{K}{4(\ell'+ia)}. \tag{6.56}$$

Integrate the two derivatives from 0 to $\ell = x+ut$ and from 0 to $\ell' = x-ut$. Define the ordered subtracted correlator $D_\phi(x,t) = 2\langle\phi(x,t)\phi(0,0)\rangle - 2\langle\phi(0,0)^2\rangle$.

$$D_\phi(x,t) = 2\left[-\frac{K}{4}\int_0^{x+ut}\frac{d\ell}{\ell - ia} - \frac{K}{4}\int_0^{x-ut}\frac{d\ell'}{\ell' + ia}\right] \tag{6.57}$$

$$= -\frac{K}{2}\left[\ln\frac{x+ut-ia}{-ia} + \ln\frac{x-ut+ia}{ia}\right] = -\frac{K}{2}\left[\ln\frac{a+i(ut+x)}{a} + \ln\frac{a+i(ut-x)}{a}\right]. \tag{6.58}$$

Each logarithm is continued from $(x,t) = (0,0)$ with $a > 0$. The product of their arguments is $[x^2-(ut-ia)^2]/a^2$, but replacing their sum by a principal logarithm of the product can lose $2\pi i$ across a cut. We therefore keep the two logarithms separately. Both chiralities contribute. The covariance is proportional to K, the inverse of the density-field stiffness in $H_{\rm LL}$. For Hermitian fields the actual squared-difference expectation is instead

$$\langle[\phi(x,t) - \phi(0,0)]^2\rangle = -\operatorname{Re} D_\phi(x,t), \tag{6.59}$$

while D_ϕ retains the phase needed for ordered vertices. For the dual field,

$$\theta(x) = \frac{\pi(N_R - N_L)x}{L} + \frac{i\pi}{L}\sum_{p\neq 0}\sqrt{\frac{L|p|}{2\pi}}\frac{e^{-a|p|/2-ipx}}{|p|}(b_p^\dagger - b_{-p}). \tag{6.60}$$

$$\theta(x) = \frac{\pi(N_R - N_L)x}{L} + \frac{i\pi(s-c)}{L}\sum_{p\neq 0}\sqrt{\frac{L|p|}{2\pi}}\frac{e^{-a|p|/2-ipx}}{|p|}(\beta_{-p} - \beta_p^\dagger). \tag{6.61}$$

$$\theta(x,t) = \frac{i\pi(s-c)}{L}\sum_{p\neq 0}\sqrt{\frac{L|p|}{2\pi}}\frac{e^{-a|p|/2-ipx}}{|p|}(\beta_{-p}e^{-iu|p|t} - \beta_p^\dagger e^{iu|p|t}) \tag{6.62}$$

in the $N_R - N_L = 0$ sector.

$$\langle\theta(x,t)\theta(0,0)\rangle = -\frac{\pi(s-c)^2}{2L}\sum_{p,q\neq 0}\frac{e^{-a(|p|+|q|)/2-ipx}}{\sqrt{|pq|}} \tag{6.63}$$

$$\times\left\langle(\beta_{-p}e^{-iu|p|t} - \beta_p^\dagger e^{iu|p|t})(\beta_{-q} - \beta_q^\dagger)\right\rangle. \tag{6.64}$$

$$(s-c)^2 = e^{-2\xi} = \cosh(2\xi) - \sinh(2\xi) = \frac{A+B}{\sqrt{A^2-B^2}} = \frac{u/K}{u} = \frac{1}{K}. \tag{6.65}$$

$$\langle\theta(x,t)\theta(0,0)\rangle = \frac{\pi}{2LK}\sum_{p\neq 0}\frac{e^{-a|p|-ipx-iu|p|t}}{|p|} \tag{6.66}$$

$$= \frac{\pi}{2LK}\left[\sum_{p>0}\frac{e^{-ap-ipx-iupt}}{p} - \sum_{p<0}\frac{e^{ap-ipx+iupt}}{p}\right] \tag{6.67}$$

$$= \frac{\pi}{2LK}\sum_{p>0}\frac{e^{-ap-iupt}}{p}(e^{-ipx} + e^{ipx}) = \frac{1}{K}[g_L(x+ut) + g_R(x-ut)]. \tag{6.68}$$

$$D_\theta(x,t) = -\frac{1}{2K}\left[\ln\frac{a+i(ut+x)}{a} + \ln\frac{a+i(ut-x)}{a}\right]. \tag{6.69}$$

To complete the required mixed contractions, use $\phi = \sqrt{K}(\chi_R+\chi_L)/2$ and $\theta = (\chi_L-\chi_R)/(2\sqrt{K})$. Independence of the eigenmodes gives

$$\langle\phi(x,t)\theta(0,0)\rangle = \langle\theta(x,t)\phi(0,0)\rangle = g_L(x+ut) - g_R(x-ut), \tag{6.70}$$

$$\langle\varphi_R(x,t)\varphi_R(0,0)\rangle = (K+K^{-1}+2)g_R(x-ut) + (K+K^{-1}-2)g_L(x+ut). \tag{6.71}$$

The ordered mixed function is generally complex. Reversing the two spacetime arguments is a different correlator.

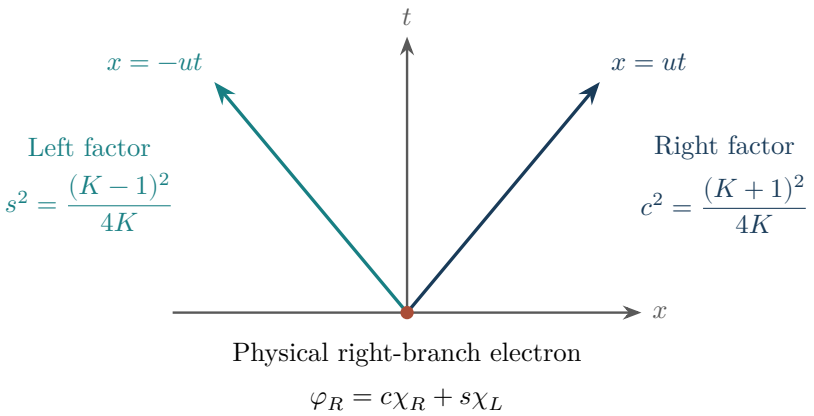


Figure 6.3: Schematic propagation of the two factors in the physical right-branch correlator. The Hamiltonian eigenmodes move at $\pm u$. Their weights in the electron's phase give the exponents c^2 and s^2; the second factor vanishes at $K = 1$. These are correlation exponents, not particle probabilities.

6.4 The bulk electron has two light-cone factors

The phase $\varphi_R = \phi - \theta$ belongs to the physical right Fermi branch. It is purely right-propagating only when $K = 1$.

$$D_{\varphi_R}(x,t) = -2\left[\frac{(K-1)^2}{4K}\int_0^{x+ut}\frac{d\ell}{\ell - ia} + \frac{(K+1)^2}{4K}\int_0^{x-ut}\frac{d\ell}{\ell + ia}\right] \tag{6.72}$$

$$= -\frac{(K-1)^2}{2K}\ln\frac{x+ut-ia}{-ia} - \frac{(K+1)^2}{2K}\ln\frac{x-ut+ia}{ia}. \tag{6.73}$$

$$\langle\psi_R^\dagger(x,t)\psi_R(0,0)\rangle = \mathcal{A}_a \exp\{D_{\varphi_R}(x,t)/2\}. \tag{6.74}$$

$$\langle\psi_R^\dagger(x,t)\psi_R(0,0)\rangle = \mathcal{A}_a\left[\frac{a}{a+i(ut-x)}\right]^{(K+1)^2/(4K)}\left[\frac{a}{a+i(ut+x)}\right]^{(K-1)^2/(4K)}. \tag{6.75}$$

As $a \to 0^+$, the two light cones lie at $t = x/u$ and $t = -x/u$. For $x > 0$, the latter occurs earlier. The finite-temperature continuation is developed next; the finite cutoff rounds both singularities.

$$\frac{(K+1)^2}{4K} + \frac{(K-1)^2}{4K} = \frac{K^2+1}{2K} = \frac{1}{2}(K + K^{-1}) \equiv p_{\text{bulk}}. \tag{6.76}$$

$$p_{\text{bulk}} = \frac{1}{2}(K + K^{-1}) \geq \sqrt{KK^{-1}} = 1. \tag{6.77}$$

The local electron correlator decays faster than the $K = 1$ result when $K \neq 1$. A g_4-only interaction can change u while leaving $K = 1$, so an interaction need not change this exponent. The two chiral weights are generally noninteger, and their sum is the local time-decay exponent. The bulk electron scaling dimension is $\Delta_{\psi,\text{bulk}} = p_{\text{bulk}}/2 = (K+K^{-1})/4$. At $K = 1$ the left factor disappears and the right factor has unit power. Figure 6.3 connects the factors to propagation.

6.4.1 Independent propagating fields

Define the unit-normalized eigenfields

$$\chi_R = \frac{\phi}{\sqrt{K}} - \sqrt{K}\theta, \qquad \chi_L = \frac{\phi}{\sqrt{K}} + \sqrt{K}\theta. \tag{6.78}$$

They are independent bosonic propagation modes; their Gaussian Hamiltonian does not make the physical electrons noninteracting.

$$\theta = \frac{\chi_L - \chi_R}{2\sqrt{K}}, \qquad \phi = \frac{\sqrt{K}}{2}(\chi_L + \chi_R). \tag{6.79}$$

$$H_{\mathrm{LL}} = \frac{u}{2\pi}\int dx\,[K(\partial_x\theta)^2 + K^{-1}(\partial_x\phi)^2] \tag{6.80}$$

$$= \frac{u}{2\pi}\int dx\left\{\frac{1}{4}[\partial_x\chi_L - \partial_x\chi_R]^2 + \frac{1}{4}[\partial_x\chi_L + \partial_x\chi_R]^2\right\} \tag{6.81}$$

$$= \frac{u}{4\pi}\int dx\,[(\partial_x\chi_L)^2 + (\partial_x\chi_R)^2]. \tag{6.82}$$

$$[\chi_L(x), \chi_R(x')] = -[\phi(x), \theta(x')] + [\theta(x), \phi(x')] = 0. \tag{6.83}$$

$$[\chi_R(x), \chi_R(x')] = -[\theta(x), \phi(x')] - [\phi(x), \theta(x')] = i\pi\,\mathrm{sgn}(x - x') = -[\chi_L(x), \chi_L(x')]. \tag{6.84}$$

6.5 Finite temperature: thermal contractions before analytic continuation

At finite temperature, replace ground-state contractions by the thermal trace of the diagonal oscillator Hamiltonian. For $\beta = 1/T$ and $\omega_p = u|p|$, let $\rho_\beta = Z^{-1}e^{-\beta H_{\mathrm{LL}}}$. For a single oscillator,

$$Z_p = \sum_{n=0}^{\infty} e^{-\beta\omega_p n} = \frac{1}{1 - e^{-\beta\omega_p}}, \qquad n_B(\omega_p) = \frac{\sum_n n e^{-\beta\omega_p n}}{Z_p} = \frac{1}{e^{\beta\omega_p} - 1}. \tag{6.85}$$

Therefore $\langle\beta_p\beta_q^\dagger\rangle_\beta = \delta_{pq}(1 + n_B)$ and $\langle\beta_p^\dagger\beta_q\rangle_\beta = \delta_{pq}n_B$. The fixed charge-sector zero modes remain separate.

$$\langle\phi(x,t)\phi(0,0)\rangle_\beta = -\frac{\pi(c+s)^2}{2L}\sum_{p,q\neq 0}\frac{\sqrt{|pq|}}{pq}e^{-a(|p|+|q|)/2 - ipx} \tag{6.86}$$

$$\times\left\langle(\beta_{-p}e^{-iu|p|t} + \beta_p^\dagger e^{iu|p|t})(\beta_{-q} + \beta_q^\dagger)\right\rangle_\beta \tag{6.87}$$

$$= -\frac{\pi(c+s)^2}{2L}\sum_{p,q\neq 0}\frac{\sqrt{|pq|}}{pq}e^{-a(|p|+|q|)/2 - ipx}\delta_{q,-p}\left[e^{-iu|p|t}(1 + n_B) + e^{iu|p|t}n_B\right]. \tag{6.88}$$

The contraction sets $q = -p$, so $pq = -p^2$ and $\sqrt{|pq|}/(pq) = -1/|p|$. Thus the momentum sum inside the prefactor becomes

$$-\sum_{p\neq 0}\frac{e^{-a|p| - ipx}}{|p|}\left[e^{-iu|p|t}\left(1 + \frac{1}{e^{\beta u|p|} - 1}\right) + \frac{e^{iu|p|t}}{e^{\beta u|p|} - 1}\right]. \tag{6.89}$$

$$\langle\phi(x,t)\phi(0,0)\rangle_\beta = \frac{\pi K}{2L}\sum_{p\neq 0}\frac{e^{-a|p| - ipx}}{|p|}\left[\frac{e^{-iu|p|t}}{1 - e^{-\beta u|p|}} + \frac{e^{iu|p|t}}{e^{\beta u|p|} - 1}\right]. \tag{6.90}$$

$$\langle\phi(x,t)\phi(0,0)\rangle_\beta = \frac{\pi K}{2L}\sum_{p>0}\frac{e^{-ap - ipx}}{p}\left[\frac{e^{-iupt}}{1 - e^{-\beta up}} + \frac{e^{iupt}}{e^{\beta up} - 1}\right] \tag{6.91}$$

$$+\frac{\pi K}{2L}\sum_{p<0}\frac{-e^{ap - ipx}}{p}\left[\frac{e^{iupt}}{1 - e^{\beta up}} + \frac{e^{-iupt}}{e^{-\beta up} - 1}\right]. \tag{6.92}$$

Under $p \mapsto -p$, the second sum becomes

$$\frac{\pi K}{2L} \sum_{p>0} \frac{e^{-ap+ipx}}{p} \left[\frac{e^{-iupt}}{1-e^{-\beta up}} + \frac{e^{iupt}}{e^{\beta up}-1} \right]. \tag{6.93}$$

$$\langle \phi(x,t)\phi(0,0) \rangle_\beta = \frac{\pi K}{2L} \sum_{p>0} \frac{e^{-ap}}{p} \left[\frac{e^{-iupt}}{1-e^{-\beta up}} + \frac{e^{iupt}}{e^{\beta up}-1} \right] (e^{-ipx} + e^{ipx}) \tag{6.94}$$

$$\longrightarrow \frac{K}{4} \int_0^\infty \frac{dp}{p}\, e^{-ap} \left[\frac{e^{-iupt}}{1-e^{-\beta up}} + \frac{e^{iupt}}{e^{\beta up}-1} \right] (e^{-ipx} + e^{ipx}). \tag{6.95}$$

The two terms can be grouped as

$$\langle \phi(x,t)\phi(0,0) \rangle_\beta = \frac{K}{4} \left[F_a(ut+x, ut-x; \beta u) + F_a(ut-x, ut+x; \beta u) \right], \tag{6.96}$$

where F_a denotes the regulated integral defined below. An infrared regulator is understood here; subtract the coincident correlator before taking the infinite-line limit.

6.5.1 From the Bose sum to Gamma functions and thermal light cones

For real lengths b_1, b_2, positive length $d_\beta = \beta u$, and $p_{\rm IR} > 0$, define

$$F_a(b_1, b_2; d_\beta) = \int_{p_{\rm IR}}^\infty \frac{dp}{p} e^{-ap} \left[\frac{e^{-ib_1 p}}{1-e^{-d_\beta p}} + \frac{e^{ib_2 p}}{e^{d_\beta p}-1} \right]. \tag{6.97}$$

$$F_a = \int_{p_{\rm IR}}^\infty \frac{dp}{p} e^{-ap} \frac{e^{-ib_1 p} + e^{ib_2 p - d_\beta p}}{1-e^{-d_\beta p}}. \tag{6.98}$$

$$\frac{1}{1-e^{-d_\beta p}} = \sum_{n=0}^\infty e^{-n d_\beta p}, \qquad p > 0. \tag{6.99}$$

$$F_a = \sum_{n=0}^\infty \int_{p_{\rm IR}}^\infty \frac{dp}{p} \left[e^{-(a+nd_\beta + ib_1)p} + e^{-[a+(n+1)d_\beta - ib_2]p} \right]. \tag{6.100}$$

An unsubtracted F_a is infrared divergent. Only the neutral combination of the two chiralities is physical. Pair terms with the same $y = ut \pm x$ and subtract the value at $y = 0$ before sending $p_{\rm IR}$ to zero:

$$\mathcal{F}_a(y) = \int_0^\infty \frac{dp}{p} e^{-ap} \left[\frac{e^{-iyp}-1}{1-e^{-d_\beta p}} + \frac{e^{iyp}-1}{e^{d_\beta p}-1} \right] \tag{6.101}$$

$$= \ln \frac{a}{a+iy} + \sum_{n=1}^\infty \ln \frac{(a+nd_\beta)^2}{(a+nd_\beta)^2 + y^2}. \tag{6.102}$$

Here $\int_0^\infty dp\, (e^{-Ap} - e^{-Bp})/p = \ln(B/A)$ for positive real parts. Differentiation with respect to B, followed by the condition that the integral vanish at $A = B$, proves this identity. Put $\alpha_a = a/d_\beta$ and $z = y/d_\beta$. The Gamma-product form of the paired, subtracted sum is

$$\mathcal{F}_a(y) = \ln \frac{a}{a+iy} + \ln \frac{\Gamma(1+\alpha_a+iz)\Gamma(1+\alpha_a-iz)}{\Gamma(1+\alpha_a)^2}. \tag{6.103}$$

To see the sign and normalization directly, truncate the product at M:

$$\prod_{n=1}^M \frac{(n+\alpha_a)^2}{(n+\alpha_a+iz)(n+\alpha_a-iz)} = \frac{\Gamma(M+1+\alpha_a)^2}{\Gamma(M+1+\alpha_a+iz)\Gamma(M+1+\alpha_a-iz)} \tag{6.104}$$

$$\times \frac{\Gamma(1+\alpha_a+iz)\Gamma(1+\alpha_a-iz)}{\Gamma(1+\alpha_a)^2}. \tag{6.105}$$

The first ratio approaches one as $M \to \infty$. This derivation fixes the positive sign of the log Gamma product and the otherwise missing coincident subtraction. Restoring both chiralities gives all four Gamma factors:

$$2[\langle\phi(x,t)\phi(0,0)\rangle_\beta - \langle\phi(0,0)^2\rangle_\beta] = \frac{K}{2}\sum_{\sigma=\pm1}\mathcal{F}_a(ut+\sigma x) \tag{6.106}$$

$$= \frac{K}{2}\sum_{\sigma=\pm1}\Bigg[\ln\frac{a}{a+i(ut+\sigma x)} + \ln\Gamma\left(1+\alpha_a+i\frac{ut+\sigma x}{\beta u}\right) \tag{6.107}$$

$$+ \ln\Gamma\left(1+\alpha_a-i\frac{ut+\sigma x}{\beta u}\right) - 2\ln\Gamma(1+\alpha_a)\Bigg]. \tag{6.108}$$

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}, \qquad \Gamma(1+z) = z\Gamma(z). \tag{6.109}$$

The dimensionless thermal arguments are $z_\pm = iT(t \pm x/u)$. They involve T multiplied by time, not by a length without the factor $1/u$. In the scaling limit $a \ll \beta u$, the thermal part of (6.103) becomes

$$\Gamma(1+iy/d_\beta)\Gamma(1-iy/d_\beta) = \frac{\pi y/d_\beta}{\sinh(\pi y/d_\beta)}, \tag{6.110}$$

$$\mathcal{F}_a(y) = \ln\frac{a}{a+iy} + \ln\frac{\pi y/(\beta u)}{\sinh[\pi y/(\beta u)]} + O(a/(\beta u)). \tag{6.111}$$

Consequently, after the greater/lesser ordering has fixed the analytic continuation, each zero-temperature factor has the thermal scaling continuation

$$\frac{\tau_a}{\tau_a+it_r} \;\longrightarrow\; \mathcal{D}_T(t_r) \equiv \frac{\pi T\tau_a}{i\sinh[\pi T(t_r - i\tau_a)]}, \qquad t_r = t - rx/u, \qquad \tau_a = a/u. \tag{6.112}$$

Equation (6.103) is the exact result for the exponential momentum cutoff. Equation (6.112) uses an analytic time shift and represents the same low-energy scaling theory up to ultraviolet matching corrections of order $T\tau_a$; it is not an exact identity between two cutoff schemes. Its $T \to 0$ limit is $\tau_a/(\tau_a + it_r)$. For a normalization that is exactly one at $t_r = 0$ in the shifted scheme, replace the numerator by $\sin(\pi T\tau_a)$. The bulk correlator follows from (6.75) by replacing its two factors by $\mathcal{D}_T(t-x/u)$ and $\mathcal{D}_T(t+x/u)$, with the same two exponents. Algebraic decay crosses to exponential decay on the thermal time scale $1/T$. Figure 6.4 illustrates the magnitude crossover using this explicitly specified shifted-time prescription.

6.6 Checking propagation directly in the oscillator expansion

The eigenfield $\chi_R = \phi/\sqrt{K} - \sqrt{K}\theta$ can be checked directly in the oscillator representation. We retain the complete positive- and negative-momentum expansion to see which coefficients cancel.

$$\chi_R(x,t) = -\frac{i\pi}{L\sqrt{K}}\sum_{p\neq0}\sqrt{\frac{L|p|}{2\pi}}e^{-a|p|/2-ipx}\Bigg\{\frac{K(s-c)}{|p|}(\beta_{-p}e^{-iu|p|t} - \beta_p^\dagger e^{iu|p|t}) \tag{6.113}$$

$$+ \frac{s+c}{p}(\beta_{-p}e^{-iu|p|t} + \beta_p^\dagger e^{iu|p|t})\Bigg\}. \tag{6.114}$$

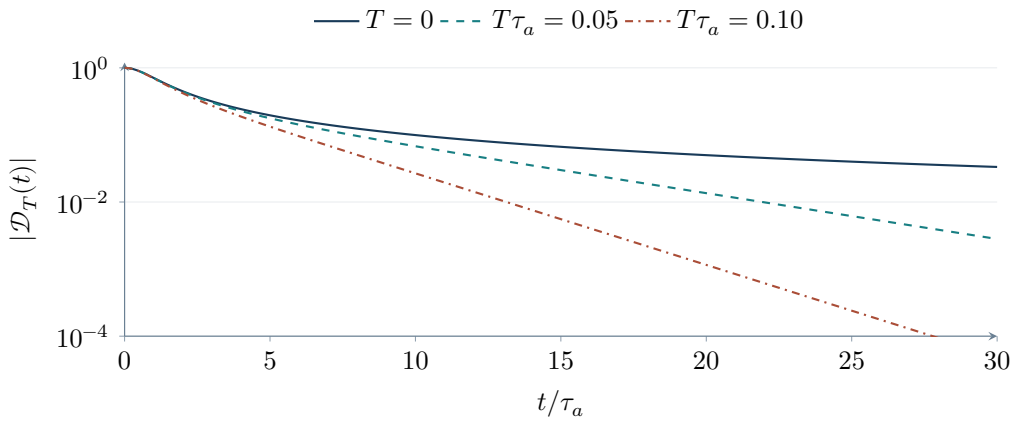


Figure 6.4: Correlation magnitude computed from the normalized shifted-time prescription $|\mathcal{D}_T(t)| = \sin(\pi T\tau_a)/\sqrt{\sinh^2(\pi Tt) + \sin^2(\pi T\tau_a)}$. The zero-temperature curve is $(1 + t^2/\tau_a^2)^{-1/2}$. The thermal time $1/T$ changes the long-time decay from a power law to an exponential. Raising these curves to an operator's decay exponent gives the corresponding local Gaussian vertex magnitude. This is a specified regulator scheme, not experimental data.

$$\chi_R(x,t) = -\frac{i\pi}{L\sqrt{K}} \sum_{p>0} \sqrt{\frac{L}{2\pi p}} e^{-ap/2 - ipx} \Big\{ [K(s-c) + s + c]\beta_{-p} e^{-iupt} \tag{6.115}$$

$$+ [-K(s-c) + s + c]\beta_p^\dagger e^{iupt} \Big\} \tag{6.116}$$

$$- \frac{i\pi}{L\sqrt{K}} \sum_{p<0} \sqrt{\frac{L}{2\pi|p|}} e^{ap/2 - ipx} \Big\{ [K(s-c) - s - c]\beta_{-p} e^{iupt} \tag{6.117}$$

$$+ [-K(s-c) - s - c]\beta_p^\dagger e^{-iupt} \Big\}. \tag{6.118}$$

Relabel $p \mapsto -p$ in the second sum. The four terms become

$$\chi_R(x,t) = -\frac{i\pi}{L\sqrt{K}} \sum_{p>0} \sqrt{\frac{L}{2\pi p}} e^{-ap/2} \Big\{ [K(s-c) + s + c]\beta_{-p} e^{-iupt - ipx} \tag{6.119}$$

$$+ [-K(s-c) + s + c]\beta_p^\dagger e^{iupt - ipx} \tag{6.120}$$

$$+ [K(s-c) - s - c]\beta_p e^{-iupt + ipx} \tag{6.121}$$

$$+ [-K(s-c) - s - c]\beta_{-p}^\dagger e^{iupt + ipx} \Big\}. \tag{6.122}$$

The two possible propagation contributions to the vacuum contraction are

$$\langle \chi_R(x,t)\chi_R(0,0)\rangle = -\frac{\pi}{2LK} \sum_{p>0} \frac{e^{-ap}}{p} \Big\{ -[K(s-c) + s + c]^2 e^{-iupt - ipx} \tag{6.123}$$

$$- [K(s-c) - s - c]^2 e^{-iupt + ipx} \Big\}. \tag{6.124}$$

For $p, q > 0$, $-p \neq q$, so the two independent oscillator sectors have no cross contraction.

$$\langle \chi_R(x,t)\chi_R(0,0)\rangle = \frac{\pi}{2L} \sum_{p>0} \frac{e^{-ap}}{p} \left[\mathcal{C}_L^2 e^{-iupt - ipx} + \mathcal{C}_R^2 e^{-iupt + ipx} \right], \tag{6.125}$$

where $\mathcal{C}_L = [s(1+K) + c(1-K)]/\sqrt{K}$ and $\mathcal{C}_R = [-s(1-K) - c(1+K)]/\sqrt{K}$.

$$\mathcal{C}_L^2 = \frac{1}{K}\left[s^2(1+K)^2 + c^2(1-K)^2 + 2sc(1-K^2)\right] \tag{6.126}$$
$$= \frac{1}{K}\left[(s^2+c^2)(1+K^2) + 2K(s^2-c^2) + 2sc(1-K^2)\right] \tag{6.127}$$
$$= \frac{1}{K}\left[\cosh(2\xi)(1+K^2) - 2K + \sinh(2\xi)(1-K^2)\right] \tag{6.128}$$
$$= \frac{1}{K}\left[\frac{1}{2}(K^{-1}+K)(1+K^2) - 2K - \frac{1}{2}(1-K^2)(K^{-1}-K)\right] \tag{6.129}$$
$$= \frac{1}{K}\left[\frac{(1+K^2)^2 - (1-K^2)^2}{2K} - 2K\right] = \frac{1}{K}\left[\frac{4K^2}{2K} - 2K\right] = 0. \tag{6.130}$$

$$\mathcal{C}_R^2 = \frac{1}{K}\left[s^2(1-K)^2 + c^2(1+K)^2 + 2sc(1-K^2)\right] \tag{6.131}$$
$$= \frac{1}{K}\left[(s^2+c^2)(1+K^2) - 2K(s^2-c^2) + \sinh(2\xi)(1-K^2)\right] \tag{6.132}$$
$$= \frac{1}{K}\left[\frac{1}{2}(K^{-1}+K)(1+K^2) + 2K - \frac{1}{2}(1-K^2)(K^{-1}-K)\right] = 4. \tag{6.133}$$

$$\langle \chi_R(x,t)\chi_R(0,0)\rangle = \frac{4\pi}{2L}\sum_{p>0}\frac{e^{-iupt+ipx-ap}}{p} = \frac{2\pi}{L}\sum_{p>0}\frac{e^{-iupt+ipx-ap}}{p} \longrightarrow \int_0^\infty \frac{dp}{p} e^{-iupt+ipx-ap}. \tag{6.134}$$

The eigenfield is already unit-normalized: the interaction factor has moved into its physical density and electron-vertex embedding, rather than into this covariance.

$$\partial_x \langle \chi_R(x,t)\chi_R(0,0)\rangle = i\int_0^\infty dp\, e^{-[a+i(ut-x)]p} \tag{6.135}$$
$$= i\left[\frac{e^{-[a+i(ut-x)]p}}{-a-i(ut-x)}\right]_0^\infty = \frac{i}{a+i(ut-x)} = \frac{1}{ut-x-ia}. \tag{6.136}$$

$$\langle \chi_R(x,t)\chi_R(0,0)\rangle - \langle \chi_R(0,0)^2\rangle = -\int_0^{x-ut}\frac{d\ell}{\ell+ia} = -\ln(\ell+ia)\Big|_0^{x-ut} = -\ln\frac{a+i(ut-x)}{a}. \tag{6.137}$$

This is a function of $x-ut$, describing rightward propagation at the collective velocity u. A vertex $e^{i\sqrt{K}\chi_R}$ has a correlator proportional to $[a/(a+i(ut-x))]^K$: its decay is faster for $K>1$ and slower for $K<1$. That statement concerns the vertex's physical normalization; the unit-normalized χ_R covariance itself has no K prefactor. The same conclusion follows from $\partial_t\phi = uK\partial_x\theta$ and $\partial_t\theta = (u/K)\partial_x\phi$: $\partial_t\chi_R = -u\partial_x\chi_R$, while $\partial_t\chi_L = +u\partial_x\chi_L$. A physical electron phase $\varphi_r = r(c\chi_r + s\chi_{-r})$ therefore excites both propagation modes when $s \neq 0$. Its two exponents are not evidence for two independent physical electron species.

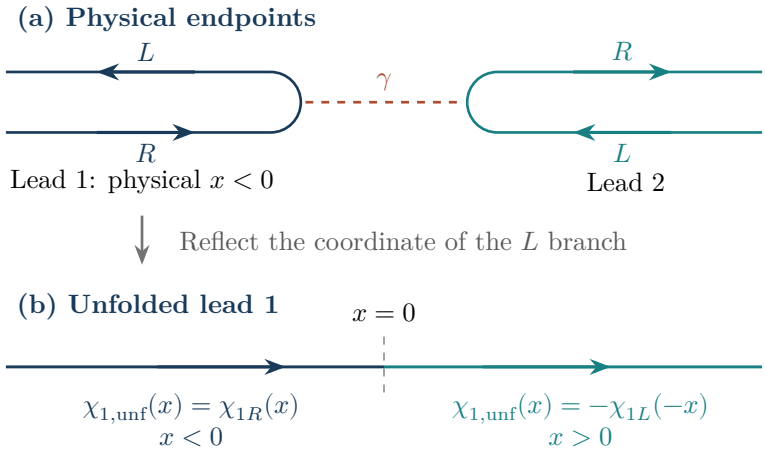


Figure 7.1: Reflecting endpoints and the unfolding of lead 1 (schematic). The two physical branches join at the reflecting end. After reflection of the left-moving coordinate and the indicated field sign, one right-moving field extends over the whole line. Its continuity enforces the pinned density-field fluctuation $\phi_1(0) = 0$. The dashed contact is the perturbation, absent in the reference Hamiltonian.

7 Reflecting boundaries and weak-link tunnelling

A reflecting endpoint is a different unperturbed geometry from the bulk. Boundary gluing and unfolding determine its electron operator. A gauge transformation places the chemical-potential difference in a tunnelling phase, from which the current is calculated.

7.1 A reflecting endpoint and its unfolded field

At a quantum point contact (QPC), the relevant electron can be a boundary operator. Merely putting $x = x' = 0$ into a bulk answer gives

$$\langle \psi_R^\dagger(0,t)\psi_R(0,0)\rangle_{\mathrm{bulk}} = \mathcal{A}_a \left(\frac{a}{a+iut}\right)^{(K+K^{-1})/2}. \tag{7.1}$$

This is the local correlator of a continuous homogeneous wire. A disconnected reflecting endpoint imposes a different relation between the two branches. Its boundary condition must enter the Hamiltonian and correlator before the local limit is taken. Figure 7.1 shows the two reflecting leads and the unfolding of lead 1. We use $x < 0$ for lead 1, with its endpoint at $x = 0$; the right-moving branch approaches the contact and the left-moving branch leaves it. Unfolding the physical fermion branches before treating their mutual interaction generally produces a coupling between reflected positions x and $-x$. The Gaussian Hamiltonian should therefore be diagonalized first. This nonlocality of a naive unfolding does not invalidate Luttinger-liquid theory.

$$H_1 = \frac{u}{2\pi}\int_{-\infty}^{0} dx\,[K(\partial_x\theta_1)^2 + K^{-1}(\partial_x\phi_1)^2]. \tag{7.2}$$

Equivalently, in the unit-normalized eigenfields,

$$H_1 = \frac{u}{4\pi}\int_{-\infty}^{0} dx\,[(\partial_x\chi_{1R})^2 + (\partial_x\chi_{1L})^2]. \tag{7.3}$$

A perfectly reflecting boundary has $\psi_{1R}(0) = e^{i\delta_1}\psi_{1L}(0)$ and zero particle current. The phase δ_1 fixes a constant value of $\phi_1(0)$, including the zero-mode/Klein convention. Shift this constant so that the fluctuating field satisfies $\phi_1(0) = 0$. At a microscopic hard wall $\Psi(0) = 0$; the contact

electron used for tunnelling is defined at a microscopic distance of order a from that wall, or as the corresponding nonzero boundary scaling operator.

$$\chi_{1,\mathrm{unf}}(x) = \begin{cases} \chi_{1R}(x), & x < 0, \\ -\chi_{1L}(-x), & x > 0. \end{cases} \tag{7.4}$$

$$\phi_1(x) = \frac{\sqrt{K}}{2}[\chi_{1,\mathrm{unf}}(x) - \chi_{1,\mathrm{unf}}(-x)], \qquad x < 0. \tag{7.5}$$

$$\theta_1(x) = -\frac{1}{2\sqrt{K}}[\chi_{1,\mathrm{unf}}(x) + \chi_{1,\mathrm{unf}}(-x)], \qquad x < 0. \tag{7.6}$$

The two squared eigenfield gradients cover the negative and positive halves of the unfolded coordinate, respectively:

$$H_1 = \frac{u}{4\pi}\left[\int_{-\infty}^{0} dx\,(\partial_x \chi_{1,\mathrm{unf}})^2 + \int_{0}^{\infty} dx\,(\partial_x \chi_{1,\mathrm{unf}})^2\right]. \tag{7.7}$$

$$H_1 = \frac{u}{4\pi}\int_{-\infty}^{\infty} dx\,(\partial_x \chi_{1,\mathrm{unf}})^2. \tag{7.8}$$

The unfolded Hamiltonian is Gaussian and chiral, while the electron vertex still depends on K.

$$\chi_{1,\mathrm{unf}}(0^-) = \chi_{1,\mathrm{unf}}(0^+). \tag{7.9}$$

$$\phi_1(0) = 0, \qquad \theta_1(0) = -\frac{1}{\sqrt{K}}\chi_{1,\mathrm{unf}}(0). \tag{7.10}$$

The density-field fluctuation is pinned at the boundary, while the dual phase remains dynamical.

$$d_{1,\mathrm{end}} \propto F_1 {:}e^{i\theta_1(0)}{:} = F_1 {:}\exp[-i\chi_{1,\mathrm{unf}}(0)/\sqrt{K}]{:}. \tag{7.11}$$

The gluing condition extends the same right-moving field through $x = 0$. Its unequal-point algebra is $[\chi_{1,\mathrm{unf}}(x), \chi_{1,\mathrm{unf}}(y)] = i\pi\,\mathrm{sgn}(x-y)$, and $(\partial_t + u\partial_x)\chi_{1,\mathrm{unf}} = 0$. The vertex has oscillator exponent $1/K$; F_1 carries the single lead-number shift and its amplitude remains separate.

$$\langle d^\dagger_{1,\mathrm{end}}(t) d_{1,\mathrm{end}}(0)\rangle = \mathcal{A}_{\mathrm{end}}\left(\frac{a}{a+iut}\right)^{1/K}, \qquad \Delta_{\psi,\mathrm{end}} = \frac{1}{2K}. \tag{7.12}$$

The decay exponent is $1/K$, rather than $(K + K^{-1})/2$ or $1/(2K)$. Pinning ϕ changes the fluctuations of θ as well: one cannot simply delete the density-field contribution from a bulk correlator. The scaling dimension is half the time-decay exponent. Unfolding works here because the diagonal fields are Gaussian, even though the electrons interact. Integrating out the interior $x \neq 0$ is an equivalent route to a boundary action and its renormalization analysis.

7.2 Moving the bias into a tunnelling phase

The bias can be represented by the occupations of two independently equilibrated leads, or by a time-dependent tunnelling phase after their energy references have been shifted. We keep lead labels $1, 2$ distinct from physical right/left branches.

$$H = H_1 + H_2 + H_T = H_0 + H_T. \tag{7.13}$$

$$\rho_0 = \frac{1}{Z}\exp[-\beta(H_0 - \mu_1 N_1 - \mu_2 N_2)]. \tag{7.14}$$

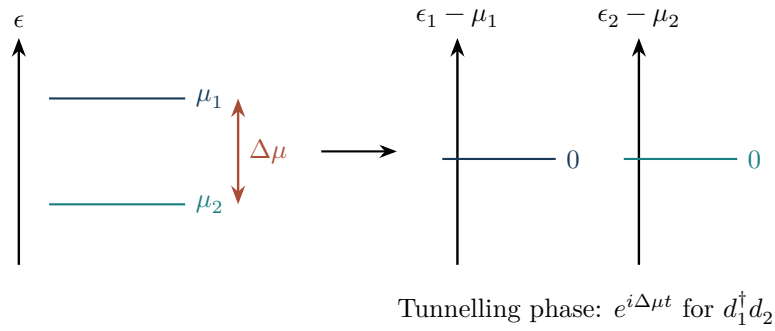


Figure 7.2: Separate reference-energy shifts (schematic). The two leads initially have different chemical potentials. Measuring each lead's excitation energies from its own chemical potential puts the imbalance into the phase of its tunnelling term. It does not remove the nonequilibrium drive of the coupled system.

The Hamiltonian fixes the excitation energies; the density matrix fixes their occupations. In particular, the disconnected leads need not have equal chemical potentials.

$$U(t) = e^{iMt}, \qquad M = \mu_1 N_1 + \mu_2 N_2. \tag{7.15}$$

$$O_{H_0}(t) = e^{iH_0 t} O_S e^{-iH_0 t}. \tag{7.16}$$

The explicit operator conjugation is $U(t)O_{H_0}(t)U^\dagger(t)$. It must be distinguished from changing the generator of interaction-picture evolution.

$$U(t) d_{1,H_0}(t) U^\dagger(t) = e^{iH_0 t}\left[e^{iMt} d_1 e^{-iMt}\right] e^{-iH_0 t}, \qquad [H_0, M] = 0. \tag{7.17}$$

$$[N_\ell, d_\ell] = -d_\ell. \tag{7.18}$$

For any operators with $[A, B] = -B$,

$$AB = B(A-1). \tag{7.19}$$

$$A^2 B = AB(A-1) = B(A-1)^2. \tag{7.20}$$

$$A^n B = B(A-1)^n. \tag{7.21}$$

$$e^{i\mu_1 N_1 t} d_1 = \sum_{n=0}^{\infty} \frac{(i\mu_1 t)^n}{n!} N_1^n d_1 = \sum_{n=0}^{\infty} \frac{(i\mu_1 t)^n}{n!} d_1 (N_1 - 1)^n = d_1 e^{i\mu_1 N_1 t - i\mu_1 t}. \tag{7.22}$$

$$e^{i\mu_1 N_1 t} d_1 e^{-i\mu_1 N_1 t} = e^{-i\mu_1 t} d_1. \tag{7.23}$$

Thus explicit conjugation gives $U d_{\ell,H_0}(t) U^\dagger = e^{-i\mu_\ell t} d_{\ell,H_0}(t)$.

$$H_0' = U H_0 U^\dagger + i(\partial_t U) U^\dagger = H_0 - M \equiv H_{\text{ref}}. \tag{7.24}$$

$$\rho_0' = U \rho_0 U^\dagger = \rho_0 = Z^{-1} e^{-\beta H_{\text{ref}}}. \tag{7.25}$$

Each lead is now described with energies measured from its own chemical potential. This is equilibrium for the disconnected reference Hamiltonian, at the expense of a time-dependent tunnelling term; it does not imply global equilibrium of the coupled biased device. Figure 7.2 illustrates the energy-reference shift. Define $d_{\ell,\text{ref}}(t) = e^{iH_{\text{ref}}t} d_\ell e^{-iH_{\text{ref}}t}$. Since $[N_\ell, d_\ell] = -d_\ell$,

$$d_{\ell,H_0}(t) = e^{-i\mu_\ell t} d_{\ell,\text{ref}}(t), \tag{7.26}$$

$$d_{1,H_0}^\dagger(t) d_{2,H_0}(t) = e^{i\Delta\mu t} d_{1,\text{ref}}^\dagger(t) d_{2,\text{ref}}(t), \qquad \Delta\mu = \mu_1 - \mu_2, \tag{7.27}$$

$$H_{T,\text{ref}}(t) = \gamma e^{i\Delta\mu t} d_1^\dagger(t) d_2(t) + \gamma^* e^{-i\Delta\mu t} d_2^\dagger(t) d_1(t). \tag{7.28}$$

The last line uses reference-picture fields. Its phase and the shifted reference occupations form one representation of the bias; the chemical-potential phase must not be inserted a second time into the same correlators. Write $V_{\text{bias}} = \Delta\mu/e$ as an energy-bias voltage. The sign of an externally specified electrostatic voltage requires its own electron-charge convention.

7.3 Weak-link current: evaluating the Fourier integral

Consider two identical disconnected spinless LL endpoints with the same u, K, a, coupled by $H_T = \gamma d_1^\dagger d_2 + \gamma^* d_2^\dagger d_1$. We compute the leading current for $E = \Delta\mu = eV_{\rm bias} > 0$ at zero temperature. Choose a contact matching convention $\mathcal{A}_{\rm end} = (2\pi a)^{-1}$. A different microscopic contact amplitude can be absorbed into the matched tunnelling coefficient γ, rather than into its dimensionless Klein factor. The exponents and Gamma-function dependence below do not depend on that choice.

$$I_{1\to 2} = e|\gamma|^2 \int_{-\infty}^{\infty} dt\, e^{iEt} \langle d_1^\dagger(t) d_1(0)\rangle_{\rm ref} \langle d_2(t) d_2^\dagger(0)\rangle_{\rm ref} \tag{7.29}$$

$$= \frac{e|\gamma|^2 a^{2/K}}{(2\pi a)^2} \int_{-\infty}^{\infty} dt\, \frac{e^{iEt}}{(a+iut)^{2/K}} \tag{7.30}$$

$$= \frac{e|\gamma|^2}{(2\pi)^2} \frac{a^{2/K-2}}{u^{2/K}} \int_{-\infty}^{\infty} dt\, \frac{e^{iEt}}{(\tau_a + it)^{2/K}}. \tag{7.31}$$

Here $\tau_a = a/u$ is the short-time cutoff, and $E_a = 1/\tau_a$ the ultraviolet energy scale. For $d > 0$ and $\mathrm{Re}\, z > 0$, the Gamma integral gives

$$\int_0^\infty ds\, s^{2d-1} e^{-zs} = \frac{\Gamma(2d)}{z^{2d}}, \qquad d = 1/K. \tag{7.32}$$

$$\frac{1}{(\tau_a + it)^{2d}} = \frac{1}{\Gamma(2d)} \int_0^\infty ds\, s^{2d-1} e^{-\tau_a s} e^{-ist}. \tag{7.33}$$

$$\int_{-\infty}^{\infty} dt\, \frac{e^{iEt}}{(\tau_a + it)^{2d}} = \frac{1}{\Gamma(2d)} \int_0^\infty ds\, s^{2d-1} e^{-\tau_a s} \int_{-\infty}^{\infty} dt\, e^{i(E-s)t} \tag{7.34}$$

$$= \frac{2\pi}{\Gamma(2d)} \int_0^\infty ds\, s^{2d-1} e^{-\tau_a s} \delta(E-s) \tag{7.35}$$

$$= \frac{2\pi}{\Gamma(2d)} E^{2d-1} e^{-\tau_a E}, \qquad E > 0. \tag{7.36}$$

For $E < 0$ the integral is zero: the zero-temperature reference cannot supply the energy needed for that direction of transfer. In the low-energy regime $|E|\tau_a \ll 1$, $e^{-|E|\tau_a} \simeq 1$.

$$I_{1\to 2} = \frac{e|\gamma|^2}{2\pi} \frac{a^{2/K-2}}{u^{2/K}} \frac{E^{2/K-1} e^{-\tau_a E}}{\Gamma(2/K)}, \qquad E > 0. \tag{7.37}$$

$$I_{\rm link}(E) = \frac{e}{2\pi} \left|\frac{\gamma}{u}\right|^2 \frac{(|E|\tau_a)^{2/K-2}}{\Gamma(2/K)}\, E\, e^{-|E|\tau_a}. \tag{7.38}$$

The last formula includes reverse bias: interchange the leads, subtract the reverse rate, and obtain an odd function of E. Each endpoint electron has dimension $1/(2K)$, so the tunnelling operator has dimension $d_{\rm link} = 1/K$. Its two-point function has power $2/K$ and its Fourier integral has power $2/K - 1$, in agreement with the explicit calculation. The regulator supplies dimensions but cannot be used to extend the LL answer to $|E| \sim E_a$.

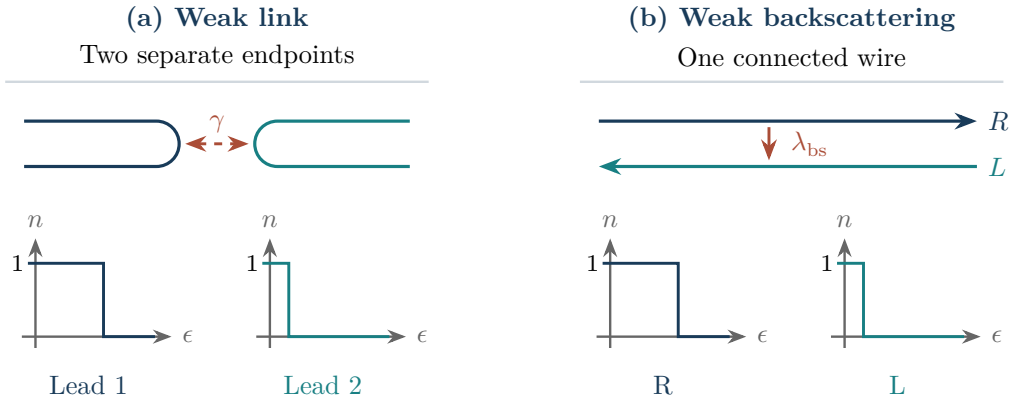


Figure 8.1: The two boundary expansions and their zero-temperature incoming occupation steps (schematic). The weak link transfers an electron between separate leads. The weak impurity converts a physical right mover into a left mover in one continuous wire. Horizontal step offsets indicate a chemical-potential imbalance; no numerical dispersion or experimental data is assumed.

8 Weak backscattering, duality, and the domain of expansion

A nearly connected wire has a different small perturbation from two disconnected endpoints. We compare their scaling dimensions, retain the source Fourier-integral reasoning, and identify where a leading-order current no longer gives a controlled approximation.

8.1 The opposite limit: weak backscattering in a connected wire

The expansion around two disconnected endpoints can fail at low energy. The opposite reference geometry is one continuous wire with a weak local impurity. For attractive interactions, $K > 1$, this connected geometry is the stable low-energy description. Figure 8.1 compares the two expansions.

$$H_{\mathrm{bs}} = \lambda_{\mathrm{bs}} \psi_L^\dagger(0)\psi_R(0) + \lambda_{\mathrm{bs}}^* \psi_R^\dagger(0)\psi_L(0). \tag{8.1}$$

Here R, L denote physical Fermi branches in the same wire; there are no independent lead labels on the local fields. A branch imbalance $\Delta\mu = \mu_R - \mu_L$ supplies the bias energy for right-to-left backscattering. Right-moving electrons can scatter into the left-moving branch at the impurity. Their incoming occupations can be fixed by different reservoirs, as indicated by the Fermi-step diagrams in Fig. 8.1.

$$H_0 = \frac{u}{2\pi} \int_{-\infty}^{\infty} dx \, [K(\partial_x \theta)^2 + K^{-1}(\partial_x \phi)^2]. \tag{8.2}$$

The relative Klein/zero-mode factor $F_L^\dagger F_R$ transfers one fermion between branches. It cancels in a neutral two-point rate after the branch-number bookkeeping has been performed; omitting it before establishing that selection rule is not justified.

$$H_{\mathrm{bs}}(t) = \frac{\lambda_{\mathrm{bs}}}{2\pi a} e^{-i\Delta\mu t} F_L^\dagger F_R {:}e^{-i[\phi(0)+\theta(0)]}{::}e^{-i[\phi(0)-\theta(0)]}{:} + \mathrm{h.c.} \tag{8.3}$$

$$= \frac{1}{2\pi a} \left[\lambda_{\mathrm{bs}} e^{-i\Delta\mu t} F_L^\dagger F_R {:}e^{-2i\phi(0)}{:} + \mathrm{h.c.} \right]. \tag{8.4}$$

The phase follows from $\psi_{L,H_0}^\dagger \psi_{R,H_0} = e^{-i(\mu_R-\mu_L)t} \psi_{L,\mathrm{ref}}^\dagger \psi_{R,\mathrm{ref}}$. Products at the same point use the common cutoff convention, with a constant short-distance phase included in λ_{bs}. Define positive backscattering by $I_{\mathrm{bs}} = e\dot{N}_L = -e\dot{N}_R$. If $B = \psi_L^\dagger \psi_R$, then $I_{\mathrm{bs}} = -ie(\lambda_{\mathrm{bs}} B - \lambda_{\mathrm{bs}}^* B^\dagger)$. This is the amount removed from forward electron transport, not the total transmitted current.

At $T = 0$ and $E = \Delta\mu > 0$, the neutral rate is

$$I_{\rm bs} = \frac{e|\lambda_{\rm bs}|^2}{4\pi^2 a^2}\int_{-\infty}^{\infty} dt\, e^{iEt}\langle :e^{2i\phi(t)}::e^{-2i\phi(0)}:\rangle \tag{8.5}$$

$$= \frac{e|\lambda_{\rm bs}|^2}{4\pi^2 a^2}\int_{-\infty}^{\infty} dt\, e^{iEt}\frac{a^{2K}}{(a+iut)^{2K}} \tag{8.6}$$

$$= \frac{e}{2\pi}\left|\frac{\lambda_{\rm bs}}{u}\right|^2 \frac{(E\tau_a)^{2K-2}}{\Gamma(2K)} E e^{-E\tau_a}. \tag{8.7}$$

The exponent is $2K$, in place of $2/K$. Accordingly the impurity dimension is $d_{\rm bs} = K$. For $K < 1$, its relative correction grows as E decreases, and the weak-backscattering expansion eventually fails. The total transmitted current is $I_{\rm trans} = I_{\rm clean} - I_{\rm bs}$ in the chosen orientation. The clean current requires a reservoir/contact model. For one spinless channel with ideal noninteracting reservoirs its low-frequency linear conductance is $e^2/(2\pi)$; an isolated homogeneous LL with a different bias definition need not have that same conductance.

8.2 The actual small parameter of a tunnelling expansion

The dimensionful prefactor in the weak-link result can be reorganized explicitly. At $E = eV_{\rm bias} > 0$ and $E\tau_a \ll 1$,

$$I_{\rm link} \simeq \frac{e|\gamma|^2}{4\pi^2 u^2}\tau_a^{2/K-2}\frac{2\pi E^{2/K-1}}{\Gamma(2/K)} \tag{8.8}$$

$$= \frac{e|\gamma|^2}{2\pi u^2}\frac{(E\tau_a)^{2/K-2}}{\Gamma(2/K)}E = \frac{e}{2\pi}\left|\frac{\gamma}{u}\right|^2\frac{(E\tau_a)^{2/K-2}}{\Gamma(2/K)}E. \tag{8.9}$$

The ratio γ/u is dimensionless because the local tunnelling amplitude has dimensions of velocity.

$$K = 1: \qquad I_{\rm link} = \frac{e^2}{2\pi}\left|\frac{\gamma}{u}\right|^2 V_{\rm bias} \quad \text{to leading order and at low bias.} \tag{8.10}$$

For $K \neq 1$, define the dimensionless perturbative ratio

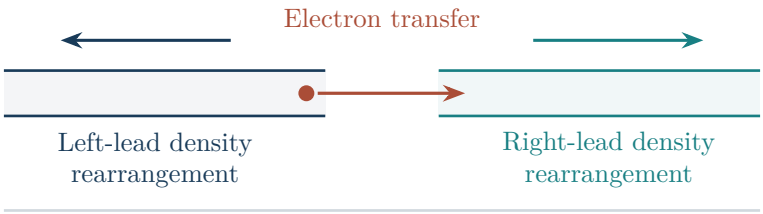
$$\mathcal{T}_{\rm link}(E) \equiv \frac{2\pi I_{\rm link}}{eE} \simeq \left|\frac{\gamma}{u}\right|^2\frac{(|E|\tau_a)^{2/K-2}}{\Gamma(2/K)}. \tag{8.11}$$

The leading expansion requires $\mathcal{T}_{\rm link} \ll 1$ at the measured energy, not merely a small bare $|\gamma/u|^2$. Its energy-dependent power is the interaction-induced renormalization factor. Away from $K = 1$, the ratio is not automatically a single-electron elastic transmission probability.

$$K < 1: \qquad 2/K - 2 > 0, \qquad (|E|\tau_a)^{2/K-2} \longrightarrow 0 \quad (E \to 0). \tag{8.12}$$

Repulsion suppresses low-energy tunnelling. A transferred electron must reorganize density fluctuations in both leads; the reduced overlap with low-energy many-body states produces the power law. The source also mentions Coulomb repulsion and dissipative environments as physical routes to tunnelling suppression. A separate environmental model is required before identifying its parameters with this closed LL model. The contact rearrangement is sketched in Fig. 8.2. Attractive effective interactions can arise, for example, from phonon-mediated coupling. Spin systems, including the spin-ladder context mentioned in the source, can also realize effective LL models. The particular Hamiltonian written next is an XXZ chain.

$$H_{\rm XXZ} = \sum_i \left[J_{xy}(S^x_{i+1}S^x_i + S^y_{i+1}S^y_i) + J_z S^z_{i+1}S^z_i\right]. \tag{8.13}$$



$$\mathcal{T}_{\text{link}}(E) \propto E^{2/K-2} \longrightarrow 0 \quad (K<1)$$

Figure 8.2: Schematic many-body rearrangement accompanying a tunnelling event. For repulsive $K<1$, the endpoint electron has a larger time-decay exponent $1/K$, reducing the low-energy tunnelling rate. The arrows represent outgoing density disturbances, not an additional microscopic transport channel.

Under the Jordan–Wigner transformation on an open chain,

$$S^+_{i+1}S^-_i \;\mapsto\; c^\dagger_{i+1}c_i, \qquad S^z_i = c^\dagger_i c_i - \tfrac{1}{2}. \tag{8.14}$$

$$H = -t_{\text{hop}}\sum_i (c^\dagger_{i+1}c_i + \text{h.c.}) + V\sum_i (n_{i+1}-\tfrac{1}{2})(n_i-\tfrac{1}{2}), \qquad n_i = c^\dagger_i c_i. \tag{8.15}$$

The XY identity $S^xS^x + S^yS^y = (S^+S^- + S^-S^+)/2$ gives $|t_{\text{hop}}| = |J_{xy}|/2$, while $V = J_z$. A staggered phase change of c_i can reverse the hopping sign on a bipartite open chain. This mapping displays an effective density interaction; the LL description additionally requires a gapless low-energy regime and does not cover every XXZ parameter. For $K>1$, the relevant exponent of the perturbative ratio is $2/K-2<0$, and

$$(|E|\tau_a)^{2/K-2} \longrightarrow \infty \quad (E\to 0). \tag{8.16}$$

The divergence is in the perturbative ratio $\mathcal{T}_{\text{link}}$, not a prediction of an infinite physical current. In fact the formal current exponent $2/K-1$ is negative only for $K>2$. Finite temperature can stop the growth at a nonzero infrared scale. If both bias and temperature tend to zero for $K>1$, any fixed nonzero bare weak link eventually leaves its perturbative domain. One must then use the connected-wire description rather than extrapolate the weak-link formula.

8.3 Temperature regularization and the linear conductance

Temperature supplies an infrared scale even at zero bias. We derive the weak-link linear conductance in the universal regime $T\tau_a \ll 1$, keeping the operator-order prescription inherited from the zero-temperature correlator.

$$I(E,T) = \frac{e|\gamma|^2}{(2\pi a)^2}(\pi T\tau_a)^{2d}\int_{-\infty}^{\infty} dt\, \frac{e^{iEt}-e^{-iEt}}{[i\sinh\{\pi T(t-i0^+)\}]^{2d}}, \qquad d=1/K. \tag{8.17}$$

$$e^{iEt}-e^{-iEt} = 2i\sin(Et) = 2iEt + O(E^3t^3). \tag{8.18}$$

$$I(E,T) = \frac{2ie|\gamma|^2E}{4\pi^2a^2}(\pi T\tau_a)^{2d}\int_{-\infty}^{\infty} dt\, \frac{t}{[i\sinh\{\pi T(t-i0^+)\}]^{2d}} + O(E^3). \tag{8.19}$$

Rescale the dimensionless time $z=\pi Tt$, so that $dt\,t = dz\,z/(\pi T)^2$. No factor u enters a rescaling of time by temperature.

$$I(E,T) = \frac{2ieE}{4\pi^2}\left|\frac{\gamma}{u}\right|^2(\pi T\tau_a)^{2d-2}\int_{-\infty}^{\infty} dz\, \frac{z}{[i\sinh(z-i0^+)]^{2d}} + O(E^3). \tag{8.20}$$

To evaluate the ordered integral without losing its phase, first compute its Fourier transform. Shift the time contour to $t = s - i/(2T)$ inside the analytic strip fixed by the ordering. Then $i\sinh(\pi T t) = \cosh(\pi T s)$ and $e^{iEt} = e^{E/(2T)}e^{iEs}$. With $q = e^{2\pi T s}$,

$$\mathcal{P}_d(E,T) \equiv \int_{-\infty}^{\infty} dt\, e^{iEt}\left[\frac{\pi T\tau_a}{i\sinh\{\pi T(t-i0^+)\}}\right]^{2d} \tag{8.21}$$

$$= e^{E/(2T)}(\pi T\tau_a)^{2d}\int_{-\infty}^{\infty} ds\,\frac{e^{iEs}}{\cosh^{2d}(\pi T s)} \tag{8.22}$$

$$= e^{E/(2T)}\frac{2^{2d}(\pi T\tau_a)^{2d}}{2\pi T}\int_0^{\infty} dq\,\frac{q^{d+iE/(2\pi T)-1}}{(1+q)^{2d}} \tag{8.23}$$

$$= \tau_a^{2d}(2\pi T)^{2d-1}\frac{e^{E/(2T)}|\Gamma(d+iE/(2\pi T))|^2}{\Gamma(2d)}. \tag{8.24}$$

The Beta integral converges for $d > 0$. The real-time expression is understood as the boundary value of the analytic function, rather than an unregulated integral across its singularity. Finite microscopic regulators modify the ultraviolet part; the displayed result is the low-energy scaling transform. The full leading current is $I = e|\gamma|^2[\mathcal{P}_d(E,T) - \mathcal{P}_d(-E,T)]/(2\pi a)^2$. It satisfies detailed balance, $\mathcal{P}_d(-E,T) = e^{-E/T}\mathcal{P}_d(E,T)$, and is odd in E. The large-argument asymptotic of the Gamma function recovers (7.34) as $T \to 0$ at fixed positive E. Differentiating at $E = eV_{\rm bias} = 0$ yields

$$G_{\rm link}(T) \equiv \left.\frac{\partial I}{\partial V_{\rm bias}}\right|_0 = \frac{e^2}{2\pi}\left|\frac{\gamma}{u}\right|^2(\pi T\tau_a)^{2/K-2}f(K), \tag{8.25}$$

$$f(K) = \frac{\cos(\pi/K)\Gamma(\frac{1}{2}-\frac{1}{K})\Gamma(1/K)}{2\sqrt{\pi}} = 2^{2/K-2}\frac{\Gamma(1/K)^2}{\Gamma(2/K)}. \tag{8.26}$$

The second form of f follows from reflection and duplication identities and is finite and positive for every $K > 0$. At values where a cosine zero multiplies a Gamma pole, it gives the removable limit. In particular, $f(1) = 1$ and the free spinless weak-link conductance is independent of T at this order. The integral above equivalently obeys $\int dz\, z/[i\sinh(z-i0^+)]^{2d} = -i\pi\, 2^{2d-2}\Gamma(d)^2/\Gamma(2d)$.

$$\mathcal{T}_{\rm link}(T) \equiv \frac{2\pi G_{\rm link}}{e^2} = \left|\frac{\gamma}{u}\right|^2(\pi T\tau_a)^{2/K-2}f(K). \tag{8.27}$$

Both $|\Delta\mu|$ and T act as infrared cutoffs. A finite system adds a scale of order u/L; this is distinct from the ultraviolet scale $E_a = u/a$. For $K > 1$, lowering all infrared scales eventually makes the weak-link expansion uncontrolled. The low-energy flow then heals the link and approaches the connected-wire fixed point. The geometry change is represented by the two fixed points in Fig. 8.3; it is not obtained by allowing a perturbative transmission ratio to exceed one.

8.4 Boundary duality and the domain of the two expansions

In their respective weak-coupling domains, the two zero-temperature formulas are

$$I_{\rm link} \simeq \frac{e^2}{2\pi}V_{\rm bias}\left|\frac{\gamma}{u}\right|^2\frac{(|eV_{\rm bias}|\tau_a)^{2/K-2}}{\Gamma(2/K)}, \tag{8.28}$$

$$I_{\rm trans} \simeq I_{\rm clean} - \frac{e^2}{2\pi}V_{\rm bias}\left|\frac{\lambda_{\rm bs}}{u}\right|^2\frac{(|eV_{\rm bias}|\tau_a)^{2K-2}}{\Gamma(2K)}. \tag{8.29}$$

The source's form $I_{\rm clean} = (e^2/2\pi)V_{\rm bias}$ applies with the one-channel ideal-reservoir convention stated above. The weak-link and weak-backscattering correction formulas exchange under

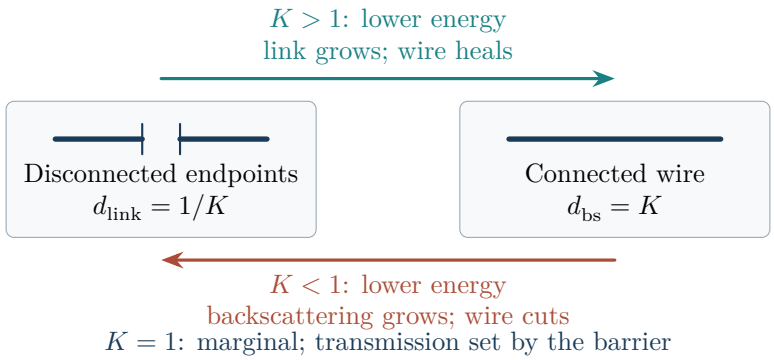


Figure 8.3: Boundary flow directions for a single local weak link or impurity in a spinless LL (schematic). The two geometries give controlled expansions near opposite fixed points. Lowering $\max(T, |\Delta\mu|, u/L)$ strengthens a perturbation with dimension below one and weakens one with dimension above one. The arrows indicate the infrared tendency, not a calculated flow at intermediate transmission.

$K \leftrightarrow K^{-1}$ and the formal replacement of their perturbing amplitudes. The reason is the boundary operator: at disconnected endpoints ϕ is pinned and the tunnelling vertex contains the dual phase θ, with dimension $1/K$. At the connected fixed point, θ cancels from $\psi_L^\dagger \psi_R$ and the impurity vertex contains 2ϕ, with dimension K. For a boundary operator of scaling dimension d, write its dimensionless coupling as $g(E)$ with a matched ultraviolet value $g_0 \propto \gamma/u$ or λ_{bs}/u. At the scale $E_{\text{IR}} = \max(T, |\Delta\mu|, u/L)$, the leading boundary flow is

$$\frac{dg}{d\ell_{\text{RG}}} = (1-d)g + \cdots, \qquad \ell_{\text{RG}} = \ln \frac{E_a}{E_{\text{IR}}}, \qquad g(E_{\text{IR}}) = g_0 (E_{\text{IR}}/E_a)^{d-1}. \tag{8.30}$$

The omitted terms matter when $|g|$ is no longer small. Duality relates the fields, operators, and boundary conditions of these two fixed-point expansions. It does not equate their bare amplitudes, microscopic cutoff matching, full current–voltage curves, or domains of validity. For attractive interactions $K > 1$, $d_{\text{link}} < 1$: a nonzero link grows as the infrared energy decreases. The system flows toward the connected-wire fixed point, where $d_{\text{bs}} > 1$ and weak backscattering becomes less important at low energy. For repulsive interactions $K < 1$, the connected wire with a nonzero impurity is unstable at sufficiently low energy. The flow approaches disconnected reflecting endpoints, where the weak-link expansion becomes increasingly accurate. The opposite flow directions are shown in Fig. 8.3. At $K = 1$, both leading scaling dimensions equal one; the transmission is marginal and depends on the barrier strength. For $K > 1$, the weak-link condition is

$$|\gamma/u|^2 (|E|\tau_a)^{2/K-2} \ll 1. \tag{8.31}$$

It requires energies above a crossover scale, parametrically $E \gg E_a |\gamma/u|^{K/(K-1)}$, while still $E \ll E_a$. The connected-wire correction for $K > 1$ instead decreases as E^{2K-2} and is controlled at sufficiently low energy if the impurity is weak in that description. For $K < 1$, the weak-link ratio decreases as $E^{2/K-2}$ and is controlled toward low energy. The weak-backscattering condition is $|\lambda_{\text{bs}}/u|^2 (|E|\tau_a)^{2K-2} \ll 1$; for $K < 1$ it requires $E \gg E_a |\lambda_{\text{bs}}/u|^{1/(1-K)}$. The two microscopic crossover amplitudes cannot be identified without additional matching. The source's qualitative bias sketches express the growing or diminishing correction near the two fixed points. Figure 8.4 plots explicit weak-link formulas over a range where their perturbative ratios remain small, so the trend can be inspected without extrapolating an invalid expansion. Whether a chosen expansion is controlled depends on the interaction parameter, the bare amplitude, and the energy at which the system is probed. A small bare coupling alone is insufficient.

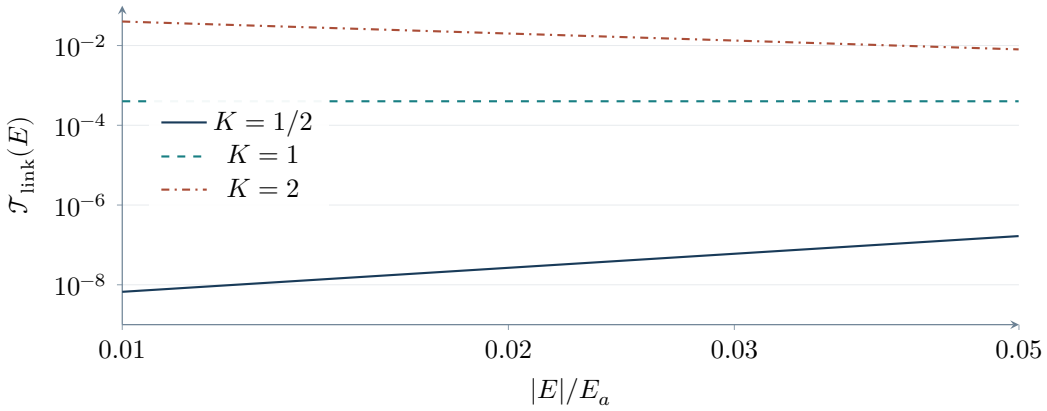


Figure 8.4: Calculated leading weak-link ratios $\mathcal{T}_{\rm link} = g_0^2(|E|/E_a)^{2/K-2}/\Gamma(2/K)$, with $g_0 = |\gamma/u| = 0.02$ and $E_a = u/a$. All curves stay at or below 0.04 in the plotted low-energy range, so the comparison stays within the weak-link domain. The ultraviolet exponential in (7.38) is omitted in this scaling plot. The attractive curve grows toward low energy and must eventually be replaced by another expansion.

9 Boundary renormalization: from fluctuations to transport

The preceding transport integrals reveal powers of energy and temperature. The renormalization group (RG) explains why those powers occur, which expansion becomes more accurate at low energy, and when a nominally weak contact ceases to be weak. We now derive the flow in (8.30), starting from the same spinless Gaussian Luttinger liquid. The reference state is the ground state of the appropriate unperturbed geometry: one connected wire, or two disconnected reflecting half-wires. The interaction parameter satisfies $K > 0$, the mode velocity is $u > 0$, and $\hbar = k_B = 1$.

The single-impurity fixed-point structure discussed here is the Kane–Fisher boundary problem [4, 5]. This addition develops the Gaussian action and the leading local flow explicitly. It does not add resonant double barriers, random disorder, spin, or a new bulk interaction to the model.

9.1 A local scatterer still feels the entire wire

Let τ denote imaginary time and $\Pi(x) = \partial_x\theta(x)/\pi$ the momentum conjugate to $\phi(x)$, so that $[\phi(x), \Pi(x')] = i\delta(x - x')$. At zero temperature, $\tau \in \mathbb{R}$. The phase-space Euclidean action is

$$S_E[\phi, \Pi] = \int d\tau\, dx \left[-i\Pi\partial_\tau\phi + \frac{u\pi K}{2}\Pi^2 + \frac{u}{2\pi K}(\partial_x\phi)^2\right], \tag{9.1}$$

$$\frac{u\pi K}{2}\Pi^2 - i\Pi\partial_\tau\phi = \frac{u\pi K}{2}\left(\Pi - \frac{i\partial_\tau\phi}{u\pi K}\right)^2 + \frac{(\partial_\tau\phi)^2}{2u\pi K}. \tag{9.2}$$

The Gaussian integral over Π contributes a field-independent determinant. Its remaining action is

$$S_0[\phi] = \frac{1}{2\pi K}\int d\tau\, dx \left[\frac{1}{u}(\partial_\tau\phi)^2 + u(\partial_x\phi)^2\right]. \tag{9.3}$$

The fields are dimensionless. Since $[u]$ = length/time, each term in the integral is dimensionless after integrating over x and τ.

For a point impurity at $x = 0$, only $\phi_0(\tau) \equiv \phi(0, \tau)$ enters the local vertex. Integrating over all other field values gives its effective action. Use the time transform

$$\phi(x, \omega) = \int_{-\infty}^{\infty} d\tau\, e^{i\omega\tau}\phi(x, \tau), \qquad \phi(x, \tau) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega\tau}\phi(x, \omega). \tag{9.4}$$

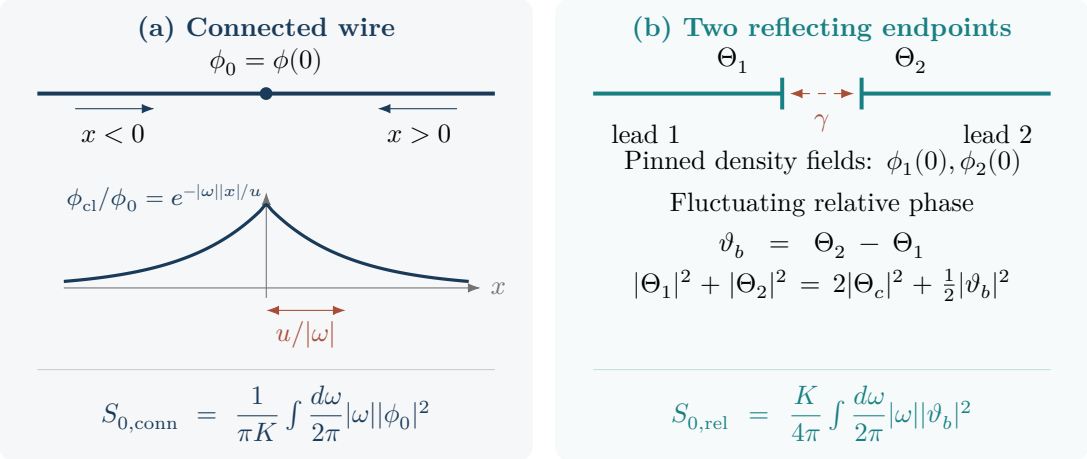


Figure 9.1: The bulk determines the boundary kernel. In (a), holding the impurity value fixed gives the calculated profile $\phi_{\rm cl}/\phi_0 = e^{-|\omega||x|/u}$ for a single nonzero imaginary-time frequency; the horizontal scale is $u/|\omega|$. Integrating both spatial halves gives (9.8). In (b), each reflecting lead supplies one half-line contribution; the common phase decouples, leaving (9.16). The device geometry and weak-link arrow are schematic. Their different fluctuating fields produce dimensions K and $1/K$, respectively.

For each nonzero ω, the stationary field at fixed $\phi_0(\omega)$ obeys

$$(-u\partial_x^2 + \omega^2/u)\phi_{\rm cl}(x,\omega) = 0, \qquad \phi_{\rm cl}(x,\omega) = \phi_0(\omega)e^{-|\omega||x|/u}. \tag{9.5}$$

It decays into both sides of the infinite wire. A fluctuation about this solution vanishes at $x = 0$; integration by parts makes its cross term with $\phi_{\rm cl}$ vanish. Its Gaussian determinant is independent of ϕ_0, so evaluating $S_0[\phi_{\rm cl}]$ gives the exact quadratic boundary action. The zero-frequency global sector is fixed separately; neutral correlators below depend only on field differences.

The two equal contributions, from the time derivative and the spatial derivative, give

$$\int_{-\infty}^{\infty} dx \left[\frac{\omega^2}{u}|\phi_{\rm cl}|^2 + u|\partial_x\phi_{\rm cl}|^2\right] = \frac{2\omega^2}{u}|\phi_0|^2\int_{-\infty}^{\infty} dx\, e^{-2|\omega||x|/u} \tag{9.6}$$

$$= \frac{2\omega^2}{u}|\phi_0|^2\frac{u}{|\omega|} = 2|\omega|\,|\phi_0|^2. \tag{9.7}$$

Consequently,

$$S_{0,\rm conn}[\phi_0] = \frac{1}{\pi K}\int \frac{d\omega}{2\pi}|\omega|\,|\phi_0(\omega)|^2 = \frac{1}{2}\int \frac{d\omega}{2\pi}\frac{2|\omega|}{\pi K}|\phi_0(\omega)|^2. \tag{9.8}$$

The kernel is proportional to $|\omega|$, rather than ω^2: the boundary coordinate is coupled to a continuum of gapless bulk modes. A slow boundary fluctuation penetrates a distance $u/|\omega|$ into the wire, as shown in Fig. 9.1. This is the origin of its long-time logarithmic variance. No additional dissipative bath has been introduced.

9.2 Two boundary operators and two scaling dimensions

For a real Gaussian variable with action $\frac{1}{2}XAX$, its covariance is A^{-1}. Equation (9.8) therefore gives

$$\langle\phi_0(\omega)\phi_0(\omega')\rangle_0 = 2\pi\delta(\omega+\omega')\frac{\pi K}{2|\omega|}. \tag{9.9}$$

Put a frequency cutoff Λ of order $E_a = u/a = \tau_a^{-1}$ on this low-energy action. At $|\tau|\Lambda \gg 1$,

$$\langle[\phi_0(\tau) - \phi_0(0)]^2\rangle_0 = \int_{-\Lambda}^{\Lambda} \frac{d\omega}{2\pi} 2(1-\cos\omega\tau)\frac{\pi K}{2|\omega|} \tag{9.10}$$

$$= K\int_0^{\Lambda} \frac{d\omega}{\omega}(1-\cos\omega\tau) = K\ln(\Lambda|\tau|) + O(1). \tag{9.11}$$

The $O(1)$ term depends on the cutoff. It changes the matching of a vertex amplitude, but not its scaling dimension. This frequency regulator is used to derive logarithms; it does not replace the ordered real-time $a + iut$ prescription in the transport integrals.

With the cutoff-matched normal ordering and number-changing Klein factors already established, the impurity operator is

$$\mathcal{V}_{\rm bs} = F_L^\dagger F_R {:}e^{-2i\phi_0}{:}, \qquad \langle \mathcal{V}_{\rm bs}(\tau)\mathcal{V}_{\rm bs}^\dagger(0)\rangle_0 \propto e^{-\frac{2^2}{2}K\ln(|\tau|/\tau_a)} = \left(\frac{\tau_a}{|\tau|}\right)^{2K}. \tag{9.12}$$

The dimension Δ of a local operator is defined by a two-point function proportional to $|\tau|^{-2\Delta}$. Hence $\Delta_{\rm bs} = K$. The neutral product makes the zero-mode/Klein selection rule explicit before its factors cancel.

Now start from two disconnected reflecting leads, with lead 1 on $x < 0$ and lead 2 on $x > 0$. Each has $\phi_j(0)$ pinned and zero endpoint number current, as in Sec. 7.1; $j = 1, 2$ labels leads. The dual field θ_j remains free to fluctuate. Its bulk action on each half-line is

$$S_0[\theta_j] = \frac{K}{2\pi}\int_{\text{lead } j} dx\, d\tau \left[\frac{1}{u}(\partial_\tau\theta_j)^2 + u(\partial_x\theta_j)^2\right]. \tag{9.13}$$

This follows by the same Gaussian integration with ϕ and θ exchanged and K replaced by K^{-1}. It is compatible with the reflecting boundary condition $\partial_x\theta_j(0) = 0$ on the unperturbed dynamical field. When computing a boundary effective action, an intermediate path integral holds its boundary value fixed; integrating over that value restores the free boundary condition.

Write $\Theta_j(\tau) = \theta_j(0,\tau)$. There is now only one half-line per lead, giving half the spatial integral used above. Thus

$$S_{0,\rm cut} = \frac{K}{2\pi}\int\frac{d\omega}{2\pi}|\omega|(|\Theta_1|^2 + |\Theta_2|^2), \tag{9.14}$$

$$\vartheta_b \equiv \Theta_2 - \Theta_1, \qquad \Theta_c \equiv \frac{\Theta_1+\Theta_2}{2}, \qquad |\Theta_1|^2 + |\Theta_2|^2 = 2|\Theta_c|^2 + \frac{1}{2}|\vartheta_b|^2, \tag{9.15}$$

$$S_{0,\rm rel}[\vartheta_b] = \frac{K}{4\pi}\int\frac{d\omega}{2\pi}|\omega||\vartheta_b(\omega)|^2. \tag{9.16}$$

The common phase Θ_c decouples from tunnelling. Inverting the relative-phase kernel gives

$$\langle\vartheta_b(\omega)\vartheta_b(\omega')\rangle_0 = 2\pi\delta(\omega+\omega')\frac{2\pi}{K|\omega|}, \tag{9.17}$$

$$\langle[\vartheta_b(\tau) - \vartheta_b(0)]^2\rangle_0 = \frac{4}{K}\int_0^{\Lambda}\frac{d\omega}{\omega}(1-\cos\omega\tau) = \frac{4}{K}\ln(\Lambda|\tau|) + O(1). \tag{9.18}$$

The relative boundary phase ϑ_b is distinct from the canonical field $\theta(x)$ and from the common endpoint phase Θ_c.

Because $d_{j,\rm end} \propto F_j{:}e^{i\Theta_j}{:}$, the weak-link operator has oscillator part $e^{i\vartheta_b}{:}$:

$$\mathcal{V}_{\rm link} = F_1^\dagger F_2 {:}e^{i\vartheta_b}{:}, \qquad \langle\mathcal{V}_{\rm link}(\tau)\mathcal{V}_{\rm link}^\dagger(0)\rangle_0 \propto e^{-\frac{1}{2}\frac{4}{K}\ln(|\tau|/\tau_a)} = \left(\frac{\tau_a}{|\tau|}\right)^{2/K}. \tag{9.19}$$

Thus $\Delta_{\rm link} = 1/K$. Separately, one endpoint has $\langle[\Theta_j(\tau) - \Theta_j(0)]^2\rangle_0 \sim (2/K)\ln(|\tau|/\tau_a)$ and electron dimension $1/(2K)$. The two independent endpoint dimensions add. This derivation also explains why deleting the ϕ term from a bulk electron correlator gives the wrong answer: reflection changes the remaining θ fluctuations.

9.3 One RG step, including its normalization

Let s label the expansion, $s =$ bs or $s =$ link. In the matched contact convention used in the previous section, define

$$g_{\rm bs,0} = \frac{\lambda_{\rm bs}}{u}, \qquad g_{\rm link,0} = \frac{\gamma}{u}, \qquad S_{{\rm pert},s} = \frac{1}{2\pi\tau_a}\int d\tau\,[g_{s,0}\mathcal{V}_s(\tau) + g^*_{s,0}\mathcal{V}^\dagger_s(\tau)]. \tag{9.20}$$

The two $g_{s,0}$ are generally complex and dimensionless; $\lambda_{\rm bs}$ and γ retain dimensions of velocity. The numerical 2π is retained rather than absorbed into g_s. For a different microscopic contact amplitude, a fixed matching constant multiplies $g_{s,0}$; the powers derived below do not determine that constant. The colons are the previously matched bosonization convention. To expose the normalization in a shell calculation, represent a matched vertex by the ordinary cutoff exponential $F_s e^{i\alpha_s X_\Lambda}$, where $(X, \alpha_s) = (\phi_0, -2)$ for backscattering and $(\vartheta_b, 1)$ for the link. A fixed microscopic matching constant is included in the already defined coupling. If $W_{s,\Lambda} = F_s{:}e^{i\alpha_s X_\Lambda}{:}_0$ instead denotes the raw Wick exponential in the unperturbed Gaussian vacuum, the dictionary is

$$F_s e^{i\alpha_s X_\Lambda} = e^{-\alpha_s^2 C_\Lambda/2} W_{s,\Lambda}, \qquad C_\Lambda = \langle X_\Lambda^2\rangle_0. \tag{9.21}$$

The same infrared regulator is understood on both sides. Neutral products have a finite limit when it is removed. The prefactor in (9.21) changes with the cutoff; it must accompany the raw Wick operator in comparing successive effective theories.

Choose $b = e^{d\ell_{\rm RG}} > 1$ and split a boundary field into $X = X_< + X_>$, with slow frequencies $|\omega| < \Lambda/b$ and fast frequencies $\Lambda/b < |\omega| < \Lambda$. The covariance is diagonal in frequency, so the two fields are independent. Direct integration gives

$$\langle\phi^2_{0,>}\rangle_0 = \int_{\Lambda/b<|\omega|<\Lambda} \frac{d\omega}{2\pi}\frac{\pi K}{2|\omega|} = \frac{K}{2}\ln b, \tag{9.22}$$

$$\langle\vartheta^2_{b,>}\rangle_0 = \int_{\Lambda/b<|\omega|<\Lambda} \frac{d\omega}{2\pi}\frac{2\pi}{K|\omega|} = \frac{2}{K}\ln b. \tag{9.23}$$

For the *ordinary* exponential of a zero-mean Gaussian variable, the characteristic function is $\langle e^{i\alpha X_>}\rangle_0 = \exp[-\alpha^2\langle X_>^2\rangle_0/2]$. Applying this identity to the matched vertices at their respective cutoffs yields

$$\langle\mathcal{V}_{\rm bs}\rangle_> = b^{-K}\mathcal{V}_{{\rm bs},\Lambda/b}, \qquad \langle\mathcal{V}_{\rm link}\rangle_> = b^{-1/K}\mathcal{V}_{{\rm link},\Lambda/b}. \tag{9.24}$$

For a raw Wick-ordered fast factor its expectation would instead be one. Using that representation gives the same result only after keeping the ratio of matching factors, $\exp[-\alpha_s^2(C_\Lambda - C_{\Lambda/b})/2] = b^{-\Delta_s}$. We are therefore not assigning a nontrivial Gaussian average to a normal-ordered exponential without its matching coefficient. The logarithm comes from a narrow frequency shell, or equivalently from short-time fluctuations on the scale being eliminated.

The effective perturbation is defined by

$$e^{-S_{\rm eff,pert}[X_<]} = \langle e^{-S_{\rm pert}[X_< + X_>]}\rangle_>, \qquad S_{\rm eff,pert} = \langle S_{\rm pert}\rangle_> + O(g_s^2). \tag{9.25}$$

After this first-order average, restore the cutoff by $\tau = b\tau'$, $\omega' = b\omega$. The Gaussian $|\omega|$ action is invariant if $X'(\tau') = X_<(b\tau')$: the boundary field needs no multiplicative rescaling. The

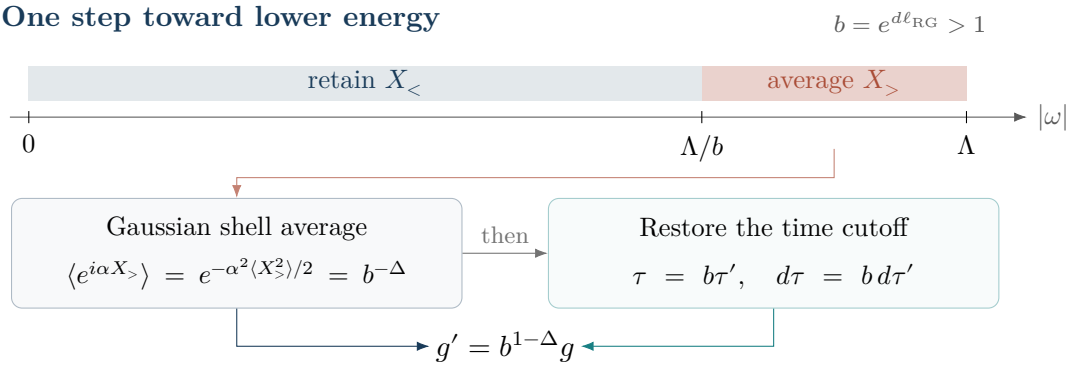


Figure 9.2: Schematic frequency-shell RG. Both signs of frequency are integrated, although the drawing shows $|\omega|$ only; the shell width is exaggerated. Gaussian fluctuations contribute $-\Delta$ to the scaling eigenvalue, while the time measure contributes $+1$. The Gaussian $|\omega|$ action retains its form. This is a boundary calculation with one integrated coordinate, giving $1-\Delta$ rather than the bulk value $2-\Delta_{\mathrm{bulk}}$.

perturbation, however, gains a factor b from $d\tau = b\,d\tau'$, in addition to the vertex factor $b^{-\Delta_s}$. Therefore

$$g_s' = b^{1-\Delta_s} g_s, \qquad g_s' - g_s = (1-\Delta_s) g_s\, d\ell_{\mathrm{RG}} + O((d\ell_{\mathrm{RG}})^2), \tag{9.26}$$

$$\frac{dg_{\mathrm{bs}}}{d\ell_{\mathrm{RG}}} = (1-K) g_{\mathrm{bs}} + \cdots, \qquad \frac{dg_{\mathrm{link}}}{d\ell_{\mathrm{RG}}} = (1-K^{-1}) g_{\mathrm{link}} + \cdots. \tag{9.27}$$

These are separate expansions around different boundary conditions. The two couplings are not simultaneous small coordinates for the entire crossover. Figure 9.2 separates the two factors in this RG step.

The number 1 in $1-\Delta_s$ is the dimension of the integration measure of a point impurity, $d\tau$. For a translation-invariant perturbation $\int dx\, d\tau\, y\mathcal{O}(x,\tau)$ of bulk dimension Δ_{bulk}, both x and τ rescale and the corresponding linear eigenvalue is $2-\Delta_{\mathrm{bulk}}$. The impurity vertex $e^{2i\phi}$ has dimension K when restricted to a point of the connected homogeneous theory, but its coupling is integrated only over time. The reflecting endpoint changes the dimension as well as the geometry; its electron operator must be derived with the boundary condition. A bulk cosine, an isolated impurity, and spatially random backscattering consequently require different RG calculations.

Writing $g_s = r_s e^{i\delta_s}$, with $r_s = |g_s| \geq 0$, the linear flow has a real coefficient:

$$\frac{dr_s}{d\ell_{\mathrm{RG}}} = (1-\Delta_s) r_s, \qquad \frac{d\delta_s}{d\ell_{\mathrm{RG}}} = 0 \quad (r_s \neq 0). \tag{9.28}$$

For one isolated contact its static phase can be moved into a boundary-field or relative-fermion phase convention. This statement does not remove relative phases between several contacts. A bias phase is time dependent: in real time it is $e^{+i\Delta\mu t}$ for the weak-link term $d_1^\dagger d_2$ and $e^{-i\Delta\mu t}$ for the backscattering term $\psi_L^\dagger \psi_R$, with the bias definitions used earlier. It is not part of the constant δ_s. The equilibrium shell calculation applies above the bias scale; the oscillating phase limits its infrared use.

9.4 What is generated beyond the linear flow?

The next term of the cumulant expansion is

$$S_{\mathrm{eff,pert}} = \langle S_{\mathrm{pert}}\rangle_> - \frac{1}{2}\left(\langle S_{\mathrm{pert}}^2\rangle_> - \langle S_{\mathrm{pert}}\rangle_>^2\right) + \cdots. \tag{9.29}$$

Two opposite-charge vertices have a short-time expansion containing the identity and derivatives of the boundary field. Two equal-charge vertices can generate a second harmonic. Around the connected fixed point, $e^{\pm 4i\phi_0}$ has dimension $4K$; around the cut fixed point, $e^{\pm 2i\vartheta_b}$ has dimension $4/K$. Charge neutrality prevents two fundamental vertices alone from producing one fundamental vertex at second order. Thus the first feedback into its own beta function is at higher order, or through other generated couplings; the ellipses in (9.27) are not specified universal coefficients.

Higher harmonics need not always be irrelevant. For example $4K < 1$ makes the second backscattering harmonic relevant at the connected fixed point. If the fundamental impurity is present and dominates initially, the leading flow remains the local asymptote for sufficiently small couplings. If a symmetry removes it, one must identify the first allowed operator and redo the stability analysis; substituting K into a forbidden operator's beta function has no physical meaning.

The two perturbations act at one spatial point. Their cumulants can renormalize boundary energies, boundary operators, and local analytic terms such as $(\partial_\tau \phi_0)^2$; they do not generate an extensive change of the Gaussian action throughout an infinite clean wire. Thus K and u are fixed bulk parameters in this single-impurity RG problem. For instance a local ω^2 boundary term is less important at low frequency than the existing $|\omega|$ kernel. This reasoning does not apply to an extensive bulk cosine, a finite density of impurities, or a spatially inhomogeneous interaction profile. Those problems can require additional running bulk or environmental parameters.

9.5 Crossover scales and when to stop the flow

Introduce the running energy $\varepsilon = E_a e^{-\ell_{\mathrm{RG}}}$, which is positive and is distinct from the signed bias $E = \Delta\mu$. Solving the linear differential equation gives

$$g_s(\varepsilon) = g_{s,0} \exp[(1-\Delta_s)\ln(E_a/\varepsilon)] = g_{s,0}(\varepsilon/E_a)^{\Delta_s - 1}. \tag{9.30}$$

An operator with $\Delta_s < 1$ is relevant: its relative effect grows as the energy is reduced. With $\Delta_s > 1$ it is irrelevant. A dimension equal to one is marginal at this linear order and requires further information to decide its ultimate behavior.

For $0 < |g_{s,0}| \ll 1$ and $\Delta_s < 1$, extrapolating the weak-coupling solution to $|g_s|$ of order one estimates a crossover scale:

$$\ell_s^* \simeq \frac{\ln(1/|g_{s,0}|)}{1-\Delta_s}, \qquad \varepsilon_s^* \sim E_a |g_{s,0}|^{1/(1-\Delta_s)}. \tag{9.31}$$

The symbol $\sim$ is essential. This estimate fixes a parametric energy dependence, not a universal numerical threshold; the weak-coupling solution is already uncontrolled near ℓ_s^*. In the two cases,

$$\varepsilon_{\mathrm{bs}}^* \sim E_a |\lambda_{\mathrm{bs}}/u|^{1/(1-K)} \quad (K<1), \qquad \varepsilon_{\mathrm{link}}^* \sim E_a |\gamma/u|^{K/(K-1)} \quad (K>1). \tag{9.32}$$

At finite temperature, imaginary time has circumference $\beta = 1/T$; nonzero Matsubara frequencies are $\omega_n = 2\pi n T$. Bias suppresses long-time contributions by oscillations at frequency $|E|$, and a segment of length L has a level or traversal scale of order u/L. To logarithmic accuracy the homogeneous-wire flow therefore stops at

$$E_{\mathrm{IR}} \sim \max(T, |\Delta\mu|, u/L), \qquad \ell_{\mathrm{stop}} \simeq \ln(E_a/E_{\mathrm{IR}}). \tag{9.33}$$

Factors such as 2π depend on which observable and finite-size geometry defines the stop; the maximum is not a replacement for the finite-temperature scaling function. The logarithmic regime requires $E_{\mathrm{IR}} \ll E_a$. Perturbation theory is controlled only if every relevant running

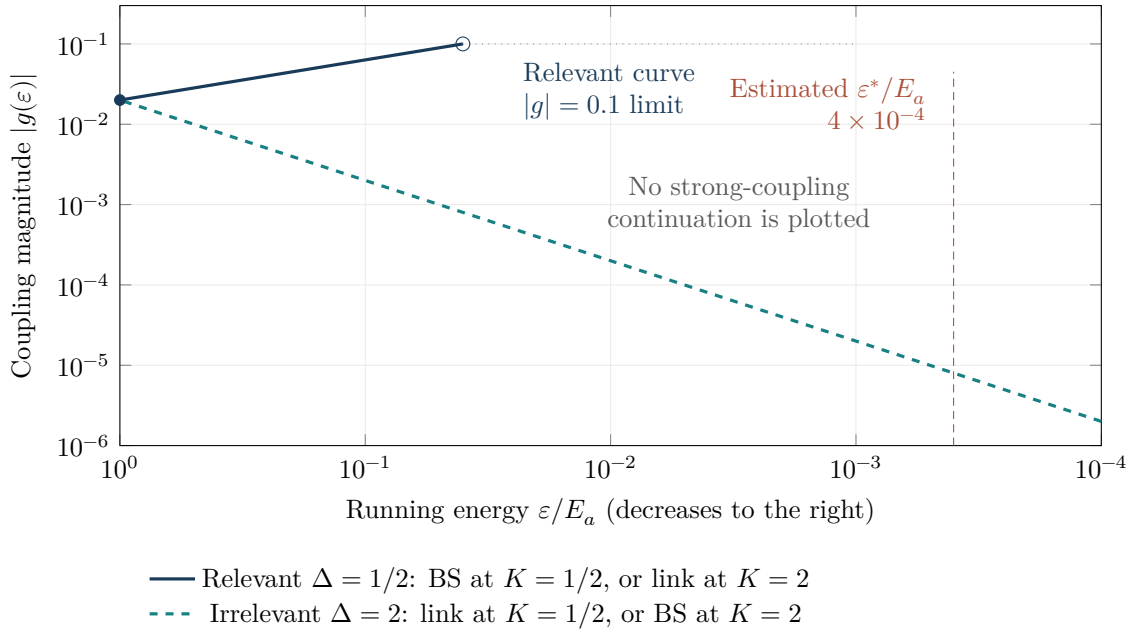


Figure 9.3: Calculated linear-flow solutions $|g(\varepsilon)| = 0.02(\varepsilon/E_a)^{\Delta-1}$, with independently matched initial magnitudes 0.02. Each curve describes either member of the indicated dual pair of examples; it is not an interpolation between their fixed points. The relevant curve is drawn only down to $\varepsilon/E_a = 0.04$, where $|g| = 0.1$. The dashed vertical marker is the *parametric estimate* obtained by formally setting $|g| = 1$; no calculated flow is asserted there. The irrelevant curve remains weak over the full displayed interval. All energies and couplings are dimensionless ratios.

coupling is still small at the stop. If it becomes large first, finite temperature does not restore the failed expansion at that same scale.

In a finite interacting segment attached to noninteracting reservoirs, u/L can mark a change of the effective boundary kernel and a crossover to lead-controlled behavior. It is then the stop of the *homogeneous LL flow*; it is not a claim that the entire open device has no lower-energy dynamics. Similarly, a charging energy or a separate island would require additional zero-mode data, absent from the present half-line model.

For a concrete comparison, take an independently matched $|g_{s,0}| = 0.02$ at the ultraviolet scale. With $K = 1/2$, backscattering has $\Delta_{\mathrm{bs}} = 1/2$ and grows as $0.02\sqrt{E_a/\varepsilon}$, whereas a weak link has $\Delta_{\mathrm{link}} = 2$ and decreases as $0.02\varepsilon/E_a$. With $K = 2$ the two dimensions and trends exchange. In either relevant example the crossover estimate is $\varepsilon_s^*/E_a \sim (0.02)^2 = 4 \times 10^{-4}$. At $E_{\mathrm{IR}}/E_a = 0.1$, the relevant coupling is 0.0632; at 10^{-5} its formal value is 6.32, which signals a failed extrapolation. Figure 9.3 draws the relevant solution only while $|g_s| \le 0.1$, a conservative display limit, and leaves its uncontrolled continuation unplotted. Equal initial numbers in this comparison do not identify the two bare amplitudes of one microscopic barrier.

9.6 Boundary stability and the meaning of duality

The signs of the two linear eigenvalues determine the local stability summarized in Table 9.1.

Table 9.1: Local stability of the two boundary fixed points for the homogeneous spinless model with an allowed single-electron perturbation.

Regime	Connected wire: $g_{\rm bs} = 0$	Cut wire: $g_{\rm link} = 0$
$K < 1$	Unstable to allowed backscattering	Stable against weak tunnelling
$K > 1$	Stable against weak backscattering	Unstable to allowed tunnelling
$K = 1$	Marginal at this order	Marginal at this order

At $K = 1$, the Gaussian spinless model can be refermionized with velocity u. A single energy-independent quadratic point scatterer then has an energy-independent scattering matrix: its transmission labels a marginal family of boundary conditions. This supplies the extra information beyond the vanishing linear beta function. It is a statement about this model; $K = 1$ alone does not forbid every microscopic interaction or additional boundary operator. In particular, g_4 may shift u while $g_2 = 0$ keeps $K = 1$.

For $K < 1$, a growing backscattering cosine tends to pin ϕ_0. Its adjacent minima are separated by π, since the vertex is $e^{\pm 2i\phi_0}$. The relation $j_N(0,t) = \partial_t \phi_0(t)/\pi$ shows that a jump by $\pm\pi$ transfers ± 1 particle. Rare transitions between adjacent minima can therefore be described as electron tunnelling across the cut. In the opposite regime, $K > 1$, the growing link locks the relative endpoint phase and the connected-wire expansion becomes appropriate. This physical interpretation identifies the operators near the other boundary fixed point; the weak-coupling calculation itself does not determine a nonperturbative crossover trajectory.

The exchange of density pinning and relative-phase fluctuations explains the dimensions $K \leftrightarrow 1/K$. It does not imply $g_{\rm link,0} = 1/g_{\rm bs,0}$, or determine their matching constants. A schematic arrow between fixed points summarizes the endpoint stabilities, not a calculated beta function for a transmission variable at every intermediate value. Resonant structures or additional symmetry constraints may have further fixed points and lie outside this conclusion.

9.7 From the running coupling back to conductance

The RG reorganization must reproduce, rather than multiply twice, the power already present in the explicit rate integral. In the cutoff-matched scheme of the preceding transport section, its zero-temperature leading current magnitude for either expansion is

$$|I_s(E,0)| \simeq \frac{e|E|}{2\pi\Gamma(2\Delta_s)} |g_{s,0}|^2 (|E|/E_a)^{2\Delta_s - 2} \tag{9.34}$$

$$= \frac{e|E|}{2\pi\Gamma(2\Delta_s)} |g_s(|E|)|^2, \qquad 0 < |E| \ll E_a, \quad |g_s(|E|)| \ll 1. \tag{9.35}$$

Here $I_{\rm link}$ is the weak-link current and $I_{\rm bs}$ is the amount subtracted from the clean transmitted current, with their previously derived orientations. The Gamma function comes from the Fourier integral; the linear RG alone determines neither it nor a microscopic contact amplitude. The ultraviolet exponential tends to one in this scaling regime.

The first line uses the bare coupling and the explicit power. The second uses the running coupling and a coefficient evaluated at the observation scale. Using $|g_s(|E|)|^2$ *and* multiplying again by $(|E|/E_a)^{2\Delta_s - 2}$ would count the same shell fluctuations twice. The leading RG improvement is meaningful only while the chosen boundary expansion remains weak.

At $|E| \ll T \ll E_a$, differentiating the finite-temperature rate gives

$$G_s(T) \simeq \frac{e^2}{2\pi}|g_{s,0}|^2(2\pi T/E_a)^{2\Delta_s-2}\frac{\Gamma(\Delta_s)^2}{\Gamma(2\Delta_s)} \tag{9.36}$$

$$= \frac{e^2}{2\pi}\,\mathcal{C}_{\Delta_s}|g_s(T)|^2, \qquad \mathcal{C}_\Delta = (2\pi)^{2\Delta-2}\frac{\Gamma(\Delta)^2}{\Gamma(2\Delta)}. \tag{9.37}$$

For a weak link G_s is its conductance. For backscattering it is the positive linear conductance correction, so $G_{\rm trans} = G_{\rm clean} - G_{\rm bs}$. Setting $\Delta = 1/K$ reproduces (8.25); setting $\Delta = K$ gives the connected-wire correction in the same local bias convention. At $\Delta = 1$, $\mathcal{C}_1 = 1$ and the leading conductance is independent of T, as required at the refermionizable point.

For E/T of order one, the full Gamma-function scaling expression derived earlier is needed. Simply substituting $T \mapsto \max(T, |E|)$ reproduces only logarithmic scales, not that crossover function or its prefactor. Likewise, the ratio $I/V_{\rm bias}$ and differential conductance $dI/dV_{\rm bias}$ differ at finite bias; they coincide in the linear limit used in (9.36).

These equations predict $G_{\rm link} \propto T^{2/K-2}$ near a cut wire and $G_{\rm bs} \propto T^{2K-2}$ near a connected wire. The first is the controlled low-temperature asymptote for $K < 1$; the second is the controlled correction for $K > 1$. A growing correction is evidence that the reference boundary condition is becoming inappropriate, not evidence for a negative total conductance or a probability exceeding one. The absolute clean two-terminal conductance still requires a contact and reservoir model. For ideal noninteracting reservoirs its one-channel value is $e^2/(2\pi)$, while the homogeneous LL exponents apply over the energy window in which the contact sees the LL boundary kernel.

The physical content of the flow is therefore a competition: time rescaling enhances a local coupling by b, whereas Gaussian fluctuations reduce its vertex by $b^{-\Delta}$. The boundary condition fixes those fluctuations, the remaining experimental scales decide how far this competition runs, and the appropriately matched rate converts the running coupling into an observable.

A Notation and useful consistency checks

Table A.1: Objects that must remain distinct.

Symbol	Meaning and domain
$\xi_{\mathbf{k}}$	Bare one-particle energy $\epsilon_{\mathbf{k}} - \mu$.
$E_{\mathbf{k}}, Z_{\mathbf{k}}$	Quasiparticle resonance energy and coherent spectral weight.
$\Gamma_{\mathbf{k}}$	Positive energy half-width; probability rate is $2\Gamma_{\mathbf{k}}$.
$\chi^{R}_{\rho\rho}$	Density response to a positive potential-energy coupling; not a bosonic field.
Ψ, ψ_r	Microscopic fermion and slow physical branch fermion, $r = \pm 1$.
ϕ, θ, Π	Canonical density/phase pair, with $\Pi = \partial_x \theta/\pi$.
φ_r	Physical fermion phase $r\phi - \theta$.
χ_r	Propagating eigenfield $\phi/\sqrt{K} - r\sqrt{K}\theta$.
$\rho_r, \rho_{\mathrm{sm}}$	Branch and total smooth excess particle-number densities.
ρ_Q, j_Q	Charge density and charge current; $\rho_Q = -e\rho_{\mathrm{sm}}$, $j_Q = -ej_N$.
$F_r, \widehat{N}_r$	Dimensionless number-shifting Klein operator and excess branch number.
b_q, β_q	Density oscillator and diagonal interacting oscillator.
v_F, u	Bare branch velocity and interacting collective-mode velocity.
g_2, g_4	Density-Hamiltonian interbranch and intrabranch couplings (velocity units).
K, ξ	Positive Luttinger parameter and squeezing parameter $\xi = \frac{1}{2}\ln K$.
a, L	UV length cutoff and system size.
β, τ_a, E_a	Inverse temperature, cutoff time a/u, and cutoff energy u/a.
ℓ, κ, r	Reservoir, contour branch, and physical chirality indices.
$\Delta\mu$	Energy bias $\mu_1 - \mu_2$; $V_{\mathrm{bias}} = \Delta\mu/e$.

In the $K = 1$ limit, $g_2 = 0$ and $u = v_F + g_4/(2\pi)$; the special fully free reference also has $g_4 = 0$. The bulk local electron exponent is $(K + K^{-1})/2$, the reflecting-end exponent is $1/K$, and the two boundary perturbation dimensions are K and $1/K$. These checks must be applied in their respective geometries. Finite-ring Hamiltonians in the interacting section use the explicitly stated symmetric chemical-potential shift $H = H_{\mathrm{ref}} - (\pi v_F/L)\widehat{N}$; this does not alter the periodic spin structure. A finite-size fermion correlator must use the same generator, with $\psi_{\mathrm{ref}}(t) = e^{-i\pi v_F t/L}\psi(t)$.

References

[1] H. Bruus and K. Flensberg, *Many-Body Quantum Theory in Condensed Matter Physics: An Introduction*, Oxford University Press (2004). Publisher record.

[2] T. Giamarchi, *Quantum Physics in One Dimension*, Oxford University Press (2003). Publisher record.

[3] J. von Delft and H. Schoeller, "Bosonization for Beginners—Refermionization for Experts" (1998), arXiv:cond-mat/9805275.

[4] C. L. Kane and M. P. A. Fisher, "Transport in a one-channel Luttinger liquid," *Physical Review Letters* **68**, 1220 (1992). doi:10.1103/PhysRevLett.68.1220.

[5] C. L. Kane and M. P. A. Fisher, "Transmission through barriers and resonant tunneling in an interacting one-dimensional electron gas," *Physical Review B* **46**, 15233 (1992). doi:10.1103/PhysRevB.46.15233.